

1. **Experiment:** ProteinMPNN-Based Sequence Design
2. **Time:** 2025.11.14-2025.11.23
3. **Member:** Zhangdong Liu, Ziyang Yu, Yuxuan Wang, Zheyuan Deng, Ziqian He, Kaili Jin, Yulongzi Piao
4. **Method:**

(1) ProteinMPNN is a state-of-the-art graph neural network model for fixed-backbone protein sequence design, built upon message-passing neural network architecture. It takes protein backbone coordinates as direct input and outputs optimized amino acid sequences that stably fold into the given 3D conformation. The model learns universal sequence-structure relationships from high-resolution experimental protein structures, enabling accurate and robust sequence design for *de novo* generated backbones, including binding proteins, enzymes, and symmetric oligomers.

(2) In this project, we applied ProteinMPNN to design amino acid sequences for the 1,200 binder backbones generated by RFDiffusion3, covering three independent binding epitopes: Part A targeting Leu19 and Tyr31 of IL-6, Part B targeting Trp249 and Part C targeting Trp296 on IL-6Rα. For each binder backbone, 10 candidate sequences were generated and ranked according to the average per-residue ProteinMPNN score, from which the top-ranked sequence per conformation was selected, yielding approximately 400 candidate sequences for subsequent expression purification and functional screening.
5. **Result:**

Table 1. ProteinMPNN-designed candidate sequencing Targeting Leu19 and Tyr31 of IL-6.

ID	Sequence
part1_1p9m_11_model_9_1	MELIKLLKEIYKVVKDKEQKALILLIGEEYYKLGKLEEAKEYFKEALKLAKEAEKEGELMKMIKESLEKVEKEK
part1_1p9m_39_model_6_1	MTVEELKKEEIEKLEKEIEAAKAAGKPAEAAHLQEKLVLKSFLKELEEDKENKEALSISLRGTLALIDKEAAGEL
part1_1p9m_13_model_2_1	MDVEKKIEEIEKIEKEAEKAKKEAEKYKEEAKKLKEEKAPEEKVKELEKKAKEAEKEAKKLEEEIFKYLKEAAV
part1_1p9m_24_model_7_1	MELEKLIKEIFEKVEKYLLKKENASEEELKKLEEEIEKLVEEAYKAAKKEGKLAKFQEALLEIDKLL
part1_1p9m_6_model_8_1	TKEEELKKEALKAAASFIVENKKIAPDLVDEKLEKALKDYYIKEISEKFGKPKEEVEEIEKEIEKLEKIGGKVTEEA
part1_1p9m_20_model_8_1	MEEELKKKAEELAKKAEEAREKGDKKKAIEYLEELLKVLNEIKTDPEKEEKIKIAEVELAWEKGDWKAYLEALEKL
part1_1p9m_20_model_5_1	MIEEEAEKLAKEELEKVEEKFELVLKGPEFAIEVEKLAKEEAKKKIIEYLIEKKIPKEEAEKIAEKVLEKLK
part1_1p9m_37_model_7_1	KEELEKEMEKIMKEIEEAEEKGDKEKARELMKKLIELEEKYLK
part1_1p9m_0_model_6_1	IDPEKLKKEIKKLKEEKLEEFEPKYKELIKELAKKNLPEEEVLKEVEKFKYKEFEKEFEKEIEKLKK
part1_1p9m_4_model_3_1	SKEIEKEAEKAFEEKKYEEAIKLYEEALKKDPSKKLYYKEGIAFAYYELGKKKEEARKLFIEIAEEL
part1_1p9m_0_model_7_1	DLEVEMLKRLAEAAEKEGNWEELEGIGLELKKLGYEEEAKKVLERAIKGYKEKGLEEKAKKVVEIL
part1_1p9m_20_model_1_1	ILEKIKEYIKEKLRKGEVISLEELWKIAKEKFGITKEEAEEELKEAHEEAVFEVIREKAKEGASVEELIEFLEKAE
part1_1p9m_4_model_0_1	EEEEKRKLKREYQELIIEYEKAEKEEGKIEEAVKLLLEEILKLAEKLGLPKEEIKSIKEELEKLKKKI
part1_1p9m_39_model_1_1	DIFEEVLKLLREYNITEEQKEKIKELEEEYKALKEAGTKDNEELRNFFKEKVLDLIKEINVPEEIKKKILELKP
part1_1p9m_11_model_7_1	AVLEEFRRLLAELRALPLGTEEFAATLQRLVDLPEVPSEEDLAELERLLEAHLLEAGLTPEEARIIARGIARDIA

part1_1p9m_29_model_5_1	SALEELLKKEAEELAKKFQEKYPVPNLEKVKEKVYKKLKELEKKNEEISEEELIEELKKEL
part1_1p9m_5_model_2_1	EALKEIEPLKKELEEAEKAKDWRKAQELLDKLEELQKEAGISEEEA
part1_1p9m_20_model_7_1	DEEWEALEPVLAAADCPLETLRAARALRAERAAALPAAAPEDRPALLAAFRARADELLAAVKNPVTADYLRAWILDG
part1_1p9m_10_model_2_1	KIEELKKKAEEELIKNLDLSLEKIEEIEKKLKEIEKLKKELG
part1_1p9m_16_model_7_1	MEEVIKLAKEKGITLEEIEEAYKELEGVLSEEQRLEVLKELV
part1_1p9m_21_model_0_1	TTLTEEEIKEIEEIEIERLKKA AEITEDEEEAKEIEEIEIEKYKKLIEEAK
part1_1p9m_17_model_2_1	MSELAEIAASEEERARLALALVEALDLPPEEAAKVALELLKLSVEELRAIIAALPPEVQAQA
part1_1p9m_22_model_4_1	MELQKLFEYEEKKAEEIKSKNLP EEELKKALEKLEKEFE EEELKKL
part1_1p9m_20_model_6_1	YEEEEKKLIEELAKEIQEGAKKLGVELSEEDAKKIAEGLIEAFRKGEEINAELIEKLLIEVGIDPETAKKIAEFLV
part1_1p9m_9_model_0_1	MEILRKIWEEELAKIPGVTEEQKEKVIAALIAAEQTERAGGDAAEAQA AVDKVLLELFGIPA
part1_1p9m_34_model_3_1	LDELIEKLI EYVSKKLSPEDAEAFKRIVEY LKKHP EEAKEFLKELSKEDKEMGELANKILKAL
part1_1p9m_10_model_5_1	EEEEIRKEAEPEIEVIEKYDPEKAEELRKEVEKEVEKIKKE
part1_1p9m_12_model_7_1	SEGIEEAAEGLVYYDIGKVEEAVKKFLKGVELLLKEKVDEKEKK SALKEAIKACEAAGKKEEAEKIKLL
part1_1p9m_25_model_9_1	KMT EEEKELIDLVTEDYLLNGKIDQRRIAKKL NISPTTVSRRIKKVFKKYP
part1_1p9m_33_model_6_1	AEAIRAAVDALIAEEGTTLTLRELAERLGVSRSTLSRILG
part1_1p9m_18_model_6_1	MSLEDIENLLKADKELKKIPKEKRKEFIESLKDKLSEDDLELLKKEL
part1_1p9m_23_model_3_1	LEEVPVEKLI EILKEAAKLAKELQIIEEDFEKIEEAIEKLNGEISKEEAIELLPEPVREIVLEALKE
part1_1p9m_9_model_6_1	SELREVLEKIEELAKKAEEAKKAGNLEEAKKLEELKKLAEEAIKNSDDPELKELAE EVLKN
part1_1p9m_7_model_6_1	EELEKLKEEVAKAAE ESSKKFAEMLKKIDSMDKETQDYIESVK
part1_1p9m_30_model_9_1	ELEKKKEEALKLLKELLEKNIKDP EELEKALKEAEP AVKELLDLLAVRGSGLEEELEEA IKEFAKK
part1_1p9m_34_model_2_1	LEEILKELKKECGKVLTEEELKEFE EILKLPGYSLEEKLDLILEAAKNKELKEYIEKKKKEVS
part1_1p9m_26_model_4_1	EERKKEILKLLIEEEKSQKEVAKELGISQSTLSRILKSYK
part1_1p9m_14_model_3_1	TLVEEIELLEELKPLLSEEQFKEYLEEVEKELKKTKEDPEKYNVTKEELEEAEKRLKEL
part1_1p9m_10_model_8_1	KIEELLKLKEELKKLN AEELFEELFEIEKKKLPKEVVEKIKE
part1_1p9m_13_model_5_1	SEALEKLYQFLKEQGIDLEKV KPKLEEALKLMKEGKFEEAEKLVDEALKEAGVP EEKRKEILPEIEKLLLEAAV

part1_1p9m_24_model_8_1	KFESLSPRERALYEKILELIEKSKKFAVTQREIAEATGISKGTVSRLLPRLLEKRGIIERLGNGYVPVI
part1_1p9m_0_model_6_1	IDPEKLKEEIKKLKEEKLEEFEPKYKELIKELAKKNLPPEEVLKEVEKIFYKEFEKEFEKEIEKLKK
part1_1p9m_20_model_1_1	ILEKIKEYIKEKLRKGEVISLEELWKIAKEKFGITKEEAEEELKEAHEEAVFEVIREKAKEGASVEELIEFLEKAE
part1_1p9m_20_model_5_1	MIEEEAEKLAKKEELEKVYEEKFKELVLKGPEFAIEEVEKLAKEEAKKKIIEYLIEKKIPKEEAEKIAEKVLEKLK
part1_1p9m_6_model_7_1	EESRILDLIKDPQMKAVEEAIEYAERVAATMDAETAARIDEIVLEALREAREKGLSGEELVERVRERLEALLAEAA
part1_1p9m_19_model_6_1	MEEFLEKLKEIAKEDPEKAVEIVVELYKKGELSEEEVEKFAEILLELGKEELVEELAKRLNL
part1_1p9m_2_model_3_1	ELEKMSLDELCELLTKASGDSLEEVRREECLKDPEACRVLAAGVLSE
part1_1p9m_2_model_5_1	TVFDEVLEGWESGSWTEEEILEKLKESGLSEEEAREIVETLKILKK
part1_1p9m_38_model_4_1	MKEFEEAILKVLDLVRSDAEQYFKIMELLEKSIEEGFIEELVKAVLEAAKKAGIKEEVEAAIKKLEKEL
part1_1p9m_28_model_1_1	SIQQKAQEMLRDGLIEAYGLDEEAATAAVNAATAAEAEAILRAAGLDAATAARLAAACVAARKVTTTEEMAEVLEAAE
part1_1p9m_34_model_9_1	LEEYEKNSEELSKRLKLLLEAAEEIEKLKKEGKEEEVKKVIKELAKKLGISEEEVRKELEEML
part1_1p9m_31_model_4_1	AAAEVEAAVARYEAAAERYLAAKRAGAITPEQERELARAYNELVALGDPAAAAARLAELDARFAAEAAAAAAAA
part1_1p9m_33_model_0_1	EEEKLEEAIKDIKSGVSQKEIMEKYGLSRSSISFLRRLVN
part1_1p9m_5_model_9_1	MEEEYRKFAEEMAKILKEKGEDPKEAAERLMKEGVPGWACALLEKL
part1_1p9m_16_model_5_1	ELEEILEKCEKGECTLEDLEKALKLEGYTPPEAKKEAKEILE
part1_1p9m_26_model_9_1	EEEELEKEAEKRLEEALKKNPEKPEEAIKEVEKEYKELKE
part1_1p9m_25_model_6_1	MRYSVAEELAEISGESRSTVSRRIKELEKEGFVERVRVKTPRGVVVYVRPKL
part1_1p9m_20_model_9_1	MKLEEILKKKQPEIDEFYELSYNQLKLITGDEEYAKKKAELKKELIEKVKEDPENFEKIGEELSKKFDELVEKLR
part1_1p9m_12_model_3_1	SHPVEEKLEEVLLKKLEELEKEEKKLKEELEKAKEKQKENPEEAKKKAELEKEIKKLEEEGEKLIEELKK
part1_1p9m_8_model_5_1	EAEKAKKKIEELEKKGDEAAALDFAAKLIKAYAKYKDKKEELKKKVEELIKEF
part1_1p9m_34_model_5_1	DEQYRKVLEFLEKNRGKWFTLSELAKEVGLSRATLSRVLPIILLKEGIVEEEEVINGVKYYRPTP
part1_1p9m_38_model_1_1	HMTLEEIKEELKLIGVSEETIEKVIKYLKEIEKIDKEKVEEIVEEVFKIALEGKEEEMIEYMEKELKKL
part1_1p9m_29_model_2_1	VEEYIKEKKYEEAKKAVKEAKEELEKLKEEFEKEGVAAVKYLEEQIEELKELEKKIEELE
part1_1p9m_29_model_7_1	ELEKIFKEYKEIIKKYKEGKLTKEEYEKELDKLLEEIQKLLKDDIEKQIEAFEELEKIEKE
part1_1p9m_0_model_5_1	MIENLKKLIEAAKKAKDPNLGYQYLVLIDFLLKENKKEEAKEVLEEAKKDPNVNEEDKKKMEELLK

part1_1p9m_31_model_2_1	KPFVVEIEIPKCKSKEEEEELIKIKIEIASKLNIKIEIKKVSIEDKILLLITSNDAEDIHKFLEELSKLPYIE
part1_1p9m_38_model_0_1	MEEEFKKFEELSEELFEKYKELAVKGKEEEALELEKKEFLKKVEEYLKEKGFSEEEVEKKIKELEENL
part1_1p9m_10_model_7_1	KEEELVEKILEEYAKNPNWSQREIAEKLGISLSTVSRLIKKY
part1_1p9m_38_model_6_1	SREEVLLENLLRLIEWEKKHGGDPEWIVEALMEEFGWSVEEVIEYLRKLGFSSEEEIERARKKAEELEKKR
part1_1p9m_36_model_0_1	SEAEKTIELQLKELKPLLEKKAEEKLAELPEVTEEAKEEAKKWIEEQLEKYKEALKLEFEGKYEEAEEILEKELE
part1_1p9m_1_model_4_1	MINEKAAEAILEAVKRKKEEDPSLDEEKLIEEVERMKKEWNVDEEEVLKIIKEK
part1_1p9m_18_model_8_1	MPSFEELKEQQKKDEELEKEIEKLKEELKTDPSKEEELKKAEEEEANK
part1_1p9m_35_model_5_1	SYEELKKEFEKIEEAEKLLLEEAKKASKEEREKILKKIEEE
part1_1p9m_39_model_8_1	GHMSRLEYIAIEIIEELKKKNPNLILSLDELIDILSEKTGESREKIRKAIERLVKEGILEEIKNENGVTYISLKE
part1_1p9m_7_model_8_1	SEKKKKELEEKKEEIEKKLEEVIIEKGDIEELLKLLKEAEKLG I
part1_1p9m_29_model_6_1	LEEALKKAKELGVDPRIVEEIIARVLALGKPINQREIAKRVGRSQSTVSRIRILREAGILE
part1_1p9m_19_model_3_1	VSEEEVEKLAKELLELVEKLLKEGKSLEEALKEALKYLDEKNVPLEVKEEALKLVEKRLKAK
part1_1p9m_0_model_4_1	LEDLPLEEQFKITLKAREKLAKKYNVDPNDVDAIKAMEENKEKLTPEEVLALEQLKEFKKKGILP
part1_1p9m_39_model_4_1	EAVELAEAEKELVKKGKYEEAKEKFKEACKVAREKGEEVELIPKLDELAEVKKIPDKEKAKELEKAMEEAW EVP
part1_1p9m_8_model_4_1	STSRLIQEIIIEYEKLMKEKKKNKDAKKIVEIIEKKKHEELCKNEEIKKTLEELAK
part1_1p9m_21_model_9_1	ELREKIVELVLKSGKSASEIAKELGISRNTCVKILAEAFGVSRKTIYEL
part1_1p9m_4_model_3_1	SKEIEKEAEKAFEEKKYEEAIKLYEEALKKDPSKKLYYKEGIAFAYYELGKKKEEARKLFIEIAEEL
part1_1p9m_3_model_8_1	PAPPTAEQEAAIAEILAAHPQAWPALLEALDAGDVAALEAALRAAGLPAEDVARIVAIVR
part1_1p9m_35_model_3_1	KEEIEKEAKELEEEELKKEGLGALAKEVAEEYRKEKLEEL
part1_1p9m_1_model_8_1	KVEELKKEAEEEKKKWEEIEIKFLEEQGAEEELAKKLKEEAEEKKEEWRKKIEELKK
part1_1p9m_8_model_9_1	DKTVEEIEKEIKLEEEVKKTNIPEFKEALEKCLKELYENKKYEEAKKLIKEGVL
part1_1p9m_6_model_9_1	IKLSKEEIEELMKEVEKLPSIKKAEEIYKKLKEKFEKKYKISPYIEKAVLVFLSYNIDVPYEELEPLVEKLILSQKK
part1_1p9m_0_model_9_1	DLEERLKELEKELEELSKNEKAKERCKELAEKLQEVQRLLDEGKEEEAEKLEIEEEVKKLKEEI
part1_1p9m_22_model_5_1	ELEVVKKVIKECKKEGITDP EEILKRVEELGVKLSPLLKELIKEE
part1_1p9m_24_model_9_1	MSMEEIKKEILEILKEKGEIHLDELVELVKERTGASEEEIGEAL EELLEEGKVELIVKEDGIYVRSKE

part1_1p9m_37_model_7_1	KEELEKEMEKIMKEIEEEAEEKGDKEKARELMKKLIELEEKYLK
part1_1p9m_24_model_5_1	TRLLPVDEAAIAALVDEELAPIRASLATADPATAAAAYLAELAAEGRAPLEAAAAAARAAGLTEFEVEL
part1_1p9m_39_model_7_1	EEEEFEKKLKELEEEIEIEIEKIMEKNKFDLLKEVLELWEKTKEEVKKLIEEGKKEEALKLLEETVKKMKELLE
part1_1p9m_33_model_7_1	EAAAAKKIEEEFEELLKKVEELPEEEKGELAKEVEEELLP
part1_1p9m_11_model_2_1	KYINVSSKNPEALKIALEEVEELYEELVEKKADPKELKPKALELIEELKVLAEASGDPAFEELIKFVEELVKKYF
part1_1p9m_37_model_4_1	EEKIREAMDEAMMEAKPDEEDAIELALKKVGVTKEEIKEVLGL
part1_1p9m_12_model_4_1	MEFIEAQQKVDEILEKAEKAKKEGNLEEAKKLYEEAIKILKEVLPKVKDEEFKKNVEELIKELKEEIKKL
part1_1p9m_39_model_0_1	IEELKEKLEKYEKELNELLEKKPIPPSQLQKLIELLEELQKIIEEIEKNNLPLEEKKPEIEKIEKELDELFKKAK
part1_1p9m_19_model_8_1	EEEREKKLKELEELIEKYLASLSPEEQEYKPTAELLKEIAKDPETHEEEVEELKELVKKGL
part1_1p9m_29_model_3_1	DEIMKEIEKIEEEINKEIEEEKKIKEEAPKEKKEETEKEIKKLKEEKKEEFFEKLKELL
part1_1p9m_19_model_1_1	INLEELEKERELEKVLYMKWAAEPPDLPAEEIKERAKKEAEEITEKKLEELKKKEELEKLK
part1_1p9m_15_model_7_1	KAEEIVKKIEELEKKYKERLKTEPQEKLDKEELEEELEKAF
part1_1p9m_8_model_0_1	DELNIEEKLKKNEEESKKIIEELLEKSNKKEYVKEIKEELEKEKKEAEEIHKKLK
part1_1p9m_6_model_2_1	SEIEELKKKVVEALKNGDLDKAAEYLGEEIHELLLEEKKYEEVIEVLEKLIKVEEKAGKEEEAEELKEELEALKKAVE
part1_1p9m_15_model_6_1	SSVVKSVEEVRKNPEESKKALLEEGYTEEEAEILVKACLA
part1_1p9m_23_model_5_1	AKKIREEAEKVAEKLKLNLNKEEAKKLVEEFLKKVVEEAKRKAKPEEVIEIEKILEEIREKIIELTGWESA
part1_1p9m_13_model_0_1	EINPSIEAIKKATGVSDKAAKKLLKEWKEAKKLLEEAKKLYEEELKNHPENAKPLKELIEELKKEVEKAEKITK
part1_1p9m_35_model_6_1	SLLKEAKEKIEELKKTDKDATQEKIDAIDEYIKKEVEEL
part1_1p9m_34_model_6_1	PEEMRRAAEKAEDPLEREIVLRLLELRERRGRKTVTEEDIEEVAEELGLTPEEVKERAVRAIN
part1_1p9m_31_model_6_1	MTPEEKKETEEKIKKIIEYMEKTGKIPSQSELAKLLNISEKTASKLLKKLFDKDPESVNLLISIVQAEQKRK
part1_1p9m_37_model_8_1	GELEKKGEELLKELDEKEPTLKGLEFIKLMEEKEEEMKKFIEE
part1_1p9m_12_model_6_1	SELEKLLKELEAVLKELRALPVQPASFPELEKAAKEAKERAKVAENPEIVKEEIKAIKAGKAALKEAKA
part1_1p9m_36_model_7_1	ASEEEEEILFKIYKEMLEEEAKRLLKEGKSVEEVKKEIKKLVEELKKEAEKEGRKILVEKLEKYKPELEALIKKLA
part1_1p9m_18_model_7_1	LEEEIEKRIEEAIKEAIAKKNLDASEEELLKALEEAIEKVKKEIEKEL
part1_1p9m_21_model_4_1	IEELLKELEKEVEIGNPWAKETYEEVKKKIEEAIASGDKEKLEKLLKEL

part1_1p9m_15_model_5_1	KEEELKKLEEEIDKKAKELTGEELQKYLEEAEKKAAEEIKK
part1_1p9m_13_model_9_1	SEPIELTEERKELIEKIKKAIEEKKYEELEKLIIELIKLLPEEHAKKAKEELEILKKLNVEFMKELAEALIEDI
part1_1p9m_32_model_4_1	SIAESPVGAIAARAGVSEEQYRELERRVWEAIESAETPEELEARLAAAFAEVGLSPELAAEVRAALAPEIATRKEAL
part1_1p9m_5_model_5_1	GEEIEKKLEKELEEEELDKAMSKANGEELQKLIEEAEKKIEEKKKEI
part1_1p9m_19_model_7_1	SEEEIEKKAEEELANKYIEALKEGKIEEADKLLKEIKELAKKDPELAKKCEEILKKLLGTKEL
part1_1p9m_15_model_0_1	SRIDEAIDKIIKEGGLEKLTAEKLAEVAGISKSTASRILR
part1_1p9m_5_model_7_1	PLKEQELKLMREMAETAPPEKKPAFAAAALKALEAGDWEKYKEYAKA
part1_1p9m_22_model_7_1	SKLEEILKKEEEIPKIEKLAKEEGKEKLKEAEKIKEEIEKIIKSI
part1_1p9m_25_model_7_1	PSEEALKELEKILEELAAEAAKGLTEEELEKKAEEIAKKLPPEEVRELLKKV
part1_1p9m_14_model_2_1	MEEAKEKAKKLGKELAKKFPEDPSIVEVEKELKKISEEYGIPLEELSEWVSEGFDEEI
part1_1p9m_3_model_6_1	KIAELRRELEELKEKIKELEKEIKKAKEEKNEEKLKELEKKLEEAKKKAEEVEKKLKELE
part1_1p9m_35_model_1_1	PVILPLEECLADPELLAKCKPAAKVEFKPDGTLVETPVCV
part1_1p9m_6_model_4_1	KELLKKKAEEAKKKAEELKKKVEELKKKGTDSELEAAIKLRLEAAEAEKEFAKKLEELAKTVSEEEAEEAIEKLLKEL
part1_1p9m_36_model_5_1	MEEEIKKVEEELKEVEKLVPKTHPLLQEAKEELIKAENYNIYIKEYNEAPPEIIEQMEKALKEAKEKIKEAKKEVE
part1_1p9m_38_model_5_1	KELSLFGKALLKCLETEKLPSEEIFDRAVVEAAFALSKEREGVIEALCKMTGASREEAEKLIKLLGYKF
part1_1p9m_16_model_4_1	KEKEERIREEAEKKLEELLKSGDVEKAIKEAEYEEKLLKKEY
part1_1p9m_27_model_1_1	LEDNELEKKIKEELLKNPFASDEEIKKMVEKLVVEFIEKEAGDIELPESARKKLKEELIKELLPKALKLKEELRA
part1_1p9m_28_model_4_1	ELEKKVEELLKKLEEEIELLEELIKVQKDDKKRVEELQKEIEKLLKKLEEEALKEYKENGDKETLEKAIETEKF
part1_1p9m_3_model_9_1	KCEEMLKEVEKLIKIEGKLEEALKALEEALIECKDFEQYQRLVKLQEEVSKKLEEKKKKKEE
part1_1p9m_18_model_5_1	DLLDELIRYAAEEEEKLSATEIAKVLGLSKSTISRRFKEITGVSLRDY
part1_1p9m_34_model_0_1	MLEEVAKKFGISPEAVALVTMLQAGFSREEAIRAATRLEGVDPEKARAVIEAVENGEIPKDV
part1_1p9m_20_model_8_1	MEEELKKKAEEELAKKAEEAREKGDKKKAIEYLEELLKVLNEIKTDPEKEEKIKIAEVELAWEKGDWKAYLEALEKL
part1_1p9m_10_model_3_1	MSKEEIKKKIEELKKEYEKVKPEDKEKREKVEKEIKELEKEY
part1_1p9m_27_model_6_1	YLEKVKEAAKRIHELRELLASLPDSSIPREELEEEIEKLEREYAEAEARLLGKRPEEAEKLLKEIEEFADKYGV
part1_1p9m_5_model_8_1	EELIKKEIEKAKEEGDEEKVKELEELLELSKKSEEEKEFAEIFVKH

part1_1p9m_14_model_1_1	EVEEYLEKVKKGKNEKQKELYKYFEELYEKSCLKALEKLLEDLKKDPRVNGKDLAEVEAE
part1_1p9m_21_model_6_1	MEEEMLKKKIKLEIHKEMLKNGKSQNEIAKELNISKTTTSRYIKEIKEEE
part1_1p9m_8_model_6_1	KEEEARKLAIELIKEYPDIARNYPEALAKAVAKETGIGEKKALKIVKEVLKELE
part1_1p9m_35_model_2_1	KEELKEKVELLYLHTKDQKEIAKKLGISQSTVSRLIKEIE
part1_1p9m_30_model_2_1	AVEILEALAEAKRAEAERLAAEDPSLAAEAAALRAEAEALKAEALERLKEKDYEKEIKILEELAKK
part1_1p9m_7_model_3_1	SFEKKKEKVYREGLDKLLEGKDPEEVIKEIDEEIAKLEKEKEE
part1_1p9m_33_model_2_1	DIFPDPDIEKLEKQIEELDLPEDFREELLKELEKRKKELE
part1_1p9m_4_model_9_1	MEEAEKLKKEAQEKERKAELEIEIFRSSPEEAELKKKAEEKKKAKELAEKAMKLSGMSEEEVK
part1_1p9m_5_model_6_1	DTLEILEEIAKESEELKKKGKSKEEIDKIIKEKLKEKIPGLPESVL
part1_1p9m_17_model_7_1	MKELERLQLLLKAAEAIAEEFGGKPVTTREEAVKVVM EKLGLSREEAEKVVEEAVKEGLIKLK
part1_1p9m_15_model_2_1	SVEEKLQAVDDLLEAEELAKPDGNKEEA EKRKEEA EKIV
part1_1p9m_7_model_4_1	KIEEALKLLPEAEQKVALEELKEIEKKDPSKIEEEIKKLIEEL
part1_1p9m_4_model_6_1	SLDEIFEDIDSLPESAREGFRKVIAEGIAKELFPEDEEKQKELEEEELTNASKEEIREIHKKLKEER
part1_1p9m_2_model_9_1	EKA EKGDP EVKEKEEEELFEKYWKELGGEPFEEVLKKIEEELKKFAE
part1_1p9m_32_model_2_1	MLEEVEALEEEV RTATPEEVGP LLERVLELLKSKDP ELIEKVNEVLALLEDVPVSPEAAAVWQEIIKELQKLNVEGK
part1_1p9m_37_model_1_1	KKEECEKLREEVEKLCELVPNPGFKKVLFEELDKYCKKKLEEE
part1_1p9m_38_model_3_1	LLDEETLKKALELAKLLSTAPDEEVLALLRDPEQRELLLKLSDEELQKAVREREKEEEEEKKKKEELLK
part1_1p9m_12_model_0_1	KQTPLSSLEPEEREVL DVLGELQLPRGTAPLAEIAARTGIPRARLDALLARLEARGLVEVDREAGTVRGL
part1_1p9m_23_model_6_1	SGELYLEMLLLEDEKEDNLSPELKKLLDERLAAAEKTPEAILEVLEELKEEIP EEDMPEIMVYVLKKY
part1_1p9m_8_model_1_1	DALAKKYYELALEYIKEGNIEAAKEIVEEAKKKFGKHKQIEEMEKL LKEIEKKK
part1_1p9m_10_model_6_1	KEKEELKKKIEELKEKMLNAKSLEEYEELLKKA EELAKKLGE
part1_1p9m_24_model_0_1	MMEEQEA EERILAALEAAGDRLLTAE EIAERSGLSLPTVRRVLRRLEEEGLVERVRLSARRPLWRLIP
part1_1p9m_23_model_8_1	REELRRRVLELLAEGARTARELAERLGVSRPTLSRVLRELEALGLVRRSERPGVWELAVPLMEAERLLA
part1_1p9m_20_model_0_1	MEEQKKEAIKAGLEAFEKTLKETGDVLEALKAGIDAVLDK CIEKKLSIEEMEEVVKEAAEILAKKTNIPVEEILKL
part1_1p9m_33_model_4_1	KKKEEVKEAVLEERIIYKGV TQRELAEKYGVSRKTISRYAK

part1_1p9m_17_model_0_1	EVVVAAYGGESAEEVKAAVLEAKKKYPDKKVVVQALPGADPRVPEAAKEAGAELVEVEEELE
part1_1p9m_22_model_0_1	SNLKLEVLEVLNEAGEKKISAEELSKKTNISKKTARKILKEFNSL
part1_1p9m_27_model_8_1	SALTEEMVKTIKELIKLLTDDPEIIEGETEVIELDPERVLPPLIVEAIEKNVDKIKDPEAKKLAEAEIALLYKKKKA
part1_1p9m_13_model_2_1	MDVEKKIEEIEKIEKEAEKAKKEAEKYKEEAKKLKEEKAPEEKVKELEKKAKEAEAAEAKKLEEEIFKYLKEAAV
part1_1p9m_36_model_1_1	AIEKAKELLKKPSLENAIKAAELIEKVVESLKNKKNKEKIDKLITEMLDIKFENPHENQEEALKLLEEVEKLIKED
part1_1p9m_27_model_5_1	ENEKKAKEAYEKGKKYLEEKKYDEAIKYFEEAIKLNPNNSKIASSEYYLIADIYMKEKKLPEALELLLKANAIPNQ
part1_1p9m_35_model_9_1	IEEDLELLKQLKLFNEEEIEEIKKLIAEKGEVVFKGIKFT
part1_1p9m_26_model_8_1	EELKKRCEEYIKEAEKVSPDFAEVIKEECEKEIKKAEEEEK
part1_1p9m_2_model_2_1	MEEKLEEIGKLFKEGKFEEAEKLLKELFGDEEEVKVLLEAAKLEAE
part1_1p9m_1_model_5_1	SRVEEVGRKIAELLEEGKVVEAEKLYEEFKKEASEEERKKLDEIWEELKPYAERI
part1_1p9m_4_model_1_1	MEEVKKLIEELIKIQEEAVKKGIIISKEELEETKKFLEEQLVSESEKEKLLDYLKEGIIKKLKELI
part1_1p9m_25_model_3_1	MTEEEMLEFVRKYLIETEDEETAEAIVEEYKKSKDLELLKKDYEEAKKKKE
part1_1p9m_37_model_0_1	DADLPPAVKAALAAAAAAAAAAAAAGLSPEERAKIESAVIDEVL
part1_1p9m_28_model_5_1	LLETVEVNRANLDAVAELEARYGLPVDREQLERVRRERIGDEPTLVFVEEEDGRLVLTLPTEINPEMTPEAARRLAAA
part1_1p9m_27_model_0_1	ASIQEQLEKVEELLEKAEKETDPEKIEKLLKEIDKLVEELLKELESIDPERAKEFKEYYEKEKKKIEEKREKLLA
part1_1p9m_8_model_8_1	IEKAEELLKKFSEELKNFNEEKIEEIIKELEKTGYSVFPYIVKESYEEKKKELE
part1_1p9m_7_model_9_1	VLERLEELAKKAPEKQEEIIIEIDKAKTLKELEELVKKLEKEL
part1_1p9m_17_model_3_1	MKPSKEAEKILKEIEKYIDENLSNLTYEQLEEIEKLIEKAAKSKDIEEIKKIAEEIKEILKK
part1_1p9m_4_model_2_1	IEEVIKLLKEAFKTGDEELIKKAKELLLKKIKEELKELGLENEEDLERIAEKLSEEIWEEILKEVK
part1_1p9m_1_model_7_1	AAAEERFRAAIRAGDPEALRAALPLVSPEVRRRLAAEEVPRLPEVEEAVLAVIPE
part1_1p9m_39_model_9_1	LPEWTPPEYKIFAYILQKVESGELSIDEKILVSELAEVGVSKSTLYRSLKKWEKEGLVEIVKEGRKVYIRLIPG
part1_1p9m_29_model_8_1	MEKWEALEKAKKEGNEKEALKLEYEILLEIKEGMVTEKTEELEKEIEELKKKAKELGLYEI
part1_1p9m_13_model_1_1	LPGEGLNDEEIKKVAEKLGVDPPELKKILEEYYRLEVLKELTGDKMSEEKLRFEELKKKVDEIKETAAKIKAE
part1_1p9m_5_model_1_1	LEERLEKLLEEEKKEIEEKNLPEEKKKKKELEELEKEVEELKKRIKE
part1_1p9m_14_model_7_1	MEKEIEELKKELKEAIKEFDKILEENEKDEKGNLKEVQKAQEKIEELEKKIKELEKKKE

part1_1p9m_29_model_9_1	SSDEAWAIVMAALDAEWAAISREERRALLEAMGLPADTVEAAVARQGALSAEVRAALRAAG
part1_1p9m_23_model_1_1	EVNNHPEVKKLKEELEKAKAEGDEEKAKKLEEELEKKSFELELLSLGLSKEEAEKLAELYIESLKKEKE
part1_1p9m_20_model_3_1	MEERLKKLKEEIKKFPDGKEEYKKFEELEKELEKILKEMNGNEELKKKLKNYLIELLEEYIEEKKKKVLSKGLDPY
part1_1p9m_24_model_6_1	MEEAEKVAKKIVESALGPEIKKLEEKVKEYTEKGDFENAEKAKKELEEKKKKYEEELKKLLEEVKELL
part1_1p9m_39_model_6_1	MTVEELKEEIEKLEKEIEAAKAAGKPAAEAHLQEKLVLKSFLKELEEDKENKEALSISLRGTLALIDKEAAGEL
part1_1p9m_0_model_8_1	RRERILAVARELLEKAPDPRKVTLDEIAAAAAGVSRGTLYHHFPSKDALLDAVQDAVIEEELAKLLA
part1_1p9m_30_model_7_1	DVEAAARELARLYKAGDWSGAGEPLARLAAAPADEARAALVAAILEVMKDLTPEQAAAAIAEAILAA
part1_1p9m_28_model_7_1	KIEEYIEKLLKLEELKKAEEEGNEEKIKKLKKEKEELLKKKDEIAGEAAVELIKKMAKEGKDPEEIVEAIEEMEKKL
part1_1p9m_12_model_8_1	TLEEEAYELGKNPTEESIEKLEWKKTNPAAKAYKEGLKEYLIKELLSLNVDKEDAEKIATKLVEDIAE
part1_1p9m_27_model_9_1	MTEELIKDIEEELAKEALKDPSNIEKVKKHIEELKKKYEPIDKKLCELLDEELEKIEKAVELLKKGEKVELDI
part1_1p9m_38_model_8_1	LDVPAEAIDIALKLLFFQKKVNLPEDVFEQYVKFVAEEYGIKYEDFKKIFEWAEKNKIEEEEIPEIAKK
part1_1p9m_37_model_5_1	SGMEKYKEEIEKEAEELLKEGKITKEEYEKAKKEMEELVKEFE
part1_1p9m_35_model_4_1	EVEEILKKLEEAKKKLEELKKKGGTKEEQDKLLDEVLREI
part1_1p9m_3_model_3_1	MEKEEVKKKVEELIKKIEEENKNTSNNEAVENLIKKLKEEKEKIKTLEELEELLKKLEKE
part1_1p9m_1_model_0_1	DDIELIKKSLADPNLTREEKELLKKFLKELEKNKDNPEKRKELLKKLEEEMEKL R
part1_1p9m_9_model_4_1	TIEEEVKKAKEVLDPELLSTGNEAAAKYALELLEEEIKKVDPELAEELKKLKEEFLKKLKELK
part1_1p9m_37_model_2_1	AAEEYKKFAIEEALKEKEEKGVSEETVKKKIEEALKPIEEAKK
part1_1p9m_9_model_3_1	ELEALRARLLAAAAAGDPAAVAAVAELLAAALAEFSQYPDPALAAAYVRETVAEEVAALLAA
part1_1p9m_13_model_4_1	SDREPLAPSTRRVLEVLAE RPVLTA AEIAARTGLPLSTVNEILRELERMGIVRRVGRRYVIDRERLAAYLAEHA
part1_1p9m_6_model_3_1	MEEEEELKKLYEEVKKKALEAGLSEKEAEELAKKALEEYKLEKEFDKALEEFKKGGENVEEVVKKLKEIFEKLEKKE
part1_1p9m_26_model_1_1	YLEKAKELEKEKKYEEILKLLDEAAKENPEEYIELLQKIK
part1_1p9m_2_model_1_1	KAEELKKKAEEKRKEAEKAEKAGKKEEAKKLAE EAAELEREA EKLS
part1_1p9m_24_model_1_1	SELERKKVELEVAKAISEELEKKNMSITEEKL EELTELALKKSLEEVLKEALKEVEELKKKDPNLKTI
part1_1p9m_16_model_9_1	SKKYEEIVEELKKKYPELPEEVIREEVEKIKDYEEAKKKVLE
part1_1p9m_21_model_5_1	VEELAKKLKEEVAEELEKKGWLLDLEKKEPEIKKLEEEAKKKA EKLYKE

part1_1p9m_33_model_5_1	KLEETLKKLIEELKKEYPELSEELAELKAKIFLEDNPELK
part1_1p9m_25_model_8_1	VKLTDREKEVLVYAELFGESDESLEKLSKKLGLSVEEIRELLERAKARDTS
part1_1p9m_9_model_5_1	SEERLKEAIKLVLELIKKNPVTKHEELKEYLKSkgfSEEEAEKVIEEVADILAKEGIFDVKE
part1_1p9m_29_model_0_1	KKKPTEEEKEEYKEYEAAKERIKLAEELNNEEEKEKFEKELKEILEEVKESDKELYEELK
part1_1p9m_30_model_6_1	LEEEYEKRIEELAKESGISKEKVKELLERYTKLLVEGKEEEADEEVFKELVEAGLSPEKAKELVEK
part1_1p9m_11_model_5_1	EEFEKLIKELKKLEEEIEKLLEEYKKALEEGKYEEAAKYRKIEELEKEIEKKLKELEKLAEKYPEKKPEVEKVK
part1_1p9m_18_model_4_1	EKEEALELLKKKKSEEEVKEAEKLIEEAekMGdKEYVKAIIIEIKKI
part1_1p9m_12_model_9_1	DELTKEELKELLEHYKEINISPEELDEILILVAKSGDKELVELVLEFLKEEGIEPSEEALAAAEKVfKEA
part1_1p9m_8_model_3_1	SIFLETLEELEKRHPLTEEQREKLREARERVEAEKDFLGKLEALEEAAEALGVE
part1_1p9m_25_model_0_1	EEERKLLREIAEAILNKNWTEEKMLKWIKELYPDLSEEERKKIAKEELEKL
part1_1p9m_32_model_7_1	MEALEEALAAfRAALEAADVEALEAALAEILAHLGSPeAQAAALRAFLAEHRPLLEALLAAAAPDSPLARFLRDLLAE
part1_1p9m_18_model_0_1	AEELRKKVESLPPEEQKKLIDEIMDKELKSPNAPEALKKLEEIfPGL
part1_1p9m_9_model_8_1	SEIAKKYYKELEKKLEELAKTETVEEFLPKEEEELVKEVLEKCLKKAKEIGDEKIKEECKKLA
part1_1p9m_35_model_8_1	EELKKfSEELLKKIEEEAKNRDIQELLEEAeKEYEKKKKKE
part1_1p9m_37_model_3_1	SPKSEEIRKRCEKLGldEKTLEAIaKDPSITDEEELeLIAKYEK
part1_1p9m_33_model_1_1	SfKVYEEAKKELEKKLKEAKTKEEKEKVKKELEEELEKIKK
part1_1p9m_16_model_8_1	MEpILMTEEDIEKLKEEAeKAETEaERKLLEELVEEAkkRYE
part1_1p9m_10_model_9_1	EELLEeYKKLVAEKpNTPEAKEVANKIADKYGIQYVIEKKLP
part1_1p9m_16_model_3_1	EEEEeIRRLKELNLSEEELKKALEELNERAAERYIEELRKKI
part1_1p9m_34_model_4_1	MTSVERYLYLAeQYEkNKDYEGAiKAYEEALKEKKsKEILKKLAELYKKAGNKEGYKKYKKEA
part1_1p9m_31_model_1_1	AGEQLEKFLEAIeKINEAIKKKDVeeVKkYmKELEKLREEVLEKLGPNSelVELAEKTLKEaKEYIKKEEEK
part1_1p9m_3_model_7_1	KLDREKVLDEeYDKVLKEVAKEMAKLTDPEEALKLYEEALKKLDEELPKKIEELEKKKKEE
part1_1p9m_14_model_5_1	LKEYAKKTLEEAIKAAYeYLKKGDkSKALEALFAAILKALNEGKEEEAKKLLEIAEKLG
part1_1p9m_28_model_8_1	LPeLPQELLEALSklSLEERQKLfELIEEEGERfRKEKLpVEEALALFEERAAKELSAEAQPVALALARLEFEAMAAS
part1_1p9m_17_model_8_1	LEAVRALfGEERAEeIfEKILEARALGLTQEEIAKYAGISPGTIRRIDAAALKSEGVTWEELL

part1_1p9m_33_model_8_1	EKEKLYEECKKEATELYKKMDLPEEKIEELVKEECEKAVK
part1_1p9m_27_model_7_1	NLEKEREKDKKEVHKIVAEGEEALKKLVLLEILKEEIEARGELSELVKERVKEILADYDIEVSDELLEEAFFEEAKK
part1_1p9m_11_model_4_1	EEEQERADKLVVMYVNAELEGNKELMKLLKKLYKKVNVKDEELKEVWKKLLEAEGDPEKILKILTEYEEKKKKKKE
part1_1p9m_21_model_1_1	ELEKDLEKIKKEETKGVYEEVKKKIEENPPKTKEELEKIIDEAIDNAKA
part1_1p9m_32_model_6_1	MEAEELLKKAKEALKEGKYEEAIELGKKAYEKELEKENAPPEKKAEALWIIAEAYEKLGLKLEAQEYLLKKANELAPDS
part1_1p9m_5_model_0_1	KIEELEKKALEEAEEKKIEEALKKGEDLEKVKEEALDEAIKRIVKEL
part1_1p9m_0_model_2_1	LCKDCSPGCRAGCEEVLRALEALDAGDLAAAEHLARAEALAAQHPDCPELRYGLELVRGKLALLK
part1_1p9m_17_model_9_1	EVEELVKEAKKLMEEKKYDEAIEKIKKAVEKSDDPEIKDLYYEALLRAGKVEEAQEFAEKNI
part1_1p9m_13_model_7_1	SLDNKYDPFAVMKRYLLDKKEEVKAEDIVKYLVTSNIPREEAEELVKELLKEGGIKPNEKGYYSKDVDLKAVDK
part1_1p9m_2_model_4_1	LTEEQKKKLEEKGKEILEKLEKPNLTAEERLELLKELEKIEKEAKE
part1_1p9m_24_model_7_1	MELEKLIKEIFEKVEKYLKLKKENASEEELKKLEEEIEKLVEEAYKAAKKEGKLAKFQEALLEIDKLL
part1_1p9m_23_model_2_1	IEERVAKYVREAVEEAFSKLKTTEEEMEKALKEFEEYVKELEEAHGTGDAGRKAAKEAYEKELKKKKKEE
part1_1p9m_16_model_2_1	AKEKELEELRKKAKEIVDEPVTTYEKLLEKQKIEEELAKILE
part1_1p9m_28_model_6_1	MEKVKKLLAEGKEEEALEELEKLVDALANRHSGKAKEKLKEAAKAAKLIANVKDPKLAEEAKEAEKTLKALIEALE
part1_1p9m_15_model_9_1	VDLEKVEKVKKELEKLNIEQGGKKIIELEKNEPELFGLS
part1_1p9m_31_model_5_1	MKIDFPIRNELELKVLEYIEKAEEAKKEKKYEEAIKLLKEALKIAKEANNEELVKLIEEALKKLEEEAGIKS
part1_1p9m_10_model_1_1	KPKTLEEKRKLLAEIILKKLEELKGSPERSEIFLPLVEEYEE
part1_1p9m_29_model_1_1	YEEIYEELKKAALDPENFEENLDKIYKETITKSEDPLKAIEDIEKAIEAVVKELGLDIPP
part1_1p9m_15_model_8_1	IREEVLKKLEEDLSVFNTEEEKKLIEKIAEEVVKELEEEK
part1_1p9m_31_model_9_1	KQKEVLEEIKKYIEEKIEKAKEELKDKDFEEIKKEYEKEISDYLYQKAKEYNLPIDIIKLEEEASKKLEEL
part1_1p9m_3_model_5_1	AAAEVAALRAELEALLADPTTPLAELEAAAFARLVDALALGDRAGLAEALALAEAVRAAR
part1_1p9m_2_model_8_1	VEEEIKKAIEEAELLKELPDPVVKEILEEELKELAKKYGKPEALK
part1_1p9m_17_model_4_1	LEEYEKKVKEIEEELEKLKKEGKKEEALKVVREKGEELLEKALSEEEIPILEKLEELEKKAK
part1_1p9m_12_model_1_1	LTSEEIEESINLLIEFKAYDEVLELLEDVNGKDKDNEEKLKKIKEKLEELKKLKPPELKEKIDKILKKIN
part1_1p9m_6_model_0_1	DTAEAWLAKGKELYAQGKVAEAAEAFLKAVALDPDNREALANAEFVLEQNAELYGDAAAKAALRALAAARGWPV

part1_1p9m_38_model_7_1	GIEKMKKEALKLMKEGKLEEAKKILEEAIKKAKDPAEKGELKSLLVDALEKAGKLQEAELELLQEAE
part1_1p9m_31_model_8_1	MEKETREALEKARKAIEAGNLLAELEYLQKAIELNAPEHLLLEAELEFSAIGREEEAKELREKA
part1_1p9m_2_model_6_1	KELEKKLEELEKELDKLLASKNVKPEEIKKKYEEAEKLEKEIEKAK
part1_1p9m_35_model_7_1	SSLSKENLKKRIRKVFSEEEIKRLKEQGITLDDLVEEELK
part1_1p9m_34_model_1_1	GLSATIQKILEWLEKNVKPGEKFRMSDICKELGISRTTLSRALKELEKEGIVEIEGREVRLIP
part1_1p9m_17_model_6_1	MVEKFKKELIEIVREELEKKKIPNVEEYVEKLERLEKLTSTIDEYLVKEVEKFEEELKKEL
part1_1p9m_23_model_9_1	MEVTLEKAEELLREAVKRFIEKNAEKAGSSELTEEEVSEMILECLKEIAKERNLDIEELLKLLKEKKLI
part1_1p9m_14_model_4_1	SKVEELVKRIEEKEKELEKELEELKKKYPELTEEELEKIKEERSEIIEKELIDEATKES
part1_1p9m_3_model_1_1	ELEKLVEELKKADKALYEVKPLVEALKAAKAAGDLVKEQELLAKLEEAVKEAEKVIKAA
part1_1p9m_1_model_2_1	MEKLEEVLDKLLAAAFEKKDFDTPWKPEEFLELAKELGLEFDEEKFREIIEPIVS
part1_1p9m_5_model_4_1	ELEEKAFELYEEDPELMLELLADEELLKKKVKEKAKKLKEEEKK
part1_1p9m_31_model_3_1	SLLTDAEIRALVAAQSTPEEADAVLALIARAAAAARGPEAREAFALALREAEALLAGRPVAEVR
part1_1p9m_14_model_8_1	DEIEAILKDPEGIEEALKIAKEEREVFKKENVKVGVERYDKLIKAAEEAKKKRKEEE
part1_1p9m_22_model_2_1	SEEEELLRAIAELIREKPNITEEDIEKLIKELKVDEKKARKLAGK
part1_1p9m_23_model_0_1	EDLEKRREYMIKFGKELNEISGKKYEFENLSLEELEKKLDELIEEALKEDPKTRAKLQELLKK
part1_1p9m_30_model_5_1	EAEIEKKMLEEVYKEKTGEEMDWSKEEDVEKACKVLKEVLVEEWEVDSEEAARKFCEYCKE
part1_1p9m_9_model_1_1	MLNEEELEKLLKETEEKLKKAKTEERKKVIKEFEEETLKLNLQQRKKAEEYLEKLEKELEE
part1_1p9m_16_model_6_1	EEKKKKLEEELKKAKEELPKKSGIEKEYLEELIKEIEEALKK
part1_1p9m_28_model_0_1	EWEEELLKALDALWAKKKEEAKKYWEKALELAKKEKIISSEEAELKKLFEKLIASSEDAEK
part1_1p9m_30_model_0_1	SMVEKQYEEIIELCNELNKPELAIEKVEKLKKEALEAEKSRDPEALIAKLEEMEKL
part1_1p9m_28_model_2_1	YEELAKKLKEKGFEKAAEVLEEYLKKGKLEEAKKEAEKAAAELEEEERKKALS
part1_1p9m_22_model_3_1	VEVKEDGTIDKESVLKLLKEEGIENEELAEKLAEDIVKELSNKKI
part1_1p9m_20_model_2_1	SSEEDKTCELLEKYATTEEQKDILAEMLLDLAEYIEEGNVEKAEELCKKLKEIKSESVEKRI
part1_1p9m_36_model_9_1	MKKEIATATPEKRVALLDLEFVKEVVSEDLFEKIKKEVEEEYIDDAEKARKKMKELVLKEA
part1_1p9m_26_model_0_1	VTPEEALKRFEEYKKIMEEAIPNQPEVIEEIEKELLEKLY

part1_1p9m_4_model_8_1	STTQKIKDAILDLVLEKGGLDAVNQESIAEKAGVSKSTVSRAIKKLSEEEKEYLKTLDAIRKEAGYK
part1_1p9m_11_model_0_1	AEEERQELLELLANAVEDDPEAQAELKALIAQPGVDVDEVIDWLREYLPKLSPEEWEEFNSILDSIRSLLRRLT
part1_1p9m_0_model_3_1	EEKETVKKVIELLEKYLKKNPEDAELALEAIELAKKYGLEISEELRELALKALREAGKEELAKELE
part1_1p9m_11_model_3_1	DREKVBREAIHRLKELGEEQLVDYVDKYIGEEVLKEFEKRLKELGIDVEDVLEGKISEEWEKVKEALGEAFEKA
part1_1p9m_4_model_4_1	LREELLAAASPEDREAVARVIDRVIPRKGPLPPEERLAATRVRVLEGLVAEGLISPEVARAIVAELE
part1_1p9m_38_model_9_1	SELQLDKALALYEEGKYEEALKALEERIKEEGKATAEEYELLAIEIYKALGKEEEAKKAKEKAEEVKKKE
part1_1p9m_19_model_9_1	SECLKEYEKKVKESKKEFEEIKKIHIEEIEDEEARKRLLEYEKKLEEEELKKLEEELEKCLKE
part1_1p9m_19_model_5_1	VSREEIEKLVTEILEKEGKKYNVPKEYFTKKFIEEFKPKIVEKAVEANVKPEEELEEFIKKEA
part1_1p9m_14_model_6_1	NPLDEASEREFEILALREYLGLSQREIAKYLGISQSTVSRSVRRLELLASAGITLEDL
part1_1p9m_28_model_3_1	ALEKAKELIKEALKLEEEGKYEEAIKLYKEAAELDPSLKGECFLISLCYKLKGKRKEAAESLEKALPDLVKESPYDI
part1_1p9m_22_model_9_1	EREERLEETEELLEELGLSLEEARELLRAAGYSEEEIERALERIA
part1_1p9m_6_model_6_1	FEEIAKSLASLKSPEELLKELKELSLYIDLEEIPKVIEAVSKLSKLSKETLKKIEEEVIKYHKEKSVSPSVLIKLP
part1_1p9m_8_model_7_1	LEELEKLAETLGEKLKEAVEKVKEIAEELKEALSPEEAKEVTTRELAEEALKEVL
part1_1p9m_39_model_5_1	SESEKLKEELEKLAEAAEKLHLKNKEKVPEQSEKTLEEIEKKLEEAAKLIEEGKLEEAKKLIKEIKKMIKEMEEL
part1_1p9m_31_model_0_1	EETEQEIEALLNLIEYYKKIGNLDEAAKVLIEALKLAKEAGKEELAKKIEKELKEIMKNCSEETKKLIEKAL
part1_1p9m_4_model_7_1	LSNEEGIEKTIEALKKEGLPEEEIPKAIEEAKKIVIEEKKKLEEALKSPTADKESLEPLLEEAKKL
part1_1p9m_29_model_4_1	MKKEELKEALELAKFFYKQRSNVELVLEVLEECIKEAEELNEPEKAKELKKWLEEIKKEA
part1_1p9m_24_model_4_1	SGQTRYVEALIDGARGEESFREIVELSGLPRSTVSRHLRRLEEEGKIELVGVPETRVRYRSLLEP
part1_1p9m_19_model_4_1	KEKEKLCKLTKLILENKSQEEIAKALGVSKETISKFFKELFEKASKSEPLEEILEKKLKECE
part1_1p9m_17_model_1_1	KEEELKKELEEAATELLYEAALKGKEEEAIETLEEAALKLEEEKSLEEALKLIKEKIKELKEE
part1_1p9m_9_model_7_1	MEEAKEKLEAAKKAVEEGKAEEAYRLALEAVELAAENKDFELLQEALEFLKEIEEFNLPPG
part1_1p9m_31_model_7_1	MLCKPDESLEECKKKLTEILTYLEASGEEEAKEIENLAKQQPLLEELAKELWKEITEKKDKELFEKLEKE
part1_1p9m_36_model_8_1	DVETAVKVQTEILLDQLEEILKKKIEEGNPPEAKPKCEEAKKKLEEAKKKALKIENPEEAVKYLDELAKEEELEKE
part1_1p9m_38_model_2_1	LDELIDYLLSKGRLKEAIEVIKEIEEKEKELEEEEEKKLKELGKGKNIKEEKEKRLKELIEKKKKELIEK
part1_1p9m_6_model_1_1	DEEEIKLLLEEAKKALEEKKYEEAEKLAREVIEKSTDKELIEEAAEILLIARLALAKSAEEAIEAVEEELEKLVKEKL

part1_1p9m_15_model_3_1	MKEIVKELAEKLGIDEEVLLSYIKENLPEISVEELLKCLK
part1_1p9m_27_model_2_1	EKGEKLLKEAEEHAKKAEELKKEAKKTKDPKKKEELLKKAKEHLEKAEAEALLKAMVYYPEEYEELLKKLEELEKI
part1_1p9m_13_model_6_1	PTEEEVKKLEEEVEKLLLEEVYENPDEEKLEEFVKKAKELAKKCLKELGKEKEAEELKELKEEKERLELLLLKCLK
part1_1p9m_11_model_1_1	VEKLGEAVEKAKKEFEKIVKKAKANPGEGYKEALKEIEKVVDKAMEEVKTAEEELVALIELAKKAEKEVKKAAEA
part1_1p9m_21_model_7_1	MEEEIRKKVKELEKFIKENPELNIDIEKIKEEAFEKILEEEIKKVVEEK
part1_1p9m_32_model_3_1	SEEEERRRKAVRYAAKALQLESKSGKVEEALEYAKKALEEDPNHAEAWALLAEIYKILGKEEEAEFYLKAKELGL
part1_1p9m_18_model_3_1	SLEKKIEEAKKKIEEELEEKNSYEELEEYIEKLPKELKEAAKKYLK
part1_1p9m_3_model_0_1	AKEEEERKKGLEEMEKKLLEEFKKEVKELLKENPNATID DIAEKLKQSKSGISKILKELK
part1_1p9m_26_model_6_1	EAEELARKIAIEESA EIARKPGGEEFEKKVKEITEKLKK
part1_1p9m_26_model_5_1	SELEEEIKKKLEELKKERLEELSKTIPREEAEKLVKKEA
part1_1p9m_32_model_0_1	WFEEFVARLRELIEAGRYAEALELLRRAAEESPSPAVREYLR AALPELEALAADPEAFRAGVRALLEELERLHKIAE
part1_1p9m_0_model_7_1	DLEVEMLKRLAEAAKEGNWEELEGIGLELKKLGYEEEAKKVLERAIKGYKEKGLEEKAKKVVEIL
part1_1p9m_17_model_5_1	SELLKKLKEKLEEYKLLKEMDDKSKPHTYEKLL EIEKIAEEMKKLAKELEKKGYKVSKEVY
part1_1p9m_26_model_3_1	LTETEKKIIE LAEKNGLSQTEISKITGVSKTAISR AFKKA
part1_1p9m_21_model_8_1	EKVKETIKKIIEENLKSGDIEKAKEIAIKELIEKYNISKEEA EKIIIEI
part1_1p9m_3_model_4_1	EEELQKEVAEILPKYMKEKKDYNEALKAAAEDLNKEWVKKTPYKLVYDEETNEIVVVPKS
part1_1p9m_25_model_2_1	VEEELEKAIESGDKEAVEKAIEKYVEATPLESEEEEREILRKILLYFEEKS
part1_1p9m_34_model_8_1	AEALVAELKALLAQLKAGDKSAIPAALATIRALLPLPATAESIEALEEAVAALAAVDPSIKPE
part1_1p9m_26_model_2_1	DTSYTVKNPEDKEKLDEILKSILEGKELTPEQMKT LVEIF
part1_1p9m_25_model_1_1	DMEEVLEKAKKLIIEGKVVEAEELLADYAARSQDEEAEKAKELAKEAGVE
part1_1p9m_7_model_5_1	EKEKKFKEGNFEYLIKKGEDKVTVT LKHLESGKEEVVEVEKPE
part1_1p9m_30_model_8_1	MKRKEAE EYKKGKEALKEGKYEEAIEYFKKSLELDPTNLEAWKGLLEALKKAGNKEELAKVEKEI
part1_1p9m_4_model_0_1	EEEEKRKLKREYQELIIEYEKAEEEGKIEEAVKLLEEILKLA EKLGLPEEEIKSIKEELEKLKKKI
part1_1p9m_6_model_8_1	TKEEELKKEALK AASFIVENKKIAPDLVDEKLEKALKDYIYEISEKFGPKPEEVEEIIKEIEKLEKIGGKVTEEA
part1_1p9m_8_model_2_1	KLEEAVEKALEKALKKEIERILTSEEFKKLVIEELKKLVPLTDEEAEKIVKEYE

part1_1p9m_10_model_4_1	ITEEQLEKIKELLAKGKSQTEIAKELGISKSTISRIIEKLLK
part1_1p9m_11_model_8_1	SEEAKLKEIEGDKYYLEKKYEEAIKEFKEAIELSEGEEEKARYYAAIARAYKKAGKEEEEAKKYLEIAKKIDPNVKL
part1_1p9m_36_model_6_1	TAEELKKEAEKAIKEGNVEEALKLIEKIAELGDYETALELYKKLSKLLKIDEETLKLELALLAIAAGKEEEEAREIL
part1_1p9m_15_model_1_1	VEEVIKKIEELQKKIKEAKDP EEKEKLKKEIEKLWEELEK
part1_1p9m_22_model_6_1	EEERRKKLEEEARRLEEEIREARKNGDPRVPELEKKLKEVEKELE
part1_1p9m_14_model_9_1	AEEERLRARLALGAADHPDPDLPPELQALRARLAAGADEEALVAEVLARDTSILDAPV
part1_1p9m_13_model_3_1	SEEIGEKILETLKELGVATVRELSERSGINKTTVSRVLNEMVKENFVKKSRANDPLGRSTVYSVNEEADEEELEK
part1_1p9m_30_model_4_1	RREEAKKLLEEAELKLYKAGKLEEAKKVLYKALELIFPTEEEVKAAKEFIDENPEKNLKEAIEGLKE
part1_1p9m_6_model_5_1	EVEELLKKIEELEEKFLERVRAKADPEEREKIKEEAREELKPLLEKLAELLGGSPGEFTEKQLELAEERLERVWKES
part1_1p9m_23_model_7_1	SLIEELEEKVKKALEKIKDCPPEQIEYFLEKMEEAAKLLEEGKIEEAKEVLESVKEKYPEIGSILLEEL
part1_1p9m_24_model_2_1	MTEKEELLKLFEEAKEELEKLKKEGKIPEEEYEELEKLLKEGLEAAEEGKIDEALKILQKIISPIEL
part1_1p9m_34_model_7_1	GVAEKILQILDENPETPFTVSELAKKVGISQPTCSRILERLAKEGKVEILVNEKGTKLVRAKL
part1_1p9m_18_model_1_1	MKEIEKIKEEFYEELKKNPTLEAIKEIEKKVKELEKKLKEEGINIKL
part1_1p9m_7_model_1_1	DVVEAFKWDELAEKGLTPPEILKLLTQNWSEEAKKYLRDAI
part1_1p9m_13_model_8_1	SPYSWEAALEAVEAFLRAEPEAARSLEAVTAYLVETLGVPKELAEKQAEALMAVLLPLLAAGASPAEAVAAARK
part1_1p9m_11_model_6_1	KEEILQKAIDEAYESIENLEQRQKFLLVVNQVEEEYKNAPLEEKIKKIIEKLPVLSEEAYEKVKEILEKALKEI
part1_1p9m_19_model_0_1	ELADEALEAAAKGDKEKLKEIAKKIAKEIGDEEILKRLEEVAEVEEKEERIRKAIEVLEEVF
part1_1p9m_22_model_1_1	MSLSEEKRKILLEKAKSSGDLEEIAKALGVTVEEALKYLEELGIE
part1_1p9m_1_model_6_1	LEEAELKMLEELNKLKKKGKNKEEVKKLTKELVLKLYGDEEFAEKEAEYVLKNLEDL
part1_1p9m_23_model_4_1	KPPGSSTTKSNAQDAKILQYIEESGSNVQTLDELVEKTGLSRPTVSKILKEWEKEGLVTYNKETKSYIL
part1_1p9m_12_model_2_1	GKELVELRAKYDELVEKKKYEEAAKVRKELAKLEKEKGNEERLKEIIEEAEDLKNKQITKEDLEEIKKI
part1_1p9m_4_model_5_1	EEKAKKIKEKLEEYAKQLGLEILEEPKIIENGKKKIKVVRDGDKIIILTYDINGNLISEEVKEV
part1_1p9m_36_model_3_1	SEVEAEDAALAAARWAALAAALAAAGDRATAAALAAQLLPLLIESHDYARLAELIARLDAAGITHPLQPEARALIAKA
part1_1p9m_1_model_3_1	PLEKIKKEVLEEERKKGLNSEELEKLKELIEEYLNKNKSEEEKKKILEKALKELK
part1_1p9m_32_model_8_1	MVEELLKALEEVQKLLKELEERKKLEEKGDKENAEKLLKFLEELKKKIDEIFDSGLGIEEIIKKLKELIEEAKKKK

part1_1p9m_39_model_2_1	KIEEELKKELEEALKLIEELGKALNLP EEVIKEIIEEVKKEFEELLKAHPGDYEF LKKKIPELFEEAVKKLKERL
part1_1p9m_33_model_3_1	MKTEILYDDGKYVVIKIGDKVAVVEAKEIPDKEEGVKKLA
part1_1p9m_36_model_2_1	KRKEEIKKLVKELLEAVKNGDEEKA EKLGGKKLKVLPKEQYDKIEELVVKYEIAYKEGDEEAMKKYLKELEKAVEE
part1_1p9m_25_model_5_1	REERDRELDRDIAELEIEGAVKGKPLSRAEIAERLGVAPSTLSRLERKYGR
part1_1p9m_15_model_4_1	MEEKIRELQIKIEGKPESAPIEPLTEEELEMAERILKKEY
part1_1p9m_30_model_3_1	MEEELLKIAEEAIIKAAKEGKFEEAFKLAKEAEALAPLPEELANEILDIIIEKNNIEEYIRFLLTLK
part1_1p9m_16_model_0_1	NFEELKKLLEEKLGKKLP EEEARKLIEKELSVFPEDLRKYIL
part1_1p9m_37_model_9_1	LPEWLFENLEEIKKLFDEYIKKTPNLDVSKLTLEELIEIYKEL
part1_1p9m_25_model_4_1	MLEKAREFAIKILEKDGIPYANEEMIEALVEEVLKILEKNPNATFEEVYEK
part1_1p9m_20_model_4_1	EVLEAAFKLPAEYLVEAVEKIDEY LKLVEKYPEFKEFVEEVAKEVIALCEAKKGAEAIKKLEEALKKLKEKVEKLS
part1_1p9m_27_model_4_1	MEEAEKLYKKAKELYKKKKYEEALKLLDEAISLAPKNSKLLQEILSLKIEVLLELGKYQEALET LQELIKNESKE
part1_1p9m_18_model_9_1	EGEKLAKLNEKERLAVWGYESGLSLSEIAKELGISKSTVSRYLKKFE
part1_1p9m_0_model_1_1	LPPDLTPRERAILDELAKGPLTVRELAERLGLPVPTVSRHLRRLREARGLVVRLAGGRYALAVTPEV
part1_1p9m_1_model_1_1	EKEREREVLKLLQEFIERIEKEGDLEKVYEEILKEYKKEDSEIKKEFIEELEKLL
part1_1p9m_21_model_2_1	MEELLKKAKEAIIKEGNKEEVEKIVEEVIEKGDLESLLIYKECEKYGIE
part1_1p9m_21_model_3_1	MLEERIKELSEKYNIPLEELEEIAADCLVENDYDEEKALKETEKCKEK
part1_1p9m_33_model_9_1	KPREELYEKVYKELSEEYKKVGVPEENIKKYVEEVVEDLL
part1_1p9m_37_model_6_1	MTPEEFNAKLDPLISSNTAEKAEEAIIKALEAELKAKGELESAG
part1_1p9m_7_model_2_1	MEELEKLKLELELYEELKKEYEKKGEKLTEEELEKALEKKLKE
part1_1p9m_2_model_7_1	KEILTD FEDLEEAVKELEEWAKEKNVEIEISVEFEGTKVIVSAKEK
part1_1p9m_32_model_9_1	SAAAEAAIEALIDAYMADPTNDAALAALLAALAANPALRDRAAALLRERMLK RTPNEALAE EEEAAALRAAAEAAIAA
part1_1p9m_0_model_0_1	MEAYEAA LAAAKAAVAAGDLAAAEAALEAAYEAGLLAAARS GDPSLIEEVLAALAPLAERLAALKA
part1_1p9m_10_model_0_1	ISEETIEKAKELYLKGIIEGKSLKEIAKELNLNYSTLKKHLI
part1_1p9m_35_model_0_1	AKAAAIRAECEAE AARLADRPELVAEIRAECEAEIAALEA
part1_1p9m_39_model_3_1	GLEEELKRKYDAALKIDDLPTLSQLAKELGISRTTTSRILRRLEEAGYVRRPRHPKDPRRKVIEFTEKGIEAYEE

part1_1p9m_7_model_7_1	KKVVEEIEKILKEGEEKIEEAKKKGDLEKAKEIEKETLKKLDES
part1_1p9m_3_model_2_1	AEEEEKLLKKEIEKDPNNAELWAKLAKELAKAGKEEEEAAEAVEKAIELGADVCLKLILEVK
part1_1p9m_26_model_7_1	EEIPEVREKVEKELEELGLPEDFKKEIIEEEIERVKKKL
part1_1p9m_12_model_5_1	MEKDEELLKFAKEKLDKILEVMLNPEPDEESVKKLIEELEKEIEKAKNISEEVKKELKELKKILERIRKI
part1_1p9m_2_model_0_1	EEEKLKEAKEEAEKLLKKEYGEMVDLPSVKQNILDLLEEEYGVKVPI
part1_1p9m_9_model_2_1	SEEKEKLIEEKLKEAEELLKENVPAEVLVIMAEYELSKNPLLSEEDVKKAIERLKEKIEELE
part1_1p9m_11_model_9_1	MELIKLLKEIYKVVKDKEQKALILLIGEEYYKLGKLEEAKEYFKEALKLAKEAEKEGELMKMIKESLEKVEKEK
part1_1p9m_19_model_2_1	DEVIPLVEKVIEEIKKAPEEERDKLLDEAIPAILLIPVELSEKSKERLKKILKEEILKKI
part1_1p9m_36_model_4_1	KEAEELIEKGIALYGEGKAEEAIKEFEKALELDPNNAEAWLALASVYKNKLKDEENAAKYLAkakALDPelaakSE
part1_1p9m_30_model_1_1	FFSELLDEAIELFKEGKVSHETVEEVLKLVEEMSKKYKKEAKTEEEKKKIKEEALKKFKEIKEKAL
part1_1p9m_1_model_9_1	IEEELKEKIEEAIEETEEKKIKELLNKNASIEEIEELIETKEKIEEIKKEAEKK
part1_1p9m_32_model_1_1	ELKELEKEISEEIEKLWDILLKIDVEKMAELQEKMMDLSGIEAVEYLIIEYVSSLIDTEEVKKIHKLLKELLKKIKEL
part1_1p9m_27_model_3_1	MEKLLKAKELLKKMTTEEAKKIVKEMKELGMSKEEIELYVRKAIPAIEEALKSLGVVEEAPALAEAEVKEILEEA
part1_1p9m_32_model_5_1	TRVRVLADGSQEGSGRVLEAEWDSETRLLSLALYDAATNKLLEETVPPVASREELLALIRARAEALGGLRRLEVEEE
part1_1p9m_28_model_9_1	KLLELIKEGKSLEEILEEARRIAERLLEEGTLYEETKRLAKEAGDEELLKRIEEAGRDPEKLKGLYEEALLLLAAKKL
part1_1p9m_18_model_2_1	KIEELKKEIEEAKEKELALSMNDKEMAELYDKIEKAKKKIEELEK
part1_1p9m_9_model_9_1	MEIEEIAEELIEKIKEDPENAPEIVRKTIIEILEKYTEEDAELFEKVFPVLQSLGITVEVP
part1_1p9m_14_model_0_1	MEEEEKKLREEIKKKLEEIEKLEKENPVKAMEKLEEELEKWIEEKIKKYPEFKEVLEEEG
part1_1p9m_16_model_1_1	ELREFAATLPAEEREKLLAAAENPDPEERAKVLKELGLEGEA
part1_1p9m_7_model_0_1	KEELELLYKIAKEELKEEEGDDVTEEKLLIYVKKKDPELAKKL
part1_1p9m_5_model_3_1	EEEIKKEIEEKWKELEEKLKKPGEEENFKFYRETWEEIEKKEEELK
part1_1p9m_24_model_3_1	AAALARAAAAARALRDRLDTREARLAFASLP EEEREELLLFLEKWDLFPelVDELLREEVDRLEAAA
part1_1p9m_22_model_8_1	EAEIEKKLPQEEREKKCTEILDKVVEEGKNLEEYEKTLKLCeeVG

Table 2. Design of High-Affinity Binding Peptides Targeting the Trp249 Site of IL-6R α .

ID	Sequence
part2_trp249_1p9m_33_model_0_1	LAELYGAAQRALAEERYEEAIALLEEILARREEVPPSLLAAAERLLAEAREKL
part2_trp249_1p9m_19_model_3_1	ETVTTEVVEEPDRLTITRTVSGVAEPPPERIEELQRALEAEVRARGFEPVLVRAVVENGVLTVTAVGVRPL
part2_trp249_1p9m_25_model_4_1	MKKA EVEIYDEDGTLVHSEVVPVAEDGTVTYTFTVPESGKYTVKVYDEFGRLVYEETIEAKAGETYTVTVKVEP
part2_trp249_1p9m_19_model_7_1	EREALVERAVARYEEVRALLEAGDLEGFEEAIRRIAELGEAVATTIGALARNRAPAEAEARRVIARVVAAL
part2_trp249_1p9m_34_model_4_1	PELVGEAVLDGRRLRLYLRETKDEQGKPVWELEEFADGELVRRRTFASREEALAHVARLRG
part2_trp249_1p9m_28_model_7_1	LYEAVVTWEKNGQKGRAVRVYPSREEAEAAAEEIRRRLEAEGATNIRVEIR
part2_trp249_1p9m_8_model_4_1	VVRSAVSALAKTPEEALAKAKAKAEALGFKVLEVASLVKVGDKWQATVRGEKEV
part2_trp249_1p9m_14_model_7_1	VLEVAAYSVSADGKTVVNNINQTAAALGISVEEAVARLTAAPSTVIETAPDGTVYGRVPAAAFAA
part2_trp249_1p9m_36_model_5_1	KTVERVVGEVFPAGTPAEERLAAVEALAAERAPAKPARLRMELVEVLPDGSVRYRVIAIYEL
part2_trp249_1p9m_32_model_0_1	QREAVVEELERYGISRELAERLVEEAAAALGAASAVVSPEEAVDIASIHAQTLAATLPGEEAVREALRRAAEIIAEIEAE
part2_trp249_1p9m_9_model_0_1	PEVRTETITSDLPPEEVRKEVAARIAELKAQGAKEVVVIDQRVEGSSTVTTLLEVTY
part2_trp249_1p9m_6_model_6_1	RVVVEREVIEFEPIPIEEARPIAERRVEELERRLREEYPDAEIEVRASFIEEDGLVKELIVEAVVRVP
part2_trp249_1p9m_30_model_1_1	ATAVVTVDSRPRTVDIECVDPGIRLRATYPPGRHERTFEVP PGSVIRFLDPANAEGVTVTVTE
part2_trp249_1p9m_39_model_0_1	PRLRTLTYTTPKEKAEERIRIVAALAEASGLRVVDVKREQVDEDTV RITITVEE
part2_trp249_1p9m_33_model_9_1	VTVRVGPFATEEEAVAAAAEAGREVAARLGGR LAEIGLVRGEAWTGVAVVEVP
part2_trp249_1p9m_36_model_1_1	RELEQVIVLPPDAELLSVSGDCTVEVEELPDGTTRVTVRYREGQTF TITYRYRDDVVTETFSV
part2_trp249_1p9m_29_model_7_1	DTLARQEAVEAFLRARGLEPQQIELRLEPDGSWVAVAYVDAGPGRLLRVTVREVPGGGWEVVAEELLELP
part2_trp249_1p9m_31_model_0_1	PCVLPPPFTLTVLFPGDAEPRTLWLCTPVRCFTEDGVPLAEYACRSLEDPSLTGKAVLRLEGDEWVVVSVSYNRPEP
part2_trp249_1p9m_31_model_8_1	PGRTLTLTVITVSGAESVEAARAAALAQLAALKPGTTIT IAGRPYRLLSVGEYTVIEDPKTKSATVEEAVVQEL
part2_trp249_1p9m_16_model_5_1	AAELRELAALVEAAFPEWRVVGFEEDGELCVVALREL PDGRLLRALFRRREDGSVELVRVEEVPL
part2_trp249_1p9m_7_model_3_1	APRRYVITDDRELMEEILEADPSVVAITVESREAAERLAARLRAPEDVADVEVRVP
part2_trp249_1p9m_23_model_7_1	MLKKYLVARPVVEGNQPSIEIEVEAASPEEALEKAIEILREQGITGTWVISVIVDGELVERRTVTIE
part2_trp249_1p9m_12_model_4_1	MTPEQLAKADEYFAKSEEARKKAGEYLKLYDETGD PALLEKAEAYRKEADIYTAKGYKAAGDTETAKAIADKLGVTL
part2_trp249_1p9m_22_model_4_1	GMKRMRETITVAPEKVPEAVALLEELKARLPEARIEFAVDGN TITVEVEFP EEALPRLEEALAEFYARIRPLSRREYES

part2_trp249_1p9m_38_model_8_1	REEEF ² EALVAEHWP ² EIEAVYAEAA ² RTLSREETA ² AAVRAAIEALLPPEVA ² AYAIAHYAPKLE
part2_trp249_1p9m_2_model_7_1	MLKFLGLSITPAPQTEEEYEKVEKLQKEVEEKHKELSKKLDKETTLKELKKYAE ² EKAKELKLP ² AKRINI ² HVEEGE
part2_trp249_1p9m_10_model_5_1	MEVVERTLEIDAAAPEEEAVARVLAALPALPEALARA ² AVREVKANPGSEVTVGVETPERRELLTVVSR
part2_trp249_1p9m_19_model_9_1	RLDELVAS ² YAAHLAELGLAPEE ² VAAAVARIRAAAAAARAANASLPD ² PRLRERAVLDAASAEIERIGRERAA
part2_trp249_1p9m_6_model_2_1	SSKVLRYRRVGDADIDELIATVKEQLAAEGASDIRLLERSPDGKRVVLVYKDPNGEERYAIVEVVEEEP
part2_trp249_1p9m_35_model_5_1	SAALRARGLAALRAEAERLGLDPA ² AVRLVAVLRVGD ² RVVALGELPPEAYPHADPD ² DSRVRF ² EAEVPA
part2_trp249_1p9m_31_model_3_1	IPANVQVGLPDPSLLPLALKAVEVARKVVGPDATIGVTYNPETKTFSVHAQAPGQPPRYLIVAEDGTVLEE ² GEVPP
part2_trp249_1p9m_5_model_3_1	SVKVVT ² SQCTLPDGTVVTVTVSCTGPGEYTWEVSFYDADDPSNRTPPCTVTCTCSPDDP
part2_trp249_1p9m_6_model_7_1	APSAEQLAALAAEARAALLAQAAELGWTPSDLELVRPVPEGVLA ² IARATATRP ² GQPPVRLRAEALLRL
part2_trp249_1p9m_26_model_7_1	TETLT ² VNLHPGSGDTVSNISTEMVTPDGSVVRVSFDLHLGDGTVVRRSFDVQGVNGQPITVTVSREGDLVKITVTPTS
part2_trp249_1p9m_28_model_8_1	DVGERLRRAALEEGLSLEELRALAARLEAEHPDVRVRIDEESRTAE ² AERLP
part2_trp249_1p9m_14_model_7_1	VLEVAAYSVSADGKTVVVNINQTAALGISVEEAVARLTA ² AVPSTVIETAPDGT ² VYGRVPA ² AAFAA
part2_trp249_1p9m_21_model_8_1	GRWELLSE ² RFHRLPRRPTVTSTYRYVDPKNNRNLLRVVTRNAV ² TGEVLSDR ² CVEVP
part2_trp249_1p9m_39_model_5_1	SGLVVAIATAADGTRLTVGPFPTPEEAAAAAREEAARRGLTVESVEVVPVAEWR
part2_trp249_1p9m_0_model_1_1	LVVPPTAGAVFPTPEEAAAAEAAAAFYRALPDCVSVRIEESTSSSGYPVFTVVTTFRDGRVLRTVLEALDT
part2_trp249_1p9m_6_model_6_1	RVVVEREVIEFEEPIIEEARPIAERRVEELERRLREEYPDAEIEVRAS ² FIEEDGLVKELIVEAVVRVP
part2_trp249_1p9m_29_model_5_1	RELRRTVTAPSPEEAIARACDALEAEAAAQGYEADVYVLRRGKNSALLEALLRRGRERERTVHLRLVVE
part2_trp249_1p9m_31_model_7_1	MEARLEE ² GEAELVVVRVPASVPVREAVAAAAAARALHPADPAAPPVLLSLEVRPDEIVLTF ² AVPSTLLRALRERR
part2_trp249_1p9m_1_model_1_1	SGEVLEE ² VKGTEPLTEEELAA ² RQAKLEAETGKR ² WVASEEPGTLLMVDE
part2_trp249_1p9m_22_model_9_1	EREALEKVAKIIREACEGSS ² LKTRELISVAHEFISRNPFASEEEIRKIIIEAAK ² KYGASEEEKERALEVALKMRE ² VYREY
part2_trp249_1p9m_33_model_6_1	TTFEAEIMVDGEP ² QPVTVTVDES ² GVATWTVVDPVTGEPLTTTRSLEALELLL
part2_trp249_1p9m_18_model_4_1	PANTDEAALRAAAQAAYAAAAAAGLTPEEAGRVAAGGAREAGATRAELLAPLRVRVLVRLADGTVREYVLELEAP
part2_trp249_1p9m_10_model_1_1	ARAEQLLAQARAAIAAAREAAAAGRLEEARALARLAIGLAESALEVDP ² SNADAKAALAEAEALLAELE
part2_trp249_1p9m_9_model_4_1	AVPDGSRPEVVDVEVEVTEDGIVVRAVGADGEVLDEARIPDNSQEARDALLRAVAE
part2_trp249_1p9m_27_model_4_1	AGYVVDAYVIETEEGLAVRIVLRDADGNPLAERVIPVERRPSEEELEALGRAVLASL

part2_trp249_1p9m_15_model_0_1	PVVVTRTEVLANGDVLRLRETYVDGVLVAREVLELEVRRE
part2_trp249_1p9m_6_model_9_1	DEVVRVAEEAFRAALAAGLSLEEARTVAGEAASRYVASRVDLSTLSDEEIRRLTEEVERATERAVERAL
part2_trp249_1p9m_22_model_2_1	SAALAAAEALLAAVVGPARPAAPPPPAEAAAAAAAAAYAAAAAAYAAAAAILAAAGDTATAARYAASAAAHAAAAARAAAE
part2_trp249_1p9m_30_model_1_1	ATAVVTVDSPRPRTVDIECVEDPGIRLRATYPPGRHERTFEVPPGSVIRFLDPANAEGVTVTVTE
part2_trp249_1p9m_0_model_6_1	DAVAFLERHFSPEDVAAAVELVELYKAGRKEEARALAAELSVEQLSAATALALAEGLRELAELAAMARAKL
part2_trp249_1p9m_38_model_7_1	KEEEIKRAAILAACEEFGAEIVRVNGKLTVKGTTEEQLKKIEKVVELALELGGDVEKALKLA
part2_trp249_1p9m_26_model_1_1	RAAAEAAFAAAHGPVRLARAREAGAPAAELAALAAIAAGDVARVVQLLSAALNRHSSGEAAAAAARELCDALYAHLR
part2_trp249_1p9m_17_model_3_1	SPYPVVEVDALSEALTEEEVVIDTRVVRTARGFVIDYTVLLPDGTLRTYRLRV
part2_trp249_1p9m_21_model_3_1	PVTRTVTYVAEKEEYLSYEPTPGFYVPEEAAPRSNRLLKAGETVTVVELTLVDVAA
part2_trp249_1p9m_34_model_0_1	AAAVRARLAALLEARLAAAPSREEAHAILVEAAAREAGIALESDDYGAVRRALSSHLAEQLLA
part2_trp249_1p9m_6_model_7_1	APSAEQLAALAAEARAALLAQAAELGWTPSDLELEVRPVPEGVLAIRATATRPGQPPVRLRAEALLRL
part2_trp249_1p9m_3_model_3_1	REELLERLVEELREEVLRLSPDGTLTPEQVERLVRRATEEARAAGLGEEAERVAGEAVTRAIEEARREIRERLEAAE
part2_trp249_1p9m_4_model_2_1	PTRVVRVRVPAGADLEAARAAALAANPGLRLVAERPTEGRYTLLVLES
part2_trp249_1p9m_18_model_7_1	TYKVGDTVITIEVYNRGSTPIDSVKLYLGDELIAEKTNIPPNSTVTITFKADKPGTYTLVVTQGSTVNTRTFEVPV
part2_trp249_1p9m_21_model_2_1	RVVFVEATPEEIAALEPELRRRLAREEGAGTVTVTPEADPAGRAEVEALRERARAE
part2_trp249_1p9m_11_model_4_1	PVVVTLLPENADVVRVVRVVVEPRSLLARVTVVVAGKEPVTVTVLLAL
part2_trp249_1p9m_34_model_1_1	AEEAREQACALLQQAIELAEAGDLAAAIELLEQALALAREAGDPALVAEVERLLAELRARA
part2_trp249_1p9m_13_model_3_1	SGVEVVATAGPGTSVTVYDADGRPAVVVPAGHRAVYDRATGTYTVVPA
part2_trp249_1p9m_28_model_0_1	EELRAEAERLLAAGFELERRTARRAVFRNRADPSLTTEVPTELADVFEELA
part2_trp249_1p9m_35_model_5_1	SAALRARGLAALRAEAERLGLDPAAVRLVAVLRVGDRVVALGELPPEAYPHADPDDSRVRFEAEVPA
part2_trp249_1p9m_6_model_5_1	MKEITVTTTTFTPERSGTYSLVVTDKKTGQTYHLGSFTAEGETVTLTETVYVPDGLTADDLEVSVPAS
part2_trp249_1p9m_13_model_2_1	ERERLRLEDEARAARGRAEALRAAGAPPEEVAAAEEAAAALEAEVAAL
part2_trp249_1p9m_33_model_7_1	LTPEEVDSELRDGRWHVTVRVLDAEGRVALSAVGTAATEAAALALARARLAAA
part2_trp249_1p9m_31_model_5_1	TVTFDPETRTVTVDWPEAKSGRVFTTITNTAPAPTRYTVTLRWRDAEGREHELTFTVPELAPGETLTVTAGVELVP
part2_trp249_1p9m_2_model_9_1	APLPVRLEVEVKETGKGREVTIRIWVGDELVGATLRFDKRSKTTRTITVELPDGRVLTITVANDGSYTVEVRPR

part2_trp249_1p9m_30_model_0_1	EAALRAEAARRGMEILRITPWTYDPETQRWVQVAEVRLRGRPVVLQGTAASPEDTPAIAISSTPS
part2_trp249_1p9m_8_model_4_1	VVRSAVSALAKTPEEALAKAKAKAEALGFKVLEVASLVKVGDKWQATVRGEKEV
part2_trp249_1p9m_5_model_3_1	SVKVVTSQCTLDPDGTVVTVTVSCTGPGEYTWEVSFYDADDPNRTPPCTVTCTCSPDDP
part2_trp249_1p9m_20_model_5_1	TTYVDVTVTFTAPAPGKYTVTVESEDGVACGQTATADANGKITVTKTCAAIPGTTYTIVVRKEGSSEVLSTSTLTITP
part2_trp249_1p9m_1_model_4_1	MYRVVVLADGRVVAVIEVPSLEEAEAVAAAIAAADPSLTPVIVGSEP
part2_trp249_1p9m_23_model_9_1	ERATVTVTVTLPPEGAPAAEAVAQAQAQADEVAARLEAEGWTVLSRTTEVSTGADGGAVARVRIDVER
part2_trp249_1p9m_31_model_3_1	IPANVQVGLPDPSLLPLALKAVEVARKVVGPDATIGVTYNPETKTFSVHAQAPGQPPRYLIVAEDGTVLEEGEVPP
part2_trp249_1p9m_5_model_0_1	MAVRRSVRIRVKLKGSEWVEAWLVELEDGTVVVTETINPETQELVTHDLSEVVDYEVL
part2_trp249_1p9m_10_model_0_1	MVELMLTVSNPESVPGVHRTRVKTVLCRNDGLRTELAVGDGETRSRRVRCAPDGEYELEETEEVIEE
part2_trp249_1p9m_0_model_5_1	MTVTVTTLTKDASGMMRVVRTSRSERESEWRELEKGDTVTTETFEVARPGEVATVVLTKDGRVVAEGVVTA
part2_trp249_1p9m_29_model_4_1	LAEAAQQAAGYAAAGNTAGATQVIRDAVAAGADLDDVAAAAKAAAAKLAAQGLLDAAGKAALAAAIAAAA
part2_trp249_1p9m_14_model_5_1	TVITVTTPEELVALAEAAAARGARAEVSVTPTSGVVRVELDGEVRLVRLPNGAQAREFLHLVNT
part2_trp249_1p9m_24_model_6_1	TAITVEVAADDDPEAVEARVRAALAANPDATEVLVVGVPVSPANEAAILRACAEAGRTVTAP
part2_trp249_1p9m_35_model_0_1	TPFEERLAAILAEVRVAEAGQLTPETLAALVARAREAAEACGLPEEEIEDVERSIRSAADYAVSLY
part2_trp249_1p9m_23_model_2_1	MVEIEEEPSEGEgyvtviigspvagtaearlpvkgrnvvvravvdeedpaalqalaravaealaaav
part2_trp249_1p9m_2_model_7_1	MLKFLGLSITPAPQTEEEYEKVEKLQKEVEEKHKELSKKLDKETTLKELKKYAEKAKELKLPakrinihveege
part2_trp249_1p9m_10_model_5_1	GQVLSVRERDGLLTVLAGSAEAAVAEAKRLGGTDILSVEPSAELPGVWVVVTARLTR
part2_trp249_1p9m_19_model_9_1	GVIVKEFEVEDLDGHKLPCRLRVEEGSSEGVLERVNRDGSVVTQVPVNADRLLEELE
part2_trp249_1p9m_6_model_2_1	TLEESYTIYVERLPKELSEEEIKKLIIEKTGIKIEELEILEKKENSLVVLAKTTVNGLPVNIKVTIVNSN
part2_trp249_1p9m_35_model_5_1	RPLLVLIEEEIELEGTEEEERRRRMEELQREAQKENPNASVISRNDPPRLAIVRSIVPL
part2_trp249_1p9m_31_model_3_1	MAAEAEAAAAARAALAAAADAPPAERAALLDQLAAALAEAGRDAEAAAAAAAAEAARLVGDEAGAAAIRAAA
part2_trp249_1p9m_5_model_3_1	SLEKLKEELEARIAEKPDGRAYAEAEVYAELGKYAEAVAALRKAHALEPLMSPEALARRVAALEAA
part2_trp249_1p9m_6_model_7_1	KTVTFTITNTSSEDITIVKIVATFTSETDTRTFIVDEVLPANTTKTVTVTLVGKGTFTLTVVDENGNVIAATKVTN
part2_trp249_1p9m_26_model_7_1	AELDALIDEALALVGGERVGPVRVREEPPHLVLEVEALVDGVLVTAVVTIDMATGQVLSVELRPAPA
part2_trp249_1p9m_28_model_8_1	SGVRVERVEGRTGEEVAAEYEARGLEVQVFTDEETGQARVAVVRPP

part2_trp249_1p9m_14_model_7_1	VRVRTYISHASPDMEYPPPLPPPPPTLPPGTTLVEPLPATNKLTEGIVITAERITVYEP
part2_trp249_1p9m_21_model_8_1	EAEVLRHGTLEEREAVLRAAAEAAGAEVLSTERITRDGLPAVRMVWRDPAGRERTTVVAAL
part2_trp249_1p9m_39_model_5_1	MELLKKAIEKLKEGKVEEAAKLLEEAVKSGTAIPEAAMYLGRAIALKEYDLSADAPGLSAEEYIEKAEEELVK
part2_trp249_1p9m_0_model_1_1	ETVTTEVVVEEPDRLTITRTVSGVAEPPPERIEELQRALEAEVRARGFEPVLVRAVVENGVLTVTAVGVRPL
part2_trp249_1p9m_6_model_6_1	MEAILRARAAGREAATLEEVVAALKAAGLTETEEALAAIRRALAEGHIVARPDPEPDSRTLVALTRHP
part2_trp249_1p9m_29_model_5_1	MKEELEKKAEELEKQLKEDPNNAELWLELGEALAEELGKYEEALEAYRKALELDPNNRRARNALAQVCEKLGLFEEAL
part2_trp249_1p9m_31_model_7_1	LAELLATDPRGAEIARLLRERERAEALAEAREQGKEEEAARYRAEAEIAAAEI
part2_trp249_1p9m_1_model_1_1	SELEALVEQIERLLGKTTGVRVRGTRERRALTILSFEPVPEGVLVTARDVETGEVLRLLARVPETGELTVEVLE
part2_trp249_1p9m_22_model_9_1	DEELERELAIVLPRAREYPRVAISYDPTSEESRARAERMAAALLAAGVPTVVLPSEEAFAAALSPEERERALAAQPVL
part2_trp249_1p9m_33_model_6_1	PVVSRETLRMRMSADGCVAALSYEECAKRLGISPATVRRLVALARALYGPQGAG
part2_trp249_1p9m_18_model_4_1	RIRLTLRREDGSTLTVEVDAPEVAPGDRLGVRLLRSEEASALVLESVSPDGRVVRVPIELPDLSTPWELTALEFL
part2_trp249_1p9m_10_model_1_1	APCSPEAVAAVEAAAAALAAGRLAISVAPDGGVAVSGRADEADARAVGEAVAALLAERCGLTPEQREAVVETVVRAAREA
part2_trp249_1p9m_9_model_4_1	PATRATFVVADAAATAAEFGVECEVGDVVVAEIDPTTGEVVRTVVRPAA
part2_trp249_1p9m_27_model_4_1	MRAAVEECVAKAEAVKAAALAAGLSEVEAERIAQQVLLKCLREIAERALGIELPETPEL
part2_trp249_1p9m_15_model_0_1	IVFEVPEPGRTYRILVNGKEVGEVTSPASAKPGDKIEVEIPGVKPGDVVEAVDVETGEVVAGWRVP
part2_trp249_1p9m_6_model_9_1	VGAAVRTWAAREYPTPFGVVDVEQLDENRVRATVLTNGGAERETLEVDAEELLRRA
part2_trp249_1p9m_22_model_2_1	PAPPVLVDEAEPGNRYRLTYRFPNAAAAEAAAARLRAELGVEAARIERLPDGALLTVRIPK
part2_trp249_1p9m_30_model_1_1	VTVEVHFEDGTPDVPKGELKEVLSVKVTVRVDGDLATVTVEVTSRYYSKEYGAEYEGTKRVTVTVTVP
part2_trp249_1p9m_0_model_6_1	MTINKVLNRNVSREEAEEELKKLLEELKKKGTVTHSSLVIHETPEYTYATITIDVKP
part2_trp249_1p9m_38_model_7_1	AAAAEARVARVLAALREAAAESPVPGVATVTLAELARRAGISPAAARRALRRLEEEGRVGVAFRTDEETGEELATVELA
part2_trp249_1p9m_26_model_1_1	LPAAEAPAALPAILAAGPTTVTVRVDGTPEEALALARALNEAGGPTVTVTWQEGDYSTLTVRRVPP
part2_trp249_1p9m_17_model_3_1	KEELEKELKELEKKLKELTEKIAEALAAGNFKEAAEYAKKAAEAACKMAEIAKKLNDPALAAAYESLAKRYETVAKLIEA
part2_trp249_1p9m_21_model_3_1	TVPVTLTWNGEVVGTGTATITTSNGLQSVSITVTSIDAAAFAAGSASPDTPVLATINIDGTPVRVPVTSYTL
part2_trp249_1p9m_34_model_0_1	AVPPEEAEEAARAAAIAEAEAGFTLRSITTWELEGKVVTTLNFEKPGESRTVV
part2_trp249_1p9m_6_model_7_1	GLALALRHFARALALAAEAAAARLAAGEISPREALLLGEALGELAVGVGLLTRALTGDSRSPLAEALFRLAERLVR Lara

part2_trp249_1p9m_3_model_3_1	EEERRRYVEEVVERERKRGEAAVAAEKDPEARRRVRAAEEARIRSLRETGERMIEQGLSLEEIRERLEALL
part2_trp249_1p9m_4_model_2_1	MEVEEVKKKIDELMAAGKLRAAEVASKYLEELAKKYQEAAKQGDLEKVEELEKKMQEVQDHYDAAAAAYNARVL
part2_trp249_1p9m_18_model_7_1	SPSCSAQVTVTMAANSTTTFTFPACCAGGRWEVMDENGNLDTGTTDAAGKATVTNTSSVPKTVTVSVQS
part2_trp249_1p9m_21_model_2_1	RRERVLAIEELCAERGGPVSLSELARRCGISRATARRLVRELAEEGRVELREVRDGRRSRLEVLSRAELARRLAEE
part2_trp249_1p9m_11_model_4_1	TTTTVEKGPKDKTVTTTTTPLPGGKTRTVTTVIDKDGNNVETVTRERGLPDP
part2_trp249_1p9m_34_model_1_1	RALARAVGAIRAVPDEKLPEAIAIAKRLVPALADEDLRTREELIEAVVRLATPEQALAIVEALKPLLPEDARLISS
part2_trp249_1p9m_13_model_3_1	GTTTLTVTLTSPRTFANGSATVAIVTPRGHTATGTQTVSGNTVTVTVTTTLTDVLSVTATVQAADSKDRVAATTE
part2_trp249_1p9m_28_model_0_1	VFLEAESEEEALEKVREKFPDYEILSKVRRGEYITILLQNKNDPSDYLTVVVRV
part2_trp249_1p9m_35_model_5_1	MEETAKEILEEAKKIVEEVKKIKPEEAPYFEKLYEEMKKAIEAGKLDEAIAILKEMREKYNEIVKKE
part2_trp249_1p9m_6_model_5_1	VTTVTLVDTPTLRVIVHVDPSTNTVLVTTYVRLAPGVWLVVQVEYNLSSGTVTVRVSVLEADPLPVYP
part2_trp249_1p9m_13_model_2_1	TETLTVNLHPGSGDTVSNISTEMVTPDGSVVRVSFDLHLGDGTVVRRSFDVQGVNGQPITVTVSREGDLVKITVTPTS
part2_trp249_1p9m_33_model_7_1	DELVRFAAAHGIATPNRNGEITNPEDRAKIEAAYAAKAA
part2_trp249_1p9m_31_model_5_1	AAPTETVTRVLPGPCDPDSPACQETVTIDHQNRTVTTTRADGTVTTETVPLSPE
part2_trp249_1p9m_2_model_9_1	DESVTVRMSPPDSDEELEALAREAGVELETLRGARAAGLTEVTVRRLP
part2_trp249_1p9m_30_model_0_1	PRLDFRVVPGGTTVTLTSPDNRPALLVSVSGEEIVDEQVFSGTVTVTAPESKRPVTIKAYEAVVPE
part2_trp249_1p9m_8_model_4_1	VKCVFEKPGAVTVVTLRNLATGRVIATAVVPGSNVEVTVEADVTPGETLEIVMENRESLDGPVLETKTETVTVTE
part2_trp249_1p9m_5_model_3_1	RDYDALIAEAVAEFRRLFPGETSDARAQAIAAAVRAEAAARPLSDAEVRALVRARVEAAA
part2_trp249_1p9m_20_model_5_1	LTVTGLPAGSTVTLYRAHRDDPVPVPAAAPVVADADGTATFPGDVRPIPGYEFYAVVTNPA
part2_trp249_1p9m_1_model_4_1	PRLVTTTVVYDMPAEQSPVTRLAGPLAEGETADVITYNGRPLPARVRVLERSPERLVAEVTIELP
part2_trp249_1p9m_23_model_9_1	DSLPAEELERLEALRAEAEAAGITVEELTYRRALAAGLSPEEARGLAARTALLVETARYYREN
part2_trp249_1p9m_31_model_3_1	EPVLVTVPNTPEGQPTVERLLAAGYRIVSVGKETITMERR
part2_trp249_1p9m_5_model_0_1	CTASKLEAKEGESVTMTCSDDPNAHISWYKDDTLVRSTRGKSSTLEIPKVT SADAGTYTCKVTSNGKTTTGEPVTLVGK
part2_trp249_1p9m_10_model_0_1	MRVEVVYGEFSDYEEALAAAEALAAELRARDPRWERAEAGVVELAPGRYRAVAVLVASDERIAAA
part2_trp249_1p9m_0_model_5_1	PITEPQTRVVHGAEGLEGTLAHLKEVGAEVLSVEPITSPLKEEDPTEYRVTFRR
part2_trp249_1p9m_29_model_4_1	MSNVLSEKRIPVKTKEEYEEVLKKVKEEAEKLEKEEDVRVVIKSEVDGVYVVSLVVYPKN

part2_trp249_1p9m_14_model_5_1	EPGATRVTLTFEDGSTRTFVARSEEELARLLAEARAEAPVVVRVETERA
part2_trp249_1p9m_24_model_6_1	VRVEERTLTRGLTAPSRAAAEAAVPELRAQMEAELAAIAAEPGVLSVEVLAETERIEEVPGEWVVTVSRRVRLTTRR
part2_trp249_1p9m_35_model_0_1	PPHSATATIWLGEGATEEEELDARVAAVAAAAAAGAAFELLSREQLGDELLATVRVSAATPETANAVAAYAQSLSDR
part2_trp249_1p9m_23_model_2_1	VADFVAARLNATDVELLAETERTATFRVKVGDTTVVVTLARNEEGEWYVEAVE
part2_trp249_1p9m_2_model_7_1	AAVEQARALAAAAAAAAAAGDAAAAAARAAAAARALADAAAAAADPSDPAQVAELEAISAAAAAAAAAAAAAAAAA
part2_trp249_1p9m_9_model_6_1	TVTIEGDRATGTIEDPNLPGGTRITGTITVARPGEPPVTSPPFGVYNPGEPASVRRTFVLE
part2_trp249_1p9m_27_model_5_1	HEAVVAALLAAGAAEARPLGEGVEVLIRAPEEEARALVERVCAEVGAIEVGIHPLSPTLTRVRVRLLA
part2_trp249_1p9m_29_model_1_1	GREERLLERVEREDERYRVVTEVTLITDPDGRVSRRYRRVTENLETGEVIVEEAEELLS
part2_trp249_1p9m_27_model_0_1	LAENPTVTTEYEANPDTRVVVTRTLTPQTNDQGRPLFLETTTVITPRGTTTSTAWTGQYEE
part2_trp249_1p9m_19_model_5_1	PPDPAAAAAAAEAHAAARGVTVRAAVAEMHPRSGAAVVRVTLEGYAAPETVMLIPVTAPVP
part2_trp249_1p9m_23_model_0_1	EDPELAALRARLRVLSSETEVEKLPGNVIKVRSVYRGVGVGDLAGELLELTVTEVIQGGQAEIESELRLPP
part2_trp249_1p9m_12_model_7_1	MIIIEEIVYEDAPEGATVVLSYPKPKCSIEVKLNGEPLKDARIDKENGVVSFKPKPGDVLEITFRTEDETHIVTVKRK
part2_trp249_1p9m_23_model_6_1	DALDRAIALKREGGQAFAGKYEEALKLFKEAAEAAAAHDPQLAADAYYGAAEAAKKLGKEELAKELEEKAKELA
part2_trp249_1p9m_11_model_1_1	PEVRTETITSDLPPEEVRKEVAARIAELKAQGAKEVVVIDQRVEGSSTVTTLLEVTY
part2_trp249_1p9m_27_model_8_1	PEVITVTSREELVAARKEVNERLEAGYRVEERLEGDRLTITLSPEE
part2_trp249_1p9m_24_model_5_1	DPDVPVTVEVTAEDGARASAPGLNTRPVADAEALIRAAGWRGRVATVRVLRDGEVREFEVRVPA
part2_trp249_1p9m_19_model_0_1	EREERVARILREGREEHGLTREAIADDEVCRELGLSFEEALRLVERLER
part2_trp249_1p9m_19_model_3_1	SRVRVTIEGLDDPAARAATAAAVAASPRPTVSQLLAVLAAAGVDPREVRVTLELDA
part2_trp249_1p9m_23_model_4_1	PVFTLLEDTPVPAGARVLSRHTETITLSPDKSLVIESTVYELDGRVGESRRVYLLTKE
part2_trp249_1p9m_3_model_2_1	MHPAIVDVEVVIEDRGEGPVVVVRGLTRDGTLLERAEAPTLEEAEAAALAKLAAQVEAL
part2_trp249_1p9m_8_model_2_1	LAERIAALGSVVVDEAAIDPSPEARAAAAATLEAVVEELEALRAAGRLKELQPSVVSAVLE
part2_trp249_1p9m_18_model_3_1	GARAERLRALAREALELAGAPEELVERILARPDPFAAEAVRLARAELREKVRRGELPAEEAEALIARLDELAA
part2_trp249_1p9m_26_model_9_1	MTITFTVQVTDAAASAAAAADAAADALYSQVLAYARPNLSPEELAAAEKAAKAGAEVKAQIAAAATAAPATIDVTVTVDI
part2_trp249_1p9m_39_model_4_1	APYTVLPPIPGSGNVIVVPGTDTAVVYLRLEGDNLVRGPDSVPGVSNIVYTV
part2_trp249_1p9m_18_model_9_1	PTPPPPPEGSTPYTPPGWTSTTTWSFETRNGKTYGVYTTTITLPDGSTITSTARSAPVP

part2_trp249_1p9m_20_model_3_1	DSERREAILAVLQASPEPLSAAEIAARLGIAPSTASRLRELEAEGAVERDAAGRYVR
part2_trp249_1p9m_37_model_2_1	APRRYVITDDRELMEEILEADPSVVAITVESREAAERLAARLRAVPEDVADVEVVRP
part2_trp249_1p9m_5_model_8_1	PRLRTLTYTTPKEKAEERIARIVAALASGLRVVDVKREQVDEDTVITITVEE
part2_trp249_1p9m_14_model_8_1	VLEAVPAELRPGESYELRVSDELTGELLARTPTVPRGEVVRIERQRATDAEGRPVLVTTTIWREP
part2_trp249_1p9m_9_model_5_1	VEELAEAAARALGADGVVMHEGRGFTVTWEGGRARARCTLGGGIERENLPAEKEPAVRELQALAEALF
part2_trp249_1p9m_34_model_8_1	LEAELAALEAEAAARLTAHAAAAAAGDAAAAAAAAARALAALYRRAAELEYEAAGRPQLAEAYRSAAEAAEEAAKAL
part2_trp249_1p9m_10_model_7_1	LLERAREVLASLPASASREQKISAIANELEISLEEARALLDQIEAEEA
part2_trp249_1p9m_21_model_0_1	LEDLSERERLVLAVASLAPGEWISVAELAARTGLARPTTSRHVRRLEARGLVARERRDGVLYVRVNEEGLLARR
part2_trp249_1p9m_32_model_7_1	MELLEEARERVRAILRRHGALDEADRARLRAELLPRTGSLRETDEILAEALEAVRLELTPGVVPP
part2_trp249_1p9m_16_model_6_1	EPLMEVTITDKTTGRTETRVVRASEVSAVRARYTADPSLTVEVRELSS
part2_trp249_1p9m_22_model_3_1	MKKA EVEIYDEDGTLVHSEVVPVAEDGTVTYTFTVPESGKYTVKVYDEFGRLVYEETIEAKAGETYTVT VKVEP
part2_trp249_1p9m_0_model_7_1	LVEGTATLAALKAELSRLFGVEPTDEAVAALAAAGLVRRDPEDPLRYWIRD
part2_trp249_1p9m_33_model_1_1	SDSVTLPAASACRMFGLPPNSPEAIARLLAEGKVLAAWEGPGGLEVMVPRSVVR
part2_trp249_1p9m_20_model_6_1	SLTLTVPASVTLKPGESVDVTATLTDTETGESKTEVFKNIQCPMTPSTLNLTQISLNGKSASATVTLNVCYN
part2_trp249_1p9m_19_model_4_1	AVDWA AVAAAAKEIAKVLLAGRVEEARALIKAAAAATGASVAEVEAAVGRAISA
part2_trp249_1p9m_18_model_5_1	MEEARALIARAEAALEAGDYDAAIAAAAREALAAARAEDPTSPEALAAVLAAQALRLAGDREAAAAARAELEALLEAR
part2_trp249_1p9m_29_model_3_1	AAAAA AF AEISKYSRVLGVALSPEELRRRIA EFQPEVVVLPEADTPEHRAACAEAGVPWLVEPP
part2_trp249_1p9m_3_model_7_1	KSFLSWSEETVLGGLPTRV TYTLVLKDGR LVVEKTVTNLETGATTGMVEMLT
part2_trp249_1p9m_28_model_3_1	YVNATEVTAKEGEDVVLTLPGPSNPVCVIRNVTKEDEGEYTIAEESPDYKSEPVTVKLTVE
part2_trp249_1p9m_26_model_2_1	AAELRELAALVEAAFPEWRVVGFEEDGELCVVALRELDPGRLLRALFRRREDGSVELVRVEEVPL
part2_trp249_1p9m_31_model_2_1	MKWELTAEIPVENPEEAAEKIEELAKKYNPEELLITVSPTPDGYLIKVRVAFETKEKAEALAE DLKKLFPNAKVEVSEV
part2_trp249_1p9m_8_model_1_1	STVTAPAGPLTPPVEGKEVTFEATVRSSRGETVTHR VKATWPKGKRLVVTFTFRDGKLVSV EIEYVPVE
part2_trp249_1p9m_35_model_3_1	GSVRV RVTVNVTPEATTVIVEVEVVLEGSVLLAQVRAVTNNRTGETTVEAEVRLLP
part2_trp249_1p9m_10_model_6_1	MYNLTYRLNGVLRTRTFPSLEEAEAAAAELRAQGAEILSIEMVEGTQE
part2_trp249_1p9m_26_model_7_1	METATL ERVGDVAVLLPRLRAEAAAEGYRLVEEEVEALPDGRHRIRLRWERPVP

part2_trp249_1p9m_15_model_7_1	AEADALLAEGEALFAAGRYEEAAAAFEASIAAARAAGQADSLTAIRALRLLSAVYAKLGRLEEAAAAEAEAAARAERLA
part2_trp249_1p9m_39_model_6_1	ESGEEHRRILEESKKRLKEAGIEVPDEETLERIVNACIRGDYELAREIAEKAFGEKAEEVLKIIKEIIA
part2_trp249_1p9m_37_model_6_1	AAALRAQIAALRAQAAALRAQINGDPAVEAQLAALQAQIDALLAEL
part2_trp249_1p9m_23_model_3_1	MGLEEIADIVEAYEENKGTPEPILQAKIDEICKELGLTREELRALAKEIRAA
part2_trp249_1p9m_2_model_1_1	DLGPVTPEQAAAAAAFLRPYYPGITAEDLLAAGAAGLSLEEAEQRLIREAQAA
part2_trp249_1p9m_34_model_9_1	PELVGEAVLDGRRRLRLYLRETKDEQGKPVWELEEFADGELVRRRTFASREEALAHVARLRG
part2_trp249_1p9m_34_model_2_1	AAAREAAIAAGAEAVALAIPGVLAAGAQVRGRRVAVHIIADRPLTPEELAALRAAVAALSPEPLDITIEVTV
part2_trp249_1p9m_14_model_0_1	GRTYKTLTVTLTNTTPVEKEVVYKVTKATADGRVEVITEPIRVRSVPNGGSITVERVVATAWDGKTVTVEVEVASER
part2_trp249_1p9m_36_model_9_1	REEVKRRIRELLREAEERGIPLREVLEEVARFGLSLEEVRELLEEVEEVEELERRRKER
part2_trp249_1p9m_15_model_5_1	DLLAVAAARHVAEAEAAEAAAEKAEAAAGNLYAAAAYYRAAAAAYRAAARALAEAGKPEEAAALEAKAAAAEAKAEELLAK
part2_trp249_1p9m_22_model_5_1	ARNAALAARKRAYLETHYPATDIEVEELPVPGRVRVRYRYGGRWYTWE
part2_trp249_1p9m_14_model_3_1	APAPGAKVTVEQLPNGDLKVTITVPGSEFSPELVAKIEAEFTAKYGKEYVTVSVEQRSPEEPVTITVTIKA EGL
part2_trp249_1p9m_39_model_1_1	ATVSVVAAPVAPDAEVLSATATAEDGTVYPGAVDRAAGVVTLPGVPAGTTVTLSLTLRHQGRVETLERRLTVTG VLA
part2_trp249_1p9m_38_model_0_1	LSSLSPVDRLLAELEERAATAGLADRPEALARARALIA RLRELPLESPEARAALEELLALAREEGLPEVAALLERLLALA
part2_trp249_1p9m_37_model_4_1	EVIEEEVTIENPVKNTTETIPLAGGHTLEISVGETHILIRVYDPDGRCLR SERIPL
part2_trp249_1p9m_12_model_5_1	ITPEELAAIRAALAARGWTLLSADAETRTVTYRDAEGR LGSRSVEDVL
part2_trp249_1p9m_3_model_6_1	PRFPSDLLAEIVRHVRAEFPGSDILSIEVEMAPDGRTVVARVTVRLGDRVIVAEITLRVEGD RATVASTRILSE
part2_trp249_1p9m_17_model_7_1	TVTLRMPPGTPREEALARGRELVAADPSRDVLRVWMENG DVLGEYLSP
part2_trp249_1p9m_25_model_0_1	EVLERLRAEVLGTGAEGRR LGVPTGYTLIAPSAEDAALLREVAAAAGAPLTEV VPA
part2_trp249_1p9m_24_model_8_1	DAPVTTRTVELGEPYKAPNGLWRVTATLTVGGETVTFEVEAATPEEARAKALAAVA
part2_trp249_1p9m_10_model_3_1	DLERAVALADAEAAAAEAYGAAIAAALAAGLSLAEARAAAAPVGAAALAAAARAEVAAA
part2_trp249_1p9m_5_model_1_1	AVELSVSSPIPELVPEALAVARAAYEAALAAGLTPEEAEVADR AVAAWGAARGVPPELLPRAYARLR
part2_trp249_1p9m_34_model_5_1	TDEERARLQEEIEEREEAAEEELARLPPEEAERYRREREEELRRLRRAAPPPS
part2_trp249_1p9m_38_model_6_1	GDSTYRTPLQPDVAAAAARAARALCARLGMEVVEELPPRP AVNEEGQPVLRVSVVCS DGRGNTLTVESLAA
part2_trp249_1p9m_0_model_9_1	MEELKEAVWRKARLLCPEEC PSSSLLGRTVYVEEEKLEEVAK EHGVSVEEAKKALEEAAKEEGLK LKILSKEELEAEV

part2_trp249_1p9m_26_model_5_1	RSVEEEYAEVNGIRYRYTTTTVRVGRRTVVVLSTEAEEVVGE
part2_trp249_1p9m_2_model_4_1	TVYEVLVYDARGRLLEERTFATEEEALAEAERLRAVLPPATVRVVRR
part2_trp249_1p9m_9_model_0_1	RGELIEELLEEWTHDGATIRIVRVFSNRPEDARIVAEAFCPDGRLAGSRNVDPDSPFDPRALRCDL
part2_trp249_1p9m_11_model_5_1	EVEEKFRAAALELSGGDPEEAEILTRMLRIHEAEGIPVEETIELVRA
part2_trp249_1p9m_30_model_8_1	GVVAPDGTVVSGLAVGETVTFTDSATGETYTLTRQADGSFAVTNAATGEVVGWTGWVLK
part2_trp249_1p9m_13_model_5_1	SSVVTTKESVVGKAPQLGQSVTLTVTTLTYNADTGELLSQTTSQRTETF
part2_trp249_1p9m_21_model_6_1	DPGARSPEELNAALAAAYAAIADEARAIIAKRSPEVRAEYERRAAEADGDPGAELQVARSLPPEVRAAVDARLRAAP
part2_trp249_1p9m_7_model_2_1	AEKEREELRKKLRAEFEEKLAETGDFAAALNELMAKYEGDLEKQEILCEAADEIAARIAAA
part2_trp249_1p9m_24_model_3_1	ESKVVEVEVRAPDGREAAQKALAEAAKIGKRPYILSVRLADGLWLATAGAEL
part2_trp249_1p9m_38_model_3_1	GSINLPEEAYKCPVELQAEVREQVSKALGAGATPTVTVTIDGDECTVEVTVTRP
part2_trp249_1p9m_2_model_3_1	QREAVVEELERYGISRELAERLVEEAAAALGAASAVVSPEEAVDIASIIHAQTLAATLPGEEAVREALRRAAEIIAEIEAE
part2_trp249_1p9m_20_model_4_1	EEIEALARECRAAGISRTVLVHPETGELVEVTLTRTRIGAAMVFTGKGWEREL
part2_trp249_1p9m_17_model_2_1	MTLTLLSTPYVTDPLTGTSTSDVYLLTTPRRPPRYVRVVRTRVDPENPSLATETTQTYVEGELVSTTTTRTVTLP
part2_trp249_1p9m_5_model_6_1	KEAPVTVTAPEPVVGKEVTVTATNSPLGERTTPRTWPLTVPGTATVVVTFGPLKAGETQTRTVTITA
part2_trp249_1p9m_5_model_7_1	TVEVSFDSEGALVVSVTVVEPSDPDPAAPARVTRVSVIVYTDRCGKTHVRVVTERLPPLPAP
part2_trp249_1p9m_7_model_3_1	PETVVELLDAGDKASLVITVRRDGRVVARRGTPLVPKEEL
part2_trp249_1p9m_39_model_0_1	AGAAALNDAAATVADAIAAAKAALKAETGVEPEFPPLPEAIRTKTVAEAAAEGLGIGTASS
part2_trp249_1p9m_14_model_9_1	EAARRRALEEALRAARAEAAARGEDPAPAEARVREEFAAEEARRAA
part2_trp249_1p9m_35_model_9_1	AVRVLSEEPLPEDPDRLLVTLEAATRAELEARVAALAAVPGLEVESVEVSLDPETGLWRARVVLRR
part2_trp249_1p9m_2_model_5_1	MIIADSLAEAQAIRDHLLAAGIPASEPFLGEDGKWAVAVDASPEQLAAALAAF
part2_trp249_1p9m_4_model_7_1	AEELLKKAKELLKKALELAKTDPEKARELCTEARNLTNEAIREAVEAGKPVPLEEVAEIEELLREVSELI
part2_trp249_1p9m_31_model_4_1	VTVTITDVGVIYELPDGSVEITFTVTATVGDATGGKHVHVTLPGETKTVTLTFTTEDGTTVTVTLTITNT
part2_trp249_1p9m_30_model_3_1	AEAAGLAAAQRGVAAAIAAARAGASRAAVKAAVDAAVAAALALGLSPAAAAELRAAADLAIARVYG
part2_trp249_1p9m_4_model_4_1	LEERLREAVERTVALAPDPEAAAAIREYAERALAGDLEAVHRLGALASRYGTEEDARVIEEYRKAL
part2_trp249_1p9m_25_model_4_1	DAEAEALIAEGEAAAAAAGRLEEAARLYIEAAALTRGPLAGRAARLAGRAAALAGLTPEIPV

part2_trp249_1p9m_28_model_2_1	DLVAEAREDLAALERAAELLRASSPAERAAAAAELPEGVVRARVEGGATAAEFEARVAALRADLAALA
part2_trp249_1p9m_8_model_0_1	GWVVPPDPDHPIRRADPRTAAVRLLPSPGARVPPGTRRIMVDGKWYAVRLEP
part2_trp249_1p9m_25_model_5_1	EEEEAEVAALIAEARAAAAAGDWDAVEAAVLRRLRRLGRVEEADALSAEAGLGLSVSPERARELR
part2_trp249_1p9m_8_model_5_1	EPPPVMVSAADVAAVELLAVPPGADPATAAAAVRATIAAQPRLAVQADGRLGLTAPVTMLDGRRAEALLEVRVTLRP
part2_trp249_1p9m_32_model_3_1	SLAERVRALAAEIMARDGVSEEDALEAAVVRLQRELGLPPEEAERVLAELFGEEPLPG
part2_trp249_1p9m_23_model_8_1	MASVEEILERVEAHLLSQGLDPDSPETAIEILRQATNRVAARQRELEAALRAAGQPPRPLTEEEAEALLAEFL
part2_trp249_1p9m_17_model_1_1	RIEEIARRARAEGLDRVFVEGEKLSVEELQRRIIEITGRPIGVAKPAP
part2_trp249_1p9m_24_model_0_1	CLVSEEEVKKWIDALEKVLGKKLDIPVSCDPEFVERVVRLIEAQNKNP EAYF
part2_trp249_1p9m_16_model_5_1	PRILLAVGPADRREEVF AAGRETAALLRARGHEVELRELTVTGATPEELAAAAAAAAIAAARDPETPMAVVLPDP
part2_trp249_1p9m_32_model_4_1	LFSVTFEVPKEVSSSEVEPAGVEGIEPGESFTRTFSVMDPERNTLAFLTWHIDADGTATLTGVEIVPIS
part2_trp249_1p9m_6_model_3_1	VEYRQVVVSGPAAEARAALDAAAAALEAEGWRVLERAFVERGDEATGHLLAARE
part2_trp249_1p9m_9_model_3_1	MKTETWVSNGVRVTYSEVEIDGKKYYQVRAVTEEGELLTYVYTEDKEKALAAVEAAR
part2_trp249_1p9m_1_model_9_1	PTAEIVLGADDPAEFERLLAAARAAAAEDPAVLSVREEELPGLGRVVVRVTVASPEAADAFLERLGALLAEA
part2_trp249_1p9m_8_model_8_1	TTTTERTFRDGREYEKVTTVEGDKVTTKIYELGETGPVLYDEK TTP
part2_trp249_1p9m_32_model_1_1	GPWRLAAAPVRRERLPDNVVRLTFEAGDVAGEVTLVIRIQGEPPERREPIEVEAGQVLEVTVTAPPGRGVSGYLERKV
part2_trp249_1p9m_6_model_1_1	ERVRVERDRYRPGYVDVYVRTPFGVDRFTYEGATPEEARARALAEA
part2_trp249_1p9m_11_model_8_1	SAKVEALVAQVRALLPGTTITAKALPGVSPSQGVLL EATDADGNLVAVAYANVTPTTGVLDDIELL
part2_trp249_1p9m_33_model_3_1	REAE LAALERELSAARPPAGQWPPVEEALALGLAALDALVAAGRVSP EFAAEARARLEAQAAELKALGIKEAVDPDA
part2_trp249_1p9m_17_model_8_1	VTVTWDPATRTLITVEGLTEEIEVRVK TATRTATAGLPAGPSRTVTFRVEPGTTVTIELRLLGREEWKEVTTVTAT
part2_trp249_1p9m_34_model_4_1	PLTPEQRAAVERLFGEC AAAVLYNPEVDPAEAVAAAQAALAALGIGDKYTISVEGTDAPKRVT LRSLL
part2_trp249_1p9m_0_model_2_1	MTPGERLRALVEELAARP DVVAAGAGEDADGRREVGVVVRDPSTAAAI AAEVEARTGARAHVTVA
part2_trp249_1p9m_12_model_2_1	PRPLRVVELTVPG LFDGLPVVDVGARRLRGDEREINEDQLARLPDVQAEVKALIEEGVDLVRVRGGP
part2_trp249_1p9m_36_model_8_1	AITVEVTVRPDGT LVITATGPIERAEEISRIVSELAASIPDAEVLDRRIEFSPDGTTVTVTLLVRRR
part2_trp249_1p9m_32_model_2_1	PPPPVILETDSPEEVA AKLSAYFGRPVTVEEPVEIDGRVVWHVTVEGGSSFEYPAEP
part2_trp249_1p9m_13_model_9_1	IRNAVARVRAEVMERLLALGVPEEEANRLAAEVGEAVRAALEAAAAAA

part2_trp249_1p9m_25_model_8_1	DLCAEVRAQAAAVAAAPGLTAEERAAAAAEVAARAARITAETGVDDCTA
part2_trp249_1p9m_3_model_8_1	ATVKTVTVTTNQTSLTITATNVAGVVINLNGATITNLTVNMTGAKDPGVTVSGSGRVTGQRTLTVT
part2_trp249_1p9m_32_model_6_1	EEELTPTEQAALRYCRGRAEEGVSVATIAETLGVSRTASRHLRRLEARGLVTRLPDGRYAPARR
part2_trp249_1p9m_21_model_4_1	SGTAAALAAAQAARAALSAEILEANPPLEVAEAAAAAADALLAAAVAAAAAGDAAAAAAAAAAATAAARAILAA
part2_trp249_1p9m_37_model_3_1	REEEIVRLAREILARDPTRRVLTEEDLERIAEELGVSREEVLEVVLRAVIPVPEEVARL
part2_trp249_1p9m_25_model_3_1	ERDLLTLRSTRDPSRVLVVPVTANLDDAMAALDDGPVRTATLTAPDGTETPVEIVRR
part2_trp249_1p9m_1_model_7_1	KKKILIITPRVTPLVEALVERLKAEGHEVEVEVMPPDMTYEQAAEYGRKKVREGKYDGYIVIPEEASL
part2_trp249_1p9m_21_model_1_1	MHSITRSFLIEKRLTPEQVEALEAELRAIAAAHGAEITALEVEDRK PQSITVTVSSPDRAAVEAAVAALSRHCSAVNAAL
part2_trp249_1p9m_21_model_7_1	PLHPDTREAILDAVLAADAAGRVLSLRELAEECGISLPTASRHVRELEARGLVVRVPTGYRASPGFREAQAYLAAAPPDS
part2_trp249_1p9m_7_model_6_1	ADGETVKTDVTITTESGETAKCTLECVVDGKCKATGVCTMPDGTKYPVTAWEVKDGKPVLTSLSWMTPEP
part2_trp249_1p9m_10_model_4_1	AVGTLVAVRAGETPEQAAAAARPVVERDENG NPIKRTFVTPVPASEVD
part2_trp249_1p9m_33_model_5_1	CPRVSVTVTTPAGRTVTLDLGPVCRPVGA EADVGDGTYVRALPRREGDELVDVLSLVPEP
part2_trp249_1p9m_25_model_9_1	GMKRMRETITVAPEKVPEAVALLEELKARLP EARIEFAVDGNTITVEVEFP EEALPRLEEALAEFYARIRPLSRREYES
part2_trp249_1p9m_26_model_0_1	MATLAAAAASALEEVRYLRSLGREAEADAVLDNRFAALLRANPEVSPEELA AVRAAVRAEV
part2_trp249_1p9m_15_model_2_1	RPVTLKVVPVSSLEEARAAQKELGG EVLSSDETILREGRAVIAVPADA
part2_trp249_1p9m_1_model_6_1	MRENGTCLVDEPAELGEVGTRVYLD PETNIVHVWRETRSEEETTYELVDSFPIPPD
part2_trp249_1p9m_35_model_8_1	SEEEALAWFAELGLRPDRVEPVPGQLDAYGYIEGETPTVVRVSVYELDGKKELTASTRPLL
part2_trp249_1p9m_37_model_9_1	LYEAVVTWEKNGQKGRAVRVYPSREEAEAAAEI RRRLEAEGATNIRVEIR
part2_trp249_1p9m_7_model_4_1	ASDWYCHRLPRSTTVYACRRDLDLTDPEELERHC IPLPAPCTEEVL
part2_trp249_1p9m_28_model_1_1	PPNTVTVTSPDGVTCCVVKETGELVQASAPESVC PAEEVWEKCKEIAAKLP
part2_trp249_1p9m_26_model_8_1	DTRPLTPEEKARLAERLREELGLPEEEVD AIIAAATTEGELISRVADALGPEAARKVEEFLKEIRAE E
part2_trp249_1p9m_34_model_7_1	PGRTLITLVITVSGAESVEAARAAAL AQLAALKPGTTITITAGRPPYRLLSVGEYTVIEDPKTKSATVEVEAVVQEL
part2_trp249_1p9m_33_model_2_1	DRPVEVRTLPVEEAQRIAREELGFRMPVEEGDE VGVETEDRAVRVRVE
part2_trp249_1p9m_39_model_3_1	TTTTTWLEPSLPYDPSKAVLVTRTTTTPTETRVEEEVFELP
part2_trp249_1p9m_32_model_0_1	ETAEAAKAAALAAAKGKLTPEELA ALEARGDALIALAKAGKLTVAEVKAELERLLATLLEAAARR

part2_trp249_1p9m_8_model_7_1	KTITVEVTGEGKVTVTVEEDPEKNLITITVEVDGDGVTVKVYVDGELVATMTSTSGPATLVLTPL
part2_trp249_1p9m_3_model_4_1	LEVLLAGARAGELATARACAAALTAEGRLDEAAAALEAGLAAARAAGDAETAELAALLAAVRAA
part2_trp249_1p9m_35_model_6_1	SGTRTVTLTNPDPTDITIVAILDSNGNVVDTVTVPPASSKTLTIPEGYSIEYLKRPRLW
part2_trp249_1p9m_38_model_9_1	PAAPPVDPGRTAASAVVTAPTAEAEALARAAADPSLGEVTVTVQQGPDGTWSATAVVVRP
part2_trp249_1p9m_15_model_3_1	ISLATLEKAAAVIRARLEAGQVPADPEDFAAQAAVVAEALAAAGIPAADVAEIVAANAGLGAMGILGAIEERIRAAK
part2_trp249_1p9m_38_model_2_1	ATATVKVTISKNVAAAGTTVTVKVYDGDTLTDTVTVTVPRAGANVGFLIDVKGATEGKTYTVVVYDTNGNLLGTNTTTA
part2_trp249_1p9m_11_model_0_1	RRDWELLDRWVENGKRVELWRNRRTGELVRRVYSADGELLSEEPAEVP
part2_trp249_1p9m_16_model_1_1	LPPEGSSWSVSTADGSRGGSATVNPGDATKVTVTLSDGTTLTGTVTSATATSGTVKLSDGTTNLNRLRGV
part2_trp249_1p9m_17_model_9_1	MLSLEQIEAIEEAYRPLAGPDEVIEIVECESREAAERLLAALAADAELRAQGWEEVVRVEVSEVDGRYVVVVLLRP
part2_trp249_1p9m_29_model_8_1	PTVSYAEETRTVTVTSKEGVRVTVTPAPDESLEEAVARARARLAELLA
part2_trp249_1p9m_29_model_0_1	KLYRASVKTTDGDKTTTEFVGVPFTREEAELACQRACAAKGDSVRCECIVEEL
part2_trp249_1p9m_16_model_2_1	LTVPLYRVTATAPDGTTRDLDGTIAASSEEEALANARARVGDPSLTLTAEFVSTLTVS
part2_trp249_1p9m_16_model_8_1	SGECTPAPGETVVSERVGDRLRYTIVDADGRVVRTVERRL
part2_trp249_1p9m_24_model_9_1	YKVGDKVKIEIPLGKERVVLEGTVEKILSENEIVVRATLRVLDANGEVVYEVTGTATIRLDENGNIVSISFEPE
part2_trp249_1p9m_35_model_1_1	EEIELEFPGAEIVDVEYDEETNTLSVALRRNGRITWIEE
part2_trp249_1p9m_28_model_5_1	SNTALISYYGGNPEVTSVDGIEAPPIPVSELEPGKIAIALFDKETRTVTILEIVDA
part2_trp249_1p9m_36_model_2_1	MRVLTLHGVRPEEVTVEPLPGKGVRVRIREGAEGERLLGGHDAVEVVLVP
part2_trp249_1p9m_26_model_6_1	PTVTVRVHLPDAKSAEEGEARGQELARRIQRELGRRVRIVDVRETPDGV EVTLEVE
part2_trp249_1p9m_7_model_5_1	EAVVQAIANAQSREEFLAAARAYVAAYPERAAALAAGVSADSEEARLNAIAAGVLAEAA
part2_trp249_1p9m_0_model_8_1	PEFELVETTPWVTLPDGTQRQVQLWKVTDKETGEVLGYLRVEYERDAEGNWSETEGTGWSE
part2_trp249_1p9m_13_model_8_1	VTVRVGPFATEEEAVAAAEAGREVAARLGRLAEIGLVRGEAWTGVA VVEVP
part2_trp249_1p9m_17_model_4_1	LAAAARALRAAENALILEGEDNPALRAALAAAQAAL EAL
part2_trp249_1p9m_2_model_0_1	MEVVERTLEIDAAAPEEEAVARVLAALPALPEALARA AAVREVKANPGSEVTVGVETPERRELLTVVSR
part2_trp249_1p9m_6_model_8_1	GLREETERALARAEQLIAAAETPPTDPEARAALIAEAERLRARLERLRALVEDPELRARLDAAIARLDAAIAALR
part2_trp249_1p9m_39_model_2_1	SPLAELAARYERLAAEIPEDSVVDAAWAERVA AETGDPALARALLGASGRAARAALRNEAARLRALE

part2_trp249_1p9m_27_model_9_1	AEFFLRSLGFTVLEVREDTSGEPTVVVRARDAEGRLWHIVLSTEEPPEVVSLGPDPRL
part2_trp249_1p9m_19_model_2_1	RVEELRRRRAELAAQARAVKLEERRRLVEEAEALAREIG
part2_trp249_1p9m_11_model_6_1	AKPEPVPAVTLRTGESVTVTVTFNDGSTRDLTFVAVPEEPPGVVEGPAYTGADGGTSEVRPITVVEA
part2_trp249_1p9m_3_model_1_1	MGRVPDDRLAAGLAAARTPEERFDIALRALSEEEQRELLAAVEAGDAEAAALALVDRLAPRLPPAAAAALREAVRERA
part2_trp249_1p9m_11_model_7_1	MLKKYLVVARPVEGNQPSIEIEVEAASPEEALEKAIEILREQGITGTWVISVIVDGELVERRRTVTIE
part2_trp249_1p9m_14_model_6_1	DPPIVVELDRLFASREEAEAFLEKELEAGWEILDYEIEEVEVDGVPRWRLRVRARKKR
part2_trp249_1p9m_12_model_9_1	SPDDVELDLLLRSDNVLTVRLTIGDRGSTRSFKANREEFEKLYKEKLEEELEKQGYVLVSEEYYEDKSSKEIYRYALFRRR
part2_trp249_1p9m_3_model_9_1	AVPYEARRLPDRAEVTVTTFPADTDPLEAAEAAAAIAAREAFPDRVVASVRVERGADTLTAVLVPR
part2_trp249_1p9m_6_model_0_1	TTTTTTITNTSSEPVTVTCVPLNSAQTPPVTLQPGQTLTCVSTTAAGGVSPVTVTLLVDGETVSTQTVTIQGNLAL
part2_trp249_1p9m_30_model_6_1	MEERKKVAEEAAKALKEAAKVYAERGKTDPRYAEAKGYEARAKAVLENAEVLSEELKELLEEAQQLKEAKEL
part2_trp249_1p9m_10_model_2_1	SHEELQRARTAILELLEREGALPREEVVARVCARTGASRETVERAIDRLLELGVLEEDPERGLTLARRPTPEAQRPP
part2_trp249_1p9m_35_model_4_1	LAELYGAAQRALAEERYEEAIALLEEILARREEVPPSLAAAERLLAEAREKL
part2_trp249_1p9m_27_model_7_1	LEAMREAIRARIASLRAEHPEAAAAVAAVEAGDLAAAAAIAEAAGLGQLADDLRALLAAE
part2_trp249_1p9m_37_model_8_1	PPLPKVPELEGWELVSVEVLPDSVVVTWERREGDLLRTLTLTYDRDGRCVSTESSRRPL
part2_trp249_1p9m_1_model_8_1	VEDTTEVETTEITDGSKIKSVTVTTLKADGTVTRTQTVEYKDGTVETTTDTWKLP
part2_trp249_1p9m_14_model_4_1	DAARAVILAYLRAHPDLEAGDALNRAAAAAMDAGVPREEAVAARNELVALFQRLVDEGKLSPTATAAEFLAALEAEA
part2_trp249_1p9m_30_model_4_1	ASRAELVALVREGAARAPPELRARLLALLAEDPLHPEAIAAAVAVDPSLAPHIAAAL
part2_trp249_1p9m_25_model_1_1	REEALRIAEERGPEAVAELAGLVLPSPELARELSEIAGEESRRLAEQIAAER
part2_trp249_1p9m_5_model_9_1	EIPPECKVLGEAELEAESQEAAEEVVRETTFSDPNLELKSLECVERREREAVFLVVVEDKREPGKYIAVRVRVCKR
part2_trp249_1p9m_27_model_6_1	SSKVLRYRRVGDADIDELIATVKEQLAAEGASDIRLLERSPDGKRVVVLVYKDPNGEERYAIVEVVEEEP
part2_trp249_1p9m_10_model_9_1	ELIEEAEALYAAGRYLEAAELFEEAARRLRAEGKEEEARLAATLAAEAYGLAGRYEEARRARE
part2_trp249_1p9m_20_model_8_1	SCGIAAPEPPPELLRRATEAVVAACGPEITDTPEARAACAAAIAAVLAEA
part2_trp249_1p9m_20_model_2_1	EREEAIERVRRAREEGRISGREATEVLAAL EEGVEVP EELLRRALEAV
part2_trp249_1p9m_0_model_4_1	SKRFFTVEATGATKEEAIAAAKAKIPENARVVEIEVSRLPDGSYRVRIIAEVEE
part2_trp249_1p9m_37_model_1_1	MPILDAIASNAEAGNAEQALAAALAAARADPLCAGVPAELFDALEAAIKANDGAAAVAIAEEIRALI

part2_trp249_1p9m_38_model_1_1	APLDVEFVEVTPEGRERLLGRATVPSLEAAAQRVLELATTVAPPGATVRAIDAATGTLLESEATAPVP
part2_trp249_1p9m_22_model_4_1	TTTVSVSFTPPTSGTYTVQLLDENGLVASKTVTATAGQTVTVSFNIATAGTYTAKVLDSNGTLVSETTVTV PAP
part2_trp249_1p9m_34_model_3_1	GVTTVTASVTVPLTADVAAAAAEAAAKAKALHPSPDVIVDVRAEERDGKLVVTATAMSDATA
part2_trp249_1p9m_4_model_3_1	ALAAEREGLRAVATVVAELAAAGAPAEIEAAGRAVAEAHGSPGARALLEQLVAEYRRDPERGLAAARATVAAV
part2_trp249_1p9m_21_model_9_1	MVEKVEVEVSDPSVAQVLSVELSEDGKTLKITFRMLKPGRNQLTITVTAKGYLPTTIGVSVDAPAGVHTLTVPITLTKKV
part2_trp249_1p9m_24_model_7_1	ATLRVDTGDEWVDVPATDAVTVTTDADGTKLVTVTGSTPLTAAVRAVE
part2_trp249_1p9m_28_model_7_1	SELEESLALARETGEKLRKEGLEEEAAEVREEAGKLANEGAPLEEIKKVIEERVKKAREEK
part2_trp249_1p9m_11_model_3_1	PGRLLAVGTRELPGDVVVTVLEVSGDGAAAAAAAAAAAAIAAAVDKPVTVILLPVEP
part2_trp249_1p9m_28_model_4_1	KEVVITRTTETRETAERIRDALREAGYTNNAVVSYDEETGKYVVTATVR
part2_trp249_1p9m_10_model_8_1	MTPEQLAKADEYFAKSEEARKKAGEYLKLYDETGDPALLEKAEAYRKEADIYTAKGYKAAGDTETAKAIADKLGVTL
part2_trp249_1p9m_31_model_8_1	AERAAQIAEAAAEGYYTALLSKDPDGTWTADVDPAGRLLARSTGHPSLEAAADAAAAALAGLPQPVVVIGVE
part2_trp249_1p9m_13_model_7_1	RTAELLKRLVEVLKEAAKLVAEGKTEEAAKLLKKLAEEAAGVDPATAGLAERAAANAEAGDLEGIKANIERAKERLKRVE
part2_trp249_1p9m_15_model_4_1	SRTL VFELGEDGEAEVSVEDPELEILERRVITLPRRRLAVVTVGRR
part2_trp249_1p9m_30_model_5_1	SPAERAARVAELEARVKALLDAHAGRSMTAEELAELERLAAELEEAAA
part2_trp249_1p9m_14_model_2_1	APVMSPKSVTLLRGETATLTCTTADSTYTTLTVTCDHPGSSGEYALELPDGTRSSWSTLTCVEP
part2_trp249_1p9m_30_model_2_1	EREALVERAVARYEEVRALLEAGDLEGFEEAIRRIAELGEAVATTIGALARNRAPAEAEARRVIARVVAAL
part2_trp249_1p9m_5_model_2_1	MEELRALADEIIAREGATPETVRRALERYAKERGLEVALRLAGLMVSRMDEEEARRAAAVVAEFVRERLNLPRPP
part2_trp249_1p9m_24_model_1_1	GTTVQTLGGDGPDDVVVTLTASGSTCTATVTKGTGAAVTGAAGETLTVTVTDPKNPAQKYTVSLVCPAA
part2_trp249_1p9m_12_model_8_1	DDARRAALADAVSAATGAPRSVCLAIDALEETGWDLEAAARRVAALHGIPILETALAVCRAAAALA
part2_trp249_1p9m_26_model_3_1	GRLVLVDERTVTVESGSKVTPIPAEYAGKLVVVVLTDAATGEVVAQATSTGPEVTIEIPPGVSGEVVVKIYVVDP
part2_trp249_1p9m_4_model_0_1	EEATYSLVALTREEAEAAALARAKAEFAALGWTVSDVRF EKRGGLWYVVTFTLRR
part2_trp249_1p9m_29_model_9_1	APMAGRAEVLAAALVAAGVDRAEAEALLDAIYTAPTEEAALAAARALAAAAGVTATAETAATVDAVTA AVR
part2_trp249_1p9m_31_model_9_1	KTVERVVGEVVFAGTPAEERLAAVEALAAERAPAKPARLRMELVEVLPDGSVRYRVIAIYEL
part2_trp249_1p9m_4_model_6_1	LEAEMAAL EETAVRVAEALRADDPEALRAVAEEYRERLLALGREAGREREARLRAAGFEVRALQLSPEEALRLMDEAEA
part2_trp249_1p9m_17_model_5_1	DALRAAAVA AVEALLGERVTVLESVPSEGGGLALNLVTASGRRLGVLARPD PDDPTRVHLAVRDYDTGESWELTVTLT

part2_trp249_1p9m_27_model_1_1	PPLPPGWTARMEVNPDTGEEFVLATSADGSERRKAMIKDGVWGPWEPV
part2_trp249_1p9m_15_model_1_1	ATAAVVARLEAAAAASPDPVVATCLSRAASLLAAGDLAGVERLIARLREARELNPEHAAALAEAVALIEAALAEI
part2_trp249_1p9m_18_model_6_1	KPKKLLVQYGPASKTTVVVTVTNLETGETTTSTVTRKNAPKEPIEKTYLEGETIELK
part2_trp249_1p9m_15_model_9_1	EFETVVVDNLVLRSEVVLDATPRIQPGDLVRVLRVREDGSEELVEEAREAAADGSSVRVAVPRVRLGERYRVLVE
part2_trp249_1p9m_9_model_7_1	AVELTLELEAPTEEEARALAERLAELLDALPDVRVLESSVRGTLVVVRVLVVL
part2_trp249_1p9m_28_model_9_1	ERTLTIPESQEEELLAWLAREREERDVGFTVTYNPETKTYTVTVREAP
part2_trp249_1p9m_9_model_2_1	RVEEAEAELERLVAELRRAAREADPATAAALAAAAREACRLAEASRKETEEERLAALEEAIALARAALAAA
part2_trp249_1p9m_5_model_4_1	SRARLIAVAEQLEALAAAGDLDAARALARAALAEARAAGDWESVAWLEAVIGAADSPDPALAAA AVRRLAAEIRAAAAA
part2_trp249_1p9m_38_model_4_1	DTALWTALHALAKALAAALGPEAGAAAAQAIAAALTGGAVEVQVQYSGGGQVAVGSAP
part2_trp249_1p9m_33_model_9_1	TARVRISCD�REAAEAVAAEVAALPGVLSAEVGMADGTPYVLVTCADRAAAEAVAARMAAAPGVTDVRLLS
part2_trp249_1p9m_15_model_6_1	EEELIALARELLAAGKSLEEVQNEIMRRTRDLPLEERERLLQLAAELA
part2_trp249_1p9m_10_model_5_1	SSGVRVEWREEVDPETGERSLVQTSVTDPRGRTVTLPPeerERiPLPP
part2_trp249_1p9m_18_model_2_1	RAEELIREAERLVEEAERLAEEGGPEDLKRALELAARAIELAAEARLLTDDPELLERAREVEERAEAEARRRAGRRLAELE
part2_trp249_1p9m_35_model_2_1	LTLPLPTRIPGATLGGAIEVEGDHLHVTLTGADGTSVHVSLPLPVDLGGPRPARFTLDAAANTVTLE
part2_trp249_1p9m_7_model_1_1	PPPTCYRLTLECVNERTGERATLTVMGPKRGPETTVLVGDTRYRCTVVSEEEVVGC
part2_trp249_1p9m_15_model_8_1	APPAPLVASDGERELRVEALRLLRAERNRVVWEARAPDGRRGRLVIEDRPDGVTVVRLVDEAGRVLAELEGRDITV
part2_trp249_1p9m_23_model_1_1	EEERRALREAVAALAREGKTVAEIARELGLSQAAVSRYLREAGVTPPR
part2_trp249_1p9m_12_model_6_1	ELVLRLPVLPPGTRVVGEGELDGKRFTAeAISSDAPLDVGEWVDGVLVRAVQAP
part2_trp249_1p9m_23_model_7_1	TPEELAARAARELLEAEVAALGLKVVRVETSVNEDGSATVTVTDDGTTATRSLSPEAAARIAAPIRP
part2_trp249_1p9m_7_model_7_1	PVLATVTTAVELPEVLPDSTWTTPEVVLPDGTTVVVTVVEVVRVGDRSYLVTTVTVY
part2_trp249_1p9m_20_model_9_1	MMKVKVVIKAGPGLEVDEERLEEILKKAEELTGIDLSNAEITVEEDPSIVGAFKVTVTYVENGKEKTVTFILPA
part2_trp249_1p9m_16_model_3_1	ACNINEEPEWVIEPGHTLTLFSLVLDcNPdNPVTVTVTVTATEPGTYTIPITITSADGTTSTTVTLTIHVR
part2_trp249_1p9m_32_model_8_1	GVRVRVRVTRTDDGGVLVEVEVYDENGRLRSVTVLISPRLPRIVEVT
part2_trp249_1p9m_18_model_0_1	MEVIETPTTLRDTRGRLVPARVRVTRGPNATVEVFRETPEGPQPVTVVVDGVIV
part2_trp249_1p9m_12_model_0_1	LEDSPLMREIRAIARAaVEAAVAALGPDLPaPELLAAARALARPavMALGKRYGiSEEEAEALLNGEVAAAVSALLAAR

part2_trp249_1p9m_33_model_0_1	ETVTITVSPEVINGKYQPVTVTIGLRPGTYRFTLTGSKTCTTGDITLSTATTVTVTVSASGALTVDSSGNVLATCSVP
part2_trp249_1p9m_38_model_5_1	PCVLPPPFTLTVLFPGDAEPRTWLCTPVRCTEDGVPLAEYACRSLEDPSLTGKAVLRLEGDEWVVVSVSYNRPEP
part2_trp249_1p9m_5_model_5_1	GAKKAAEAAVKVAKEFNPEVGKVAEEVAKEVVKKEKDPVAAAAEIVEETEVLKLELGKEEEAKILRENREAIIIEAARKAA
part2_trp249_1p9m_9_model_9_1	RPVLWRLTVRVTLPDGRVAERVVERLFRSEAEARAALALRAAAAEAEAAALGPATVEGEVRLRE
part2_trp249_1p9m_12_model_1_1	SLLAYKVGRAVAYLDAHGRDVSPEELAAALGIPLPLAARIVAAARAGELSAGRLTPEAQRAIEAAA
part2_trp249_1p9m_27_model_2_1	RELEQVIVLPPDAELLSVSGDCTVEVEELPDGTTRVTVRYREGQTFITITYRYRDDVVTETFSV
part2_trp249_1p9m_17_model_0_1	TVRVETYTSEGGEVLVHRTVTVTWGRRTLTVVHVTYAGPGSTTPVHVVVQ
part2_trp249_1p9m_3_model_0_1	PPPPVPVGPKNVTKKLGESYTAELLIEGKKIVCNTLVKLTEEDIGTRKVV CYDKESQLTHTYTLTVLPE
part2_trp249_1p9m_6_model_2_1	LERVPLAENVFVPGGERKEITATANGKTYTFTAGNDGPDIIINAWILLENGEELEVVEVRPA
part2_trp249_1p9m_36_model_4_1	LEEEAEKLEKEFLEKCESLSATPEELSALMDELLELLARLEDYERARRMREKYTEAMERLRKA
part2_trp249_1p9m_28_model_6_1	KIIIEKEIKSDLPMEETKEYVLQYIESRAKRLNAKIVELEIEVDEEKKVVKVILEP
part2_trp249_1p9m_13_model_4_1	TAAVEAAVAEAAAALVAGLSPEQLAELAAGLAALSPALRVPSEITPEALLAAIGRIIATDPSLAPAVLALLRAVLAA
part2_trp249_1p9m_8_model_3_1	KPLQNIAGAKVEGNTIILTDLVTGETIEIKVEVKPGETGLKFVTDDGTSVENAKPGENYVLVEVDENGNIKVLSVSRMVG
part2_trp249_1p9m_14_model_1_1	LPPQEVQTVTFSEELQERSGISRETLERRLAEEAEARGIVRRLEGGRWAVTGTEAEIQRVL
part2_trp249_1p9m_23_model_5_1	LLSDPRIGRREFHLPAAEELEARLAALAAAAAAAGRPEELLALELEPEGDGVRVTLTLRLLP
part2_trp249_1p9m_2_model_6_1	PSQVYEKTETFPDGTKINVKMTAYELPGKDLVKVTVTVNGTTKTCTVTPAGYTYSVTVDRNGNATVHAVPQTPEFG
part2_trp249_1p9m_36_model_7_1	RERISIEVKKLGKYYGVYVVDKETGARRSIALFPTKEEAEAYAEANREEIVEEFEKEL
part2_trp249_1p9m_25_model_7_1	GGKWELEEELEIDGTKIKKYVSKESNKALTEITLPDGTKITEEHELPP
part2_trp249_1p9m_20_model_7_1	RERVEVLRTENDPERKVNTTLLEVTRDGVTTLRSLVLDGVLTTTEELP
part2_trp249_1p9m_4_model_8_1	VSGGSVELPDFTDKTIREVSEWLTERGLTYTVLSVDERNRTITLEVSRIVNGVPVPVGTVTAKLE
part2_trp249_1p9m_34_model_6_1	TTTVTAEPGQKITFPVDGSPRVNTTVVVRDPDGRVLATCVVRGHPDPEITCTAA
part2_trp249_1p9m_39_model_8_1	PVVGFSVCNPELGLCLQVDVVVEDGVLYLVVRLVDADGTLLARVAVPL
part2_trp249_1p9m_37_model_0_1	SPEAVVEFDEVETPDGTLVRVTAVRPPPGVEARVTVVISRPSPTGPVTREVRVEPL
part2_trp249_1p9m_12_model_4_1	ALARLDALVAALAADPAAAAAGRLTRAERRALAARLGVTEEEVDALAIVASPPEAAAVLRERAAAAA
part2_trp249_1p9m_25_model_2_1	PTEEQRKRIIELLTSGKPITLSEISEETGLSERTIQEIRREYLEEKRA

part2_trp249_1p9m_20_model_0_1	EEEEVEVEVELEEKEEKVVKLSETESIRFSRYNGWLRTSRETATRTLSSSEVEDLE
part2_trp249_1p9m_11_model_2_1	APLTVTGPASVTVEAGKSATITVTATDGDAPPVITYTFTVENRGTEPGTKTLKVTVPGPPTGVAQTATYTTTTFV
part2_trp249_1p9m_13_model_1_1	SRVERDVREAFDLLAADVRRLIAEGRAAEAAARLEQARALLAENEDKISPELLAELRARLAALEAEVAAAA
part2_trp249_1p9m_36_model_6_1	DTLARQEAVEAFLRARGLEPQQIELRLEPDGSWVAVAYVDAGPGRLLRVTVREVPGGGWEVVAEELLELP
part2_trp249_1p9m_19_model_7_1	TTITAKDGDVILLDENGNEVASIKEGETLTITEAGTYTSLSDGENPIKTFMAPAA
part2_trp249_1p9m_18_model_1_1	RLDELVASYYAAHLAELGLAPEEVAAAVARIRAAAAAARAANASLPDPRLRERAVLDAASAEIERIGRERAA
part2_trp249_1p9m_29_model_6_1	SVTVSVATDEDGTTTTVVVTWDSGATLTVVVD AEGRLLSVEFRPGGP
part2_trp249_1p9m_16_model_0_1	AIEVENAATAKPGDKVTVTVTNTSKEPVTVSGSYPVQNGDGNLYLVEIPYEVTVPAGETKKVEGTVKAP
part2_trp249_1p9m_31_model_6_1	REEEFEALVAEHWPEIEAVYAEAAARTLSREETAAAVRAAIEALLPPEVAAAYAIAHYAPKLE
part2_trp249_1p9m_33_model_8_1	MPVLHVEIGVRSYGATPEELRAAARAYVASIGGKILEEEFSETPEGRVYRALAEGTLKRDPNAPKIPVRLELTVRGRRLP
part2_trp249_1p9m_19_model_6_1	LAELYGAAQRALAEERYEEAIALLEEILARREEVPPSLAAAERLLAEAREKL
part2_trp249_1p9m_36_model_5_1	ETVTTEVVEEPDRLTITRTVSGVAEPPPERIEELQRALEAEVRARGFEPVLVRAVVENGVLTVAVGVRPL
part2_trp249_1p9m_22_model_7_1	MKKA EVEIYDEDGTLVHSEVVPVAEDGTVTYTFTVPESGKYTVKVYDEFGRLVYEETIEAKAGETYTVTVKVEP
part2_trp249_1p9m_26_model_4_1	EREALVERAVARYEEVRALLEAGDLEGFEEAIRRIAEEELGEAVATTIGALARNRAPAE EARRVIARVVAAL
part2_trp249_1p9m_1_model_2_1	PELVGEAVLDGRRLRLYLRETKDEQGKPVWELEEFADGELVRRRTFASREEALAHVARLRG
part2_trp249_1p9m_2_model_2_1	LYEAVVTWEKNGQKGRAVRVYPSREEAEAAAEEIRRRLEAEGATNIRVEIR
part2_trp249_1p9m_27_model_3_1	VVRS AVSALAKTPEEALAKAKAKAEALGFKVLEVASLVKVGDKWQATVRGEKEV
part2_trp249_1p9m_2_model_8_1	VLEVAAYSVSADGKTVVVNINQTA AALGISVEEAVARLTA AV PSTVIETAPDGTVYGRVPAAAF AA
part2_trp249_1p9m_33_model_4_1	KTV ERVVGEVVFAGTPAEERLAAVEALAAERAPAKPARLRMELVEVLPDGSVRYRVIAIYEL
part2_trp249_1p9m_13_model_6_1	QREAVVEELERYGISRELAERLV EEAALGAASAVVSPEEAVDIASIHAQTLAATLPGEEAVREALRRAAEIIAEIEAE
part2_trp249_1p9m_0_model_0_1	PEVRTETITSDLPP EEV RKEVAARIAELKAQGAKEVVVIDQRVEGSSTVTTLEVTY
part2_trp249_1p9m_32_model_5_1	RVVVEREVIEFE EPIIEEARPIAERRVEELERRLREEYPDAEIEVRASFIEEDGLVKELIVEAVVRVP
part2_trp249_1p9m_7_model_9_1	ATAVVTVDSRPRPTVDIECVEDPGIRLRATYPPGRHERTFEVPPGSVIRFLDPANAEGVTVTVTE
part2_trp249_1p9m_0_model_3_1	PRLRTLTYYP TKEKAEERIARIVAAL EASGLRVVDVKREQVDEDTVRITITVEE
part2_trp249_1p9m_4_model_1_1	VTVRVGPFATEEEAVAAAE AAGREVAARLGGRLAEIGLVRGEAWTGVA VVEVP

part2_trp249_1p9m_37_model_5_1	RELEQVIVLPPDAELLSVSGDCTVEVEELPDGTTRVTVRYREGQTFITITYRYRDDVVTETFSV
part2_trp249_1p9m_22_model_6_1	DTLARQEAVEAFLRARGLEPQQIELRLEPDGSWVAVAYVDAGPGRLLRVTVREVPGGGWEVVAEELLELP
part2_trp249_1p9m_16_model_7_1	PCVLPPPFTLTVLFPGDAEPRTWLCTPVRCFTEDGVPLAEYACRSLEDPSLTGKAVLRLEGDEWVVVSVSYNRPEP
part2_trp249_1p9m_21_model_5_1	PGRTLTLITLVITVSGAESVEAARAAALAQLAALKPGTTITTIAGRPYRLLSVGEYTVIEDPKTKSATVEVEAVVQEL
part2_trp249_1p9m_31_model_1_1	AAELRELAALVEAAFPEWRVVGFEEDGELCVVALRELDPGRLLRALFRRREDGSVELVRVEEVPL
part2_trp249_1p9m_4_model_5_1	APRRYVITDDRELMEIELEADPSVVAITVESREAAERLAARLRAPEDVADVEVVRP
part2_trp249_1p9m_39_model_7_1	MLKKYLVARPVVEGNQPSIEIEVEAASPEEALEKAIEILREQGITGTWVISVIVDGELVERRTVTIE
part2_trp249_1p9m_35_model_7_1	MTPEQLAKADEYFAKSEEARKKAGEYLKLYDETGDPALLEKAEAYRKEADIYTAKGYKAAGDTETAKAIADKLGVTL
part2_trp249_1p9m_9_model_8_1	GMKRMRETITVAPEKVPEAVALLEELKARLPEARIEFAVDGNTITVEVEFPPEALPRLEEALAEFYARIRPLSRREYES
part2_trp249_1p9m_25_model_6_1	REEEFEALVAEHWPEIEAVYAEAARTLSREETAAAVRAAIEALLPPEVAAYAIAHYAPKLE
part2_trp249_1p9m_19_model_1_1	MLKFLGLSITPAPQTEEEYEKVEKLQKEVEEKHKELSKKLDKETTLKELKKYAEKAKELKLPKRINIHVEEGE
part2_trp249_1p9m_37_model_7_1	MEVVERTLEIDAAAPEEEAVARVLAALPALPEALARAAREVKANPGSEVTVGVETPERRELLTVVSR
part2_trp249_1p9m_17_model_6_1	RLDELVASYAAHLAELGLAPEEVAAAVARIRAAAAAARAANASLPDPRLRERAVLDAASAEIERIGRERAA
part2_trp249_1p9m_32_model_9_1	SSKVLRYRRVGDADIDELIATVKEQLAAEGASDIRLLERSPDGKRVLVYKDPNGEERYAIVEVVEEEP
part2_trp249_1p9m_22_model_8_1	SAALRARGLAALRAEAERLGLDPAAVRLVAVLRVGDRVVALGELPPEAYPHADPDDSRVRFEAEVPA
part2_trp249_1p9m_31_model_0_1	IPANVQVGLPDPSLLPLALKAVEVARKVVGPDATIGVTYNPETKTFSVHAQAPGQPPRYLIVAEDGTVLEEGEVPP
part2_trp249_1p9m_22_model_0_1	SVKVVTSQCTLPDGTVVTVTVSCTGPGEYTWEVSFYDADDPSNRTPPCTVTCTCSPDDP
part2_trp249_1p9m_30_model_9_1	APSAEQLAALAAEARAALLAQAAELGWTPSDLELEVRPVPEGVLAARATATRPGQPPVRLRAEALLRL
part2_trp249_1p9m_16_model_4_1	TETLTVNLHPGSGDTVSNISTEMVTPDGSVVRVSFDLHLGDGTVVRRSFDVQGVNGQPITVTVSREGDLVKITVTPTS
part2_trp249_1p9m_36_model_1_1	DVGERLRRAALEEGLSLEELRALAARLEAEHPDVRVRIDEESRTAEARLP
part2_trp249_1p9m_13_model_0_1	VLEVAAYSVSADGKTVVVNINQTAAALGISVEEAVARLTAAPSTVIETAPDGTVYGRVPAAAFAA
part2_trp249_1p9m_29_model_2_1	GRWELLSERFHRLPRRPTVTSTYRYVDPKNNRNLLRVVTRNAVTGEVLSDRCVEVP
part2_trp249_1p9m_36_model_0_1	SGLVVAIATAADGTRLTVGPFPTPEEAAAAAREEAARRGLTVESVEVVPVAEWR
part2_trp249_1p9m_36_model_3_1	LVVPPTAGAVFPTPEEARAAAAEAAAFYRALPDCVSVRIEESTSSSGYPVFTVVTTFRDGRVLRTVLEALDT
part2_trp249_1p9m_7_model_8_1	RVVVEREVIEFEPIPIEEARPIAERRVEELERRLREEYPDAEIEVRASFIEEDGLVKELIVEAVVRVP

part2_trp249_1p9m_3_model_5_1	RELRRTVTAPSPEEAIARACDALEAEAAAQGYEADVYVLRGKNSALLEALLRRGRERERTVHLRLVVE
part2_trp249_1p9m_20_model_1_1	MEARLEEGEAEVVVRVPASVPVREAVAAAAAARALHPADPAAPPVLLSLEVRPDEIVLTFAVPSTLLRALRERR
part2_trp249_1p9m_24_model_2_1	SGEVLEEVKGTEPLTEEELAARQAKLEAETGKRWVASEEPGTLLMVDE
part2_trp249_1p9m_24_model_4_1	EREALEKVAKIIREACEGSSSLKTRELISVAHEFISRNPFASEEEIRKIIIEAAKKYGASEEEKERALEVALKMREVYREY
part2_trp249_1p9m_12_model_3_1	TTFEAEIMVDGEPQPVTVTVDSEGVATWTVVDPVTGEPLTTTRSLEALELLL
part2_trp249_1p9m_7_model_0_1	PANTDEAALRAAAQAAYAAAAAAGLTPEEAGRVAAGGAREAGATRAELLAPLRVRLVRLADGTVREYVLELEAP
part2_trp249_1p9m_1_model_5_1	ARAEQLLAQARAAIAAAAREAAAAAGRLEEARALARLAIGLAESALEVDPSNADAKAALAEAEALLAELE
part2_trp249_1p9m_4_model_9_1	AVPDGSRPEVVDVEVEVTEDGIVVRAVGADGEVLDEARIPDNSQEARDALLRAVAE
part2_trp249_1p9m_30_model_7_1	AGYVVDAYVIETEEGLAVRIVLRDADGNPLAERVIPVERRPSEEELEALGRAVLASL
part2_trp249_1p9m_8_model_9_1	PVVVTRTEVLANGDVLRLRETYVDGVLVAREVLELEVRRE
part2_trp249_1p9m_1_model_0_1	DEVVRVAEEAFRAALAAGLSLEEARTVAGEAASRYVASRVDLSTLSDEEIRRLTEEVERATERAVERAL
part2_trp249_1p9m_9_model_1_1	SAALAAAEALLAAVVGPARPAAPPPPAEAAAAAAYAAAAAAYAAAAAILAAAGDTATAARYAASAAAHAAAAARAAAE
part2_trp249_1p9m_16_model_9_1	ATAVVTVDSRPRTVDIECVEDPGIRLRATYPPGRHERTFEVPPGSVIRFLDPANAEGVTVTVTE
part2_trp249_1p9m_1_model_3_1	DAVAFLERHFSPEDVAAAVELVELYKAGRKEEARALAAELSVEQLSAATALALAEGLRELAEEAAMARAKL
part2_trp249_1p9m_39_model_9_1	KEEEIKRAAILAACEEFGAEIVRVNGKLTVKGTEEQKKIEKVVELALELGGDVEKALKLA
part2_trp249_1p9m_18_model_8_1	RAAAEAAFAAAHGPEVLARAREAGAPAAELAALAAIAAGDVARVVQLLSAALNRHSSGEAAAAAARELCDALYAHLR
part2_trp249_1p9m_19_model_8_1	SPYPVVEVDALSEALTEEEVVIDTRVVRTARGFVIDYTVLLPDGTLRTYRLRV
part2_trp249_1p9m_29_model_7_1	PVTRTVTYVAEKEEYLSYEPTPGFYVPPEEAAPRSNRLKAGETVTVVELTLVDVAA
part2_trp249_1p9m_8_model_6_1	AAAVRARLAALLEARLAAAPSREEAHAILVEAAREAGIALESDDYGAVRRALSSHAEQLLA
part2_trp249_1p9m_19_model_9_1	APSAEQLAALAAEARAALLAQAAELGWTPSDLELEVRPVPEGVLAIAARATATRPGQPPVRLRAEALLRL
part2_trp249_1p9m_11_model_9_1	REELLERLVEELREEVLRLSPDGTLTPEQVERLVRRATEEARAAGLGEEAERVAGEAVTRAIEEARREIRERLEAAE
part2_trp249_1p9m_6_model_4_1	PTRVVRVRVPAGADLEAARAAALAANPGLRLVAERPTEGRYTLLVLES
part2_trp249_1p9m_38_model_8_1	TYKVGDTVITIEVYNRGSTPIDSVKLYLGDELIAEKTNIPPNSTVTITFKADKPGTYTLVVTQGSTVNTRTFEVVP
part2_trp249_1p9m_22_model_1_1	RVVFVEATPEEIAALEPELRLAREEGAGTVTVTPEADPAGRAEVEALRERARAE

Table 3. Design of High-Affinity Binding Peptides Targeting the Trp296 Site of IL-6Ra.

ID	Sequence
part2_trp296_1p9m_38_model_5_1	ARHTLPLTVPEGETDVYVDGKLVATVPAGEQEVTVVEVPPGTHRVSLRNRKTGESRSSDVVFP
part2_trp296_1p9m_25_model_8_1	GAAARQAAALDAAAEAAEAAGDLAAAAAYAAAAALLAAHAATHPNGANLAAALAAEAAARAAALAAALAA
part2_trp296_1p9m_29_model_0_1	GFRAAVPLRVGDPVEIRLDDGTVVPGRVDRVEGDVVSVAPLEPLTPGVHRVTVRFLTEPPQTSEPRLVEFPFPRVP
part2_trp296_1p9m_29_model_5_1	APADVPLAELEAEVAAHQAAVAATPGVDRRASGETFGRALAERLGIDPEAAAGRLAGELLAAGASPAEIRALLAL
part2_trp296_1p9m_23_model_3_1	RARVVAALAPLVSPEVAREVADLLAAGLSPEEAVARVAAAHGLDPATAEAVLAAVRAVLAAAPPLPPPP
part2_trp296_1p9m_14_model_6_1	GTVVDRDALLAEVAARLGVPATVAADVDELEAAGRIHIEPLPGSDLVSVEGGVEALTAAVRAALAA
part2_trp296_1p9m_19_model_2_1	PEVVDGGEVDAlPEVPPGYRREESLDPETGQLLVTLRPEDGEAPVVYYRVPAEA
part2_trp296_1p9m_38_model_3_1	TVPAELVAAARAAGLSDDLAALLEATHPNIRVIRLPDGR LAVAGATPEEAAAAEAAVRAAI
part2_trp296_1p9m_31_model_8_1	TVEAINRDYTVTAAGVALTVTARLSPRTLQVTAATALAVGTEVTFNTKANGATVATGTVTAVSGTTATVEVNDAVLTT
part2_trp296_1p9m_20_model_5_1	TAALAAVAADPALAAEVAAALGLASTDPAEIAARILAGGAVA
part2_trp296_1p9m_9_model_2_1	GQVIHYTDADGRPRVDVQVTVDGVTLRVTLVDGRVVRVVAETGEVLL
part2_trp296_1p9m_25_model_0_1	AAAAATRAVAAGDAAALTAVINAVIAAVTDPATAPLQAAARAAAAAGDLAAALAALAVLRAAGRLTAS
part2_trp296_1p9m_8_model_5_1	GGAHDLAELEDGQVLSESDAIPDASVEVEREVDGRLARVRVTRFRSEGNRVVAHLEVLSV
part2_trp296_1p9m_1_model_1_1	GMRVHVEGDTVIARDAAGGPPVVVPLAPGESVTITSPSGETVTVTRVA
part2_trp296_1p9m_8_model_3_1	GGYGRLRDLETGELLAERLARVDLADATRPVLLTGVDADGRVLAEAVAPATAREAEVPLA
part2_trp296_1p9m_24_model_2_1	SRHTLTFEDELPLGRYRIRVSRVLEDGSEVPVAEAEGVATSPHTVEYEVVSGEELIPGVPVAGTRIEVTPLP
part2_trp296_1p9m_36_model_1_1	GEALARATLAAARAALAAGRYEEAIALAEVIRSGLLREEAAAVVAEARAAL
part2_trp296_1p9m_34_model_3_1	GLGRRVRAAAAAAYAARYNPSPAE LRAFADALAAALRAAAAAA
part2_trp296_1p9m_11_model_9_1	MARVRATCADGTVVWGEVVPLRYPGEEVEIEVELPCGEVAAEELEV
part2_trp296_1p9m_18_model_1_1	MITVTVPEGAKVEVDATESATSVTITLTNKS DKPVTVTVTVDVVGSRSRTEFTVT LAAKASKTITVT LPA
part2_trp296_1p9m_25_model_2_1	MVRLPPLGELSPEQREALAALRAAIAASGRPM SGAEARALALAAGWP EEVANAAIAALAAEGVLDPAPE
part2_trp296_1p9m_10_model_1_1	RRVTIVGGGAAGRARALALAARGEEVAVVSATGTAAIRAAAAAAGVADRVTVVTP
part2_trp296_1p9m_35_model_8_1	NEAEAAALRALLAGRAAALRAELLARFPRLTREELDAIAAELGVR LLE
part2_trp296_1p9m_4_model_9_1	TREEIERLLREARRLLAEGRP EEAARAEEALALARAAGLP ELAAEAEALLAEARAAL
part2_trp296_1p9m_34_model_5_1	GVMVEARVLVDGEELTLRGTYEEVEARVAALGAESA EFRRLP

part2_trp296_1p9m_1_model_7_1	GANRTVTADVTEGATTTVSNVVVTATNPDGTATETGPAAEAIARAELA
part2_trp296_1p9m_19_model_1_1	GRTDPLATLTLTPGATNTITTTTTATPLPDGSTAFTTTVAADGGAVTATVTDTV
part2_trp296_1p9m_21_model_5_1	SQNALAASVPRAELEAIVGPAAAAARIVATGRATPADLARLRRAA
part2_trp296_1p9m_27_model_9_1	GLIATVTVAAGATQTITYTAQAGDVLVVTTDSATGTVLSTETITVTPEQAGEQTLTITNTSATTTVTVTLGNMTVGR
part2_trp296_1p9m_26_model_8_1	MREVVEVRPIPLPLVETVTSAVDEATGERTVLRVEVRPLV
part2_trp296_1p9m_17_model_6_1	ASITAPADGVLEVRFADGRVERRRVRAGETVTFPDREIVDWRVVP
part2_trp296_1p9m_15_model_6_1	VERLERTLEGNADLDALLAAVEAEARARGATGLRVLDEACADGTCSAVAEATL
part2_trp296_1p9m_9_model_8_1	GALRAAIRALLAEDPNPTEAQLRALAERFGVSVEEVRAELARVRAELA
part2_trp296_1p9m_24_model_4_1	GLAALRLALAAPSGRLTAAEVAAALGVPRSTASRRLRRLVAAGVLAYDPETRTYSVDRERAAARLLAGEPLPEA
part2_trp296_1p9m_10_model_0_1	SAYPVIEVPAGASEEEVRRLLVAERLGVPVEDVEVERVGDRWLVRARGELYVARER
part2_trp296_1p9m_8_model_8_1	LRRAGAGVAERLRRAGRRGRRLVRELGLTLADLESGRASREQLAAAAAALGITLAEAAAGL
part2_trp296_1p9m_27_model_2_1	AGIEPDDTADAALDRARARAVAAGADPATVDAAAAAVRAELARIPAVTLEDLHAAVARFEDLIAAAEGEAVARAAATG
part2_trp296_1p9m_39_model_0_1	TSPPVRVGDTLLSRRRTADRVLLIATDLATGEERVLAEERPVR
part2_trp296_1p9m_10_model_2_1	AEPTPAAAAAKATAAALKGKVTSVTLTTSRMTPAQAAALAEALADAGVTVTVTVT
part2_trp296_1p9m_3_model_0_1	AALAALRAACARLAAVDPAAAAARLLAEGAALDAAATAALAADAAGIVALAEADPELARLRLLAGARAA
part2_trp296_1p9m_5_model_9_1	GELVSEETDPETGVTTTAVRVAGSPVLTVTTGDTTTEVEV
part2_trp296_1p9m_4_model_3_1	AVLERVAAAEARHLAAAHPTALRVRLRALDEDTIEVVAAAEAVPGDVAPAVEAELLLP
part2_trp296_1p9m_5_model_3_1	GSPEARALDAAAEARAAGDAAAAAAAAAAAAARRAAEEA
part2_trp296_1p9m_10_model_8_1	LVAGLQALVDAARALGLRATGRVTGPDTFEFVLEGATPEERAAILAAAGVTDLVS
part2_trp296_1p9m_20_model_1_1	SEAEALAAIVAILEARGLSREEAERLARELLAAGVSAEEVAARA
part2_trp296_1p9m_8_model_0_1	AVGLPATVEAGGRRVPATLVREGDPTVWALAAALDGPGLVLVLPDGARFPVLCAGRTATAA
part2_trp296_1p9m_11_model_3_1	PGLSLREAAARLSARTGVPYAAALAYLRRLEAEDPAAARALRAELA
part2_trp296_1p9m_23_model_5_1	ADPAALAAAVAAECAALVAAGLLTPEQAAAAAAAAATAAALAAAAAGATGEEALATARAVIAAAVAAAAA
part2_trp296_1p9m_32_model_6_1	RPRAPPGETVLSTECVAPNVVRVTYFDHATGREYDRLVEAEC
part2_trp296_1p9m_12_model_8_1	PVAAARAAVRAAFADKEVKVLGVRPLSPTEAVVEAAVKDGDAYSIVTAAVRDGAVTVVASEPSDVSLEF

part2_trp296_1p9m_1_model_5_1	ERVVTTRRPDGDFFVETTLVLVPGPAPDGREVVAREETRRRAPPGSSG
part2_trp296_1p9m_16_model_4_1	GRRLVTVERRTTVREFIREECAALGLDFADVTVEGVAPEDLDKPLEAGTVLEVEVDEERTIRVRVLE
part2_trp296_1p9m_15_model_7_1	GRVTCRALTPDATVAPAAEVAADPAAVLTQLSPTVAEVTNPRLQTVCTVTSVP
part2_trp296_1p9m_39_model_1_1	APRAQDLIAEGHSLAEVARLLGVSRATVSRLVARCPEARAAR
part2_trp296_1p9m_28_model_9_1	SVSIVRRGATPEEAFAAAARAEAAAALGLTIRALVVEAPDADGLWTARALLGP
part2_trp296_1p9m_36_model_2_1	GRAYREALALRDRLAEGASLEEAREIRRRFGQRVVDIENGTEVDGLALD
part2_trp296_1p9m_36_model_3_1	PRWYELLDNGQTRPVPAEELARWTPLEADAPPGVLRVRAPDGTVRWLRRAP
part2_trp296_1p9m_23_model_0_1	ETTITIRLTSGTAAAGDVTVIAGHEATVASVDGDTITVVVPAEVAATLKVGDAATVPNPDGTATGTVVA
part2_trp296_1p9m_4_model_7_1	AAYEVALAAARATLAADPAATDAEIRAAARAACAAAGLTVTDAQIERAVAEARAAAAA
part2_trp296_1p9m_38_model_0_1	AAEAAARAALGPEALALVLAFAEACPEEVRGCDWDRLEALLAERLGPEAAARAVAAIRAAL
part2_trp296_1p9m_22_model_5_1	PGDLVDRGRAALAAGDKDAALAWLDRAIEAGANSPECVAAARALRDEARATPVIP
part2_trp296_1p9m_14_model_8_1	DPDLWVVRVLVPTLAEAEAVAAELRAAGFTDVQIVPVPDGYLVSAVTRDEAAARAFAARYGGTVTRA
part2_trp296_1p9m_18_model_9_1	SKVGHGLRPATPATVTAAAGATVTVTGEVCVEGLTASTTVTAAAGTQTATVTLTCPACGATLTASVTIVGTA
part2_trp296_1p9m_9_model_5_1	SLIAAGRVLVEPLPDGFVVTVAAPDGTTRLVVRVDADGRVVSVELVPAP
part2_trp296_1p9m_39_model_8_1	GRVTFVRPVAPDRWLVRDENGTLGVVPRADLRPLAPGVYELR
part2_trp296_1p9m_19_model_3_1	LDRELLTPEEAERMARIRRLAADPSLSVREIAARCGVSPATASAARRLIAEAE
part2_trp296_1p9m_23_model_6_1	PEILSVEVEQLDEDYAVTVTSNPGITPEEAEEAALAALAAQGIDGVVVAEEVRQTPDAVVSTLTVRRK
part2_trp296_1p9m_12_model_1_1	AGSPEAIAQLAAEVAARDPALAAEVAPLLDRLREVGPDRALLEIRAIAEAHGCELCVAAMEAALAVL
part2_trp296_1p9m_12_model_3_1	NAERLAAWLRAQGEPAADAVLALAAALPPGALTRERVLAAARAAGASPAIAALADALADVTPEQLS
part2_trp296_1p9m_33_model_1_1	GEPEILALAAAGASQAEIAARMGTTPATVSRVLARAAAAA
part2_trp296_1p9m_0_model_6_1	RLVVRGPGRWELVDEREVVEDGVPYRVLTWRNVETGETLVARRA
part2_trp296_1p9m_7_model_3_1	GPRTCPVCGAPMEPVSLDREALTATYRCPVCGYETSVPLEAPP
part2_trp296_1p9m_23_model_7_1	SARAATRAAILALLRAAGDRWLSVREVAAACGVSVATASRALRELAAEGRVTLRRDEAGVLLVRASGAT
part2_trp296_1p9m_5_model_2_1	TRSEPVVVVTVDGRERVVPAGEGVELPAGRRVEIRPLEE
part2_trp296_1p9m_30_model_6_1	ARILSVERDPATRTMVLTAADGGQPPERLPIELSPEEVRALLLAAGRAV

part2_trp296_1p9m_35_model_0_1	TVGELVTLPATGELVPALEGEPNIVLREGPEPRWYRVNLTTGELTPAP
part2_trp296_1p9m_2_model_3_1	ARALARAAAAAGVPREEIVRELVRLGVPPAEAEIAAAEAYAAAAA
part2_trp296_1p9m_29_model_4_1	VGLAIALALAGDLEAARALLDRRLAAAAEAGEIPAADLRAARDELLAAAAEAAARGDAEAAAALLADAAARLDAILA
part2_trp296_1p9m_3_model_4_1	PSALPRVRARALEVARLVAAGELSREEAVALAGELIEWVTRETGTAGVLATPEEVAEDLLALGAALL
part2_trp296_1p9m_7_model_7_1	MREVVSSRVLARDGDTVVELTCADGSRVLVTVRDGRVVASRRV
part2_trp296_1p9m_32_model_5_1	GEARRAALAPRAAALEAALRAAGVDPATARALADAWLEAAAA
part2_trp296_1p9m_39_model_9_1	MRVDAVVPAADLGPDVEEVLAETPEAVVVRRLGLRVAADVTA
part2_trp296_1p9m_12_model_9_1	AGGAAAAPALARLRATLAPRLSPEAVAAALAAAADAIAAALAALPATASPAERAAAAAAVADATLAAA
part2_trp296_1p9m_1_model_0_1	GVRTTAPVGTSTREAAAAALAAALRAAGVPAAVVARTRPDGTVEYVWA
part2_trp296_1p9m_0_model_4_1	ACERVAAAVAALGGSPAERRARLEPLAAELGAPPAALDAAVAAA
part2_trp296_1p9m_9_model_1_1	GLTLAERLLGRRLTPEELAALAEALLGDATTEEAIRAAVERWLAEKAAA
part2_trp296_1p9m_8_model_4_1	MVVPLSALPALLAAGYVVVERGTVTVGGRPVEYLVLRGNDTVLCAGDAATAAALVATL
part2_trp296_1p9m_29_model_3_1	LPLSDAALAELAALAAAGAPAEELAAAAALAAARAAGVRDAEVAEAGAAALAAAIAADPSLTTPAAVAAAAAAVAA
part2_trp296_1p9m_39_model_6_1	ASLTVAAGTTVPLLPGESFTPQTPGTYTVTLSSGGTVTVTVT
part2_trp296_1p9m_21_model_6_1	TRRVLDLSGATAEEAAAAARAEAAARLRAEGVLVELVGLERVLP
part2_trp296_1p9m_16_model_0_1	ATVTTTTDGDVRTDTITTTVADAAAAIAAKAALAAAGYTVTAVTTDVAPNATPDTLATVTITATAAS
part2_trp296_1p9m_27_model_5_1	AGEPWEREFTGRVVLELDDGRVLEGEGLVLRCTPQGDEVLYTVLRACGELAAGQVIRVRDSTYDEAARVLTLRGRLL
part2_trp296_1p9m_28_model_7_1	DQRTIVAVSPAARARLAARGIRLEPGDVLPGAELERLLREAGVDPAEVTVA
part2_trp296_1p9m_24_model_7_1	CRVVGAPDLVVTAANSTVNVTLAFLDPNGNPIPGATKTVSLTVAATTTTVTISVTDPTGQTQTHNVSITLVPV
part2_trp296_1p9m_9_model_7_1	GLYSPEELAAAAAAVAAHGGDKRAAAAAALGVSPATVSDRLAAAAARAA
part2_trp296_1p9m_11_model_0_1	MQTVTIAPGQTTTITALGPGKVNVSDBGTTTTTVTLPTAGTYTITAP
part2_trp296_1p9m_0_model_3_1	EREAVLDEAVALAEAGRLEEATALLRAHGIPVEEFVAYIAAREA
part2_trp296_1p9m_16_model_9_1	SVVSPTTVTANANNTATLDLTGARGAGEAPQRITVTLTAAAAGNYTVSYDAVIAPGGVTVSNLTVTAA
part2_trp296_1p9m_25_model_6_1	MEALAREIREECVRILVEEYGLPPEEAERIAERGVAAGLAHAEAQWAATGRRDAAAAARAVALATARELA
part2_trp296_1p9m_24_model_9_1	GGTREGAIRDLGAAAAARLGLPPISLAEAEALAAPLAADVLDALVAARRAGASPEEVAAAHDALVARLAAALRA

part2_trp296_1p9m_28_model_8_1	ELEATIAALKADPRVARLEVRVHPDDLDGRSIVITATLKDGRVEERGFAHQ
part2_trp296_1p9m_24_model_0_1	GTEEGLLNAGLSPTEARVAATLRAAGGEPVSVAEIAARTGISRATASRALRRLVALGAERVPGGGYRATPVA
part2_trp296_1p9m_13_model_0_1	MPQALRDELLARLRAAGYSPAIAAAALAAAGEASAALKGAAGVDPAAARDAVLAAGLAAAAAAEAAIP
part2_trp296_1p9m_24_model_1_1	RAAARAAVERAVAALPPGGVLTPEAVAAAANGISRATAARLLAAFEAASGRAPVVTEVSSPELGPSVRALLSGP
part2_trp296_1p9m_33_model_5_1	GLAAAAAPGDVITGDDTTTLTVSTPVGTRTVTRSLPLPSG
part2_trp296_1p9m_18_model_2_1	GPTRVVRIPVRPTDSLAALEARIAAALPGISPPEARALAEERAKALLDAAGAPGTLVITVRPGATEPELELVP
part2_trp296_1p9m_25_model_1_1	RVEALEETEEERLVRISAVAENVAAARAKVDAQVAALAAEGWEVVAEERWEYFDRVGVAGALLLRRRP
part2_trp296_1p9m_4_model_2_1	TSTATTRTIPLTGDRTLTLAVDAPLAEALAAIEAAAAEVPIDLEEVKAEVAAAIAAR
part2_trp296_1p9m_14_model_2_1	MLPEVARDLVLERGEVRPELRARLSPAARAFAEELGARIRAGEMSLAELAEALGVSSERTVRRWAGLPY
part2_trp296_1p9m_2_model_9_1	MFERLTPEQIRAEVEAELEALGIDPARARITVDLERGRVEVEVEG
part2_trp296_1p9m_6_model_5_1	PDRAQVSDVAVDAAEANTVTLTVTSAPVPGTYTVTVVDPVTGAVVATADVTVDGTQTVTIPLAP
part2_trp296_1p9m_33_model_8_1	GAARLRRLLAAGTPREEVVAVACREWGVSRAEAEAAVAAL
part2_trp296_1p9m_0_model_1_1	GAERATYAAVLAAALAAAGLSEAEARAVAEPAAAAVGAALDAATA
part2_trp296_1p9m_19_model_0_1	APTDGILPGWTVVSDTQLSENTRRVVAARPGETAVVTLTTAPGSNTPTVVSREV
part2_trp296_1p9m_21_model_8_1	LTAAEVEALCAAYGLPLPAPDTVLTEEEAAEWRARLAALRAAE
part2_trp296_1p9m_3_model_7_1	GVLYVITGRDADGTLTEARVRAALAAAGLTPLRLEV RDGVVTVESRDSRVGLVLAELGVVDITVEAR
part2_trp296_1p9m_0_model_5_1	AAEAQAAASEAAAAAARAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAA
part2_trp296_1p9m_26_model_3_1	IAALSPAGRAHLGAPPAVIVAAVTADCAAAPVHPLCAAAGE
part2_trp296_1p9m_36_model_4_1	CSNNCTVDHEPKAPGDIVTVTCTSANDNTVCTWKARKQADGTLEIPEPPVCE
part2_trp296_1p9m_11_model_2_1	VEELARRLGEAALRAALVRELTEVEGLPAAEAEARAAAIVAAWRAA
part2_trp296_1p9m_25_model_9_1	LGELAVELLERLLAEARAALAAGADPAALLAAAAARLTAE LAAHAPPDVVAQVAVATAELRRELAALT
part2_trp296_1p9m_11_model_7_1	AVEVERLPDGTWRARNSDHPREWTPAHPTEAAAIAAARAAGSVAAT
part2_trp296_1p9m_3_model_9_1	MTILEAEISPDGLTLRIVARAEGDAEAAAAALVAELRAARPHRRVEVVELEVDPETGLIRLVLRFTP
part2_trp296_1p9m_31_model_1_1	PQAEIDALLEATTAALAAALDDAAAAVAAAVDAAVAALAEHLLASGMTEAEARAQAEAAARPEIARARAVAAALRAAA
part2_trp296_1p9m_14_model_9_1	MLPPLADPTLGTEAAAAALLADPAASEAALRAAALAAYAAAGLSPAEAEAAWETHDRPVLEAALAAA

part2_trp296_1p9m_0_model_7_1	PVWLSLAEAAARLGVSPATVRRRCRAAGLSTRRLSEREVAALAA
part2_trp296_1p9m_25_model_7_1	VKVTLTGLAKDTTAADLLAELAAVPGVMRELAERLVTEHGVDLATTTVAEFAALMKEKLGVDLTVEYLP
part2_trp296_1p9m_1_model_8_1	AVADLLGRPVTWEERVALIRAAAAALGVTPSDEEIAAAAMEAARRAAAA
part2_trp296_1p9m_23_model_2_1	AGVALLADPRARAAVALLADAPLSVAELAARLGLSRATVRRYVRAGVLEERDGLVSVPEAVLRRLAELA
part2_trp296_1p9m_11_model_6_1	AEELRALVERLLAAGASEEEVEAAVRARFGDTEEARRLLRELLAAA
part2_trp296_1p9m_19_model_6_1	GDLAALAAALGLPATPEVVARVARALAALGRRVGRRPGGLTPEERA AVRAAVAA
part2_trp296_1p9m_22_model_7_1	LCPAVGSTVTLTVNGVTLPCTISSVVVENGKCRVITYTCTAPDGTTLTVTEELPVL
part2_trp296_1p9m_38_model_9_1	VSSLTIPAPAPGETVTVTVVNTNNVAITVTITISVPGAQPVTVSITVAAGQTATTDVTVAPA
part2_trp296_1p9m_12_model_5_1	GAAALAAKVEPLIPGVRDVLARLPPERARAVAREILALVEAGRVDEARALAA RVAAEVAAAEAAAAAA
part2_trp296_1p9m_18_model_0_1	AAAAAAALVAEARARGLDALAAAAAAVVAEAAA RSPAAAAAAVALIPLAIAHRSGPETAAAAALRAAAAAAEAA
part2_trp296_1p9m_9_model_9_1	LNAAGIARERAHLLAAEGRGHLLAGLSDAEAAAVILALARAEGYDLTLT
part2_trp296_1p9m_6_model_3_1	AVSARVAALVAEKEAQGYTDVTADSTGSVTNNGHIKATITVTYKDPSLPEPVTEVIDVDETLV
part2_trp296_1p9m_36_model_9_1	PEVVDNSWLPAELRRILAEARALRAQGLSLAEIAARLG VAPATVRRALAAA
part2_trp296_1p9m_15_model_8_1	AGVELVRRRTGLSREAIEAALRALGVPAEEAARAAAGGEPSPRALAALARILA
part2_trp296_1p9m_27_model_0_1	PPLAPEVVVEGTVLDIRPTVVDGEPVLALTVRLEDGEELTAPVAPEAAAAAAALRVGDRVRVTIVEVDGTWVVVAVER
part2_trp296_1p9m_0_model_8_1	MSAEELLRLARERGLSAAELGRLAGNPVATVARTLRRAAARAA
part2_trp296_1p9m_39_model_4_1	GAALSGTSTTVDAETKTETTTSGVYDPETGTVATTTTTTRDAD
part2_trp296_1p9m_16_model_2_1	VTVLEARVEGDLLTILAESQSTSDDLEAIGRAAAEKAGFPDAEVLEAELLSQDGTTRLVRVLARIPAP
part2_trp296_1p9m_29_model_9_1	GLTETVTAKEGETVTLTCGGLGKVTWYKNGKEIETVEVPEELLVTLTIENVTKEDSGTYTCKSSLGTDTTNLTVE
part2_trp296_1p9m_24_model_5_1	SLAAAADAAAAAGDGRGAARHYLRERGEELAAQGDYAAAIEAYRRAAAAAADPALAAEARAALAEAEALAEA
part2_trp296_1p9m_28_model_5_1	SRPAEELVRAALPADAEILAEELDPEARTVRVTYRTPEGEVRTLT VSYADG
part2_trp296_1p9m_22_model_8_1	RTVDEIVVEVGRAHARASPEERRAAVRALCAALGLSPEETAALAAMEEAIAARD
part2_trp296_1p9m_6_model_8_1	AEEFTARVAAQFPPEERARVRRIMDAVRASDRPLTAAELAARTGEPRSFVSRVIRVAAAVAA
part2_trp296_1p9m_36_model_5_1	GYEAEARAILAAQDTTGVPAAEELEARLKEIAAALRAAERAADPSNKS ELIVE
part2_trp296_1p9m_28_model_0_1	AAAAATAAQQAELAALLAAYPELEPLVVALAQAGASPAAILAAVRAAAAAAL

part2_trp296_1p9m_12_model_6_1	AEEAAIRAEAAALRAEAAALLAEGRATEAQVAEIRAAADAAAAAAAAAGDVDAARAALRRARALLEAAR
part2_trp296_1p9m_17_model_8_1	GVDAATLAAGVAAVRAEGLTAEELARRFGLSLADARAI VAYARAA
part2_trp296_1p9m_35_model_1_1	CCLNPNARRTERLETTAAGVPVLVTTCEDPTCTAGLCRAVERTPLPLL
part2_trp296_1p9m_27_model_6_1	GAEMLALARAGRLAEAEARFAGTLAGEIAALARAAAAGDEAAAAAHRARVEAAKAGLTPEQRAYVDAVTEAALAAALA
part2_trp296_1p9m_25_model_3_1	AQAAVDAALAAAAAAAAASAAELIAAIRAAEPALLAAIRAAEADPAHAALLVAHARAAVDAAVAAIAAAA
part2_trp296_1p9m_5_model_5_1	AEAAAAAGWTLVSATSVAGTDAVVVTWRDADGRTRTSVV
part2_trp296_1p9m_14_model_0_1	MAPSVEAVPATVTIAAGATRTITYVVRAGSEPVTVTITVTTAGNTVTKTVTLGARETRTVTVVEVTNP
part2_trp296_1p9m_4_model_0_1	GAEAEVLRRLRREGSITLEELRAIAAEHGVPFEEVLRSLVESGLAVVDEEAGR VYLAE
part2_trp296_1p9m_29_model_2_1	GVTRVVTAEERDLAEELRALAAALAREAFPDAAAVVVTGAEAVPAGATIRIEAGTPAASSDSPEPPVAHVVRTRL
part2_trp296_1p9m_38_model_8_1	ETLHNVTGTVVESVPAGDTCIVIEAREGDELLVAARVPGVCHPVGSTVTVPEGRLAERP
part2_trp296_1p9m_5_model_6_1	PILLEERLVDGRWEAVVLVGGERRTLTADSRAELEAKLAA
part2_trp296_1p9m_21_model_4_1	RVRVEVRAEPLPDGTVRVTTTVRDEDDPEGLRVEERVVPAEEL
part2_trp296_1p9m_13_model_7_1	RRLFAALLAAAAAGELRPGDVLPSQAELARRFGLSRPTVSRELRRLEAGGYVRREGRRWVVT PALVA
part2_trp296_1p9m_34_model_8_1	ARYEERRVLEETETTTVVVEERTVSTVNGEPVVESRVVVEERP
part2_trp296_1p9m_3_model_8_1	RRRRRLVLEWVDAEGRRRSWEAVVELPERITPEELEDIVAE LASRELGISREAA RNDPNLRVRVEEL
part2_trp296_1p9m_9_model_3_1	TTVTFTPDGDAVLETVTFDGGRTTETRRLTDYVREVDGLRVYADSERV
part2_trp296_1p9m_38_model_5_1	ARHTLPLTVPEGETDVYVDGKLVATVPAGEQEVTVVEPPGTHRVSLRNRKTGESRSSDVVFP
part2_trp296_1p9m_17_model_2_1	AEELARALLRADPSLSEAEVVARLEALGVPRALARRAVRVRREL
part2_trp296_1p9m_13_model_4_1	GRGLRLAELLAGRPLTAAEVAAALGVS RATASRL LREAEALGLVEREGRRWRARPEAVAAARRRLA
part2_trp296_1p9m_32_model_0_1	LATLRGVCPCGTTELVTADPSLRSTRCPVCGAEVPLTRVS
part2_trp296_1p9m_19_model_8_1	GPAGRALRAQIAALEAQLAREADPAAAAALAAQVAALRAQLAALRSPTFHAKVP
part2_trp296_1p9m_19_model_7_1	LPAPSPEAEARARAAAARELGLRRLGPGAMARIQRRTGLSGAEALALRRRTGGRL
part2_trp296_1p9m_26_model_9_1	GEGVLSLERLPDGSWRLTRSVDGRPVTSVFPTEEAARAAAA
part2_trp296_1p9m_37_model_3_1	GARRAARDAVRDAALAAGLPEAEAAARAAAHEEALVAASPVFAD
part2_trp296_1p9m_33_model_9_1	GLPEQVTVEERLEGDLLHTTVTETSSAGETVTTTRTERLE

part2_trp296_1p9m_33_model_6_1	GTVSGTPVTDPSGLTVVEDPATKTLTYRAPDGTVVATVTT
part2_trp296_1p9m_18_model_7_1	APGRAFAAVVERLRKEYGAVEAEAVDLRLDLDATRGELVIRARLADGRITLLATAVYDKRTGEVLRLEVELLEE
part2_trp296_1p9m_38_model_4_1	GLGARVLALLRAHTEPLSVAEVAALGVSPAAAAARALRALEALGLVERDPRTGRYSVSPALR
part2_trp296_1p9m_3_model_3_1	PRAAAIRALLAALEARPLTPAELAAALAAELGLSPEEVAALAAAGVLRDAAGRWWAGPPSLIPAELL
part2_trp296_1p9m_14_model_5_1	GYSVLAADSPEALALALAAVEALNPVANAHTLEDLRAAIEAAPGLPPGTRIVIARDGRPVATLVIRGA
part2_trp296_1p9m_37_model_4_1	PGAPGPLRPVGPRTVEVTVPPTLVVTHPESLEAARARALALAA
part2_trp296_1p9m_10_model_9_1	GVRLERLVLEGFADPEAGAAAARAEAEARGGTVLDIEVRPLADSWRVITALVALPA
part2_trp296_1p9m_35_model_2_1	EVVFEGRGVEERRDEDGRPCEVERRLEGDRLVLTHRCPGTTVVEEFRS
part2_trp296_1p9m_37_model_8_1	GESTETVYDPETGKTYVVTTWREGDDPEPRVEIVNLETGEVTVL
part2_trp296_1p9m_2_model_5_1	MSREELLAIVREALAAGTPRENIVRGLVEDLGVDREEAEALVAEA
part2_trp296_1p9m_37_model_7_1	AAEAAALGAEAREVEARAAAASGAEKAALRARADELVARAAALA
part2_trp296_1p9m_25_model_8_1	GAAARQAAALDAAAEAAEAAGDLAAAAAAYAAAAALLAAHAATHPNGANLAALAAEAAAARAAALALAA
part2_trp296_1p9m_6_model_2_1	MVVIEGSTVEELQAKLRELLPSEDAVILDENGQPVKVEEVKPNITYYSIRDATSSLEVFTKLV
part2_trp296_1p9m_22_model_4_1	NAEIRELIDKAAKARLAGKEEEAKKYLEKAKELAAKQGISDEEVEKEVEERAKEL
part2_trp296_1p9m_3_model_6_1	GSVTVPAGTTMAAVAAAALLAEYPDWARVEFADPDATLAAGETRTVTVTVTRDGVVETRITLVTAAG
part2_trp296_1p9m_12_model_7_1	NSLAEAPVERLIKVTTLTPTATGVATTTTTSEALATAVRNEHGELLGPSTTTTTATATTYTEYVREV
part2_trp296_1p9m_35_model_4_1	EPPPVFVGRPVGSTTRVEFTPASTERVGDITVATGADGTTYVGRLVRR
part2_trp296_1p9m_5_model_7_1	GPLDAETAERVLALLAAGLSQAEIAARLGLPRSTISRIR
part2_trp296_1p9m_27_model_8_1	APAARRTVTATAPASATTAGARAEVVDAATGEVVGSAAPVDAAGVATITADMVPGRYTVRVVDPNGAVVASTDVTLEAP
part2_trp296_1p9m_32_model_3_1	AGRVRRIPLGDNLEAGRALCAALNAVPGVRASLEDGVLTIET
part2_trp296_1p9m_20_model_0_1	SELPPVVVEEPLSEDLVRVTTTRLVTVGGEPAWLSESRRVVRVV
part2_trp296_1p9m_30_model_0_1	GLAAEVTFTLTVDGVPHTVTVRVNVNPNNDGTVTVVEVYVDGVLVATVVIP
part2_trp296_1p9m_12_model_2_1	EVRTREEAYALGRAAAAAGVTDLRVRAVGDPVEDAAAAVAAGYRAAGATSATVTVVGADGSTATVTI
part2_trp296_1p9m_16_model_3_1	GTVTITIPAGSKGKTITLSLPSTGNPTTVLSASSNCSVSVTRSATTDITIKVLVDSGTCTITLSHP
part2_trp296_1p9m_38_model_1_1	TVVIEGVEDPGALAAARLEAELAARGWRLVDARALPVPSPGQATVLEVTFERDGERLVVVLGR

part2_trp296_1p9m_6_model_6_1	GEEAELPYVMALVRREAPDASEEEERRRIAREAIARIRALYEETGRPPTDEEVEAIVREVVAE
part2_trp296_1p9m_36_model_8_1	PAPADPDAVLPPGTSATFPAVAEGDTEPRTILARNTGDAPALVVDAAVALSE
part2_trp296_1p9m_25_model_4_1	IRRDHELPEALLAAVERGARAAVAAARSPEELAAAVEALTEELYARYGDEATREEIAALVRETAAEVL
part2_trp296_1p9m_27_model_1_1	LAGAVVDAARAAAAAALGLSAAQLAAIDAVLAANPTTLDASDPATAAAQALAAVRAAAAAAAGAAAADAIAAAATAALA
part2_trp296_1p9m_3_model_2_1	SERVEVELTIAEIREGLAAAGIRCETDDIEEMVRLYL RHVV RPQAREVRILSLEQEGDRVRVAEAEAR
part2_trp296_1p9m_3_model_5_1	AADGPVTESYDAATDTLTVTYTTTRRPDTPVPEEVVDAEGRRRWRLVSREVAADPDTGLYTVTLTLRPV
part2_trp296_1p9m_25_model_5_1	GTVTLTIEFEVPGAAPDARLRVRVTVTGVDGRTATAEHEVLASDPRVDVAVPESVLPVAAVDVELTLLP
part2_trp296_1p9m_8_model_7_1	MAEQLEAALAAIEAALAAAGGDEAAVAALPADVQALVRAAGGLAAARAELRARIAAERA
part2_trp296_1p9m_33_model_7_1	GAEAVDPAVDPSVAGQPLTTVYEEAVVNGRRRRVAVTRPG
part2_trp296_1p9m_2_model_8_1	MTPEEVAIAERLLELLREVSEEEVARLIAEATGRDAAEILAVLRA
part2_trp296_1p9m_3_model_1_1	PGGLVAAAAALALGAAPAEVAAAAAAATGLAPGGEYAGDPAAAPAIAAAAGAALGKAAEVTAAVAAAV
part2_trp296_1p9m_9_model_6_1	TEEAERLAREGIRAGRPPEEIVAEMMARYGVPRERAEAVVRRIRAELL
part2_trp296_1p9m_22_model_2_1	MDNPELDDRIRDLADAGRSLREIAAELGISPATVRRRLREL RARGVPIRRLPRSL
part2_trp296_1p9m_23_model_8_1	ALGTDEDPTYDEDALAARLAARYGIPVETARAIVRAAAVAEAEARLTDPEEIRRAAGEAADAAAAAAAA
part2_trp296_1p9m_35_model_9_1	ALGPVERLRAVRDGRVVAEAVRAGRDVVPNPTFSVTFPADWEVEREIE
part2_trp296_1p9m_35_model_3_1	GALSAAEREIAAAAAAAHGVALPAIDAADPAAAAATVAAVRAQLAALA
part2_trp296_1p9m_20_model_6_1	GAAEVAALRARAAAAAAAAGDAAAAAALAAAAAAEAAAEAAAA
part2_trp296_1p9m_30_model_3_1	SPRRSYTVERSTVNGKPRYTVVETVAGVRTVVATFTDEAAALARAEALA
part2_trp296_1p9m_4_model_4_1	AYTLELVTDGLNVLRARVVGPDGRVTELDLPPGVTPAEALDLAEARLGAPIARVVTVP
part2_trp296_1p9m_14_model_7_1	VADAATAAAFLARLPAAAAAAVAAALAARPISGDEARAIAVAAGADPATVDAALAACGIAPTDLVA
part2_trp296_1p9m_15_model_2_1	MSGLSLTLPEGLAPNSSYSFVVEDPLANVRFRVTVTVGADGEVTAVETETLWE
part2_trp296_1p9m_4_model_1_1	SRPVLTLLEVLPDSDLAAVAAEAAAAALAAQPDLEVRVAHAELLTPEQLDELAALTAATAA
part2_trp296_1p9m_15_model_9_1	AGTETLTISGATPATATVTTTPDGVTATVTYPNAAPKTATGFATAEAAIAAAL
part2_trp296_1p9m_22_model_3_1	GDPAEAVVEIDGVRYRVRASDPERAAVAERYPDNPRRAIEALRAEGVAVTVEPA
part2_trp296_1p9m_8_model_2_1	EGADAATLAELRARAEEAMLAAGKSPEEILAEELERLCEERGIPLTAEQWAALEAEIRAAAA

part2_trp296_1p9m_11_model_1_1	AAEAAARAELLAIPGIGAALADRALARARAEGIPLADLLARVRAEA
part2_trp296_1p9m_2_model_0_1	APGTFVRYADLAAALGVSPATVRALFRRLAAEGRVALTPEGFVVR
part2_trp296_1p9m_6_model_1_1	GAEVLVPTEEAAAAALPLLARYPVTVVERTPEGWRFGLSAEAAEAAAADLAAAGIPARALA
part2_trp296_1p9m_29_model_7_1	ASPEEAIAEAKKLMKEAEKLAKEGKKEEAIKKLEEAEEKAKAAAESARLAGNEELAKEAEKVAKEAKKKKKEVKK
part2_trp296_1p9m_26_model_5_1	AAREAREAVLAGEDVRAVAARYFGRPDLTEEEELAARLREAL
part2_trp296_1p9m_37_model_9_1	GWLAALGPEQRALLAEYEALTGRPSTARSVAELRAELAAARAAR
part2_trp296_1p9m_37_model_2_1	GPVVVTETTADGVRTTLVNPADGVTVTVEVRADGTVTVTVTL
part2_trp296_1p9m_26_model_6_1	AGPLIRALVAAGASRDEVIAAMARYGVDEALVRAETDRALA
part2_trp296_1p9m_32_model_7_1	VLARDRRTGEWRPLAAPPGLPRSALAPGAIPGLDPEVLPPPG
part2_trp296_1p9m_20_model_2_1	RLEAAMAAAAAGLAVDARLAARIYGVSEATVRRVAARYGVPV
part2_trp296_1p9m_10_model_4_1	MRTVTLEARLEDGRWRVRATVTETSGGRTTVVADETvvvveAASEEEARA AVRARL
part2_trp296_1p9m_15_model_0_1	GFEVEVSYADLTREEAEAAAREAAALARHPGARLVSSSFDEERRVYTARFVDAA
part2_trp296_1p9m_24_model_8_1	SRADALRALADEEPDRHAAILRVLAEADSEAGVVRSQAEIAARAGVSQATVSRVVRAACERLGVPVLRSLADL
part2_trp296_1p9m_15_model_4_1	MLPAHLRAAADRVAATLPPAEAARVRAAAATLSPEAFWAALPPDVRALLAAEY
part2_trp296_1p9m_2_model_6_1	VGPPRVEVEGTRGVCTLGGLTVEVEAPDVTEEKVLAAGEAAAAAA
part2_trp296_1p9m_16_model_6_1	SVRGTAETGERAPVDVLEVLGTEGDVTDVRCVALRREGDRLVRRPAVGRLRRVHPDYVDDEVVFRE
part2_trp296_1p9m_30_model_9_1	GPPPTARVLSVDPERDTVTYEHVIGDTRIVAEYNLRDGRFLRVLSEERL
part2_trp296_1p9m_26_model_2_1	PLLALEGTADGRLVAARVRS LAEAEALAAALGLTDVVVTPL
part2_trp296_1p9m_15_model_3_1	SAREAAALAAIEAARRARAAAGEDLSLFSPADIEALARAAGLSPAEARALAAEL
part2_trp296_1p9m_5_model_8_1	GPATVERVPLPDGSVALRTRTSLQSASDPEPVTWTETTP
part2_trp296_1p9m_5_model_1_1	PAVELSVTRLSATTRAVRVRVVEVDPETGHTVTSERVVV
part2_trp296_1p9m_14_model_4_1	TATVTKTATAAPGTTVNIPVSIDTGVRVAPEDERVTVTVTQDGTTVTVTITTNRGAIGPTTISVTV
part2_trp296_1p9m_31_model_5_1	APVSITCTFTAPGAAPCTVTCTATDPSGNSTLLSTLTSAGGRVSC TLTATSPGTYTCTLSTQATAGAPQSTSKTFTVP
part2_trp296_1p9m_10_model_6_1	VERYVIPTLDEEEIKKWLEEKGYTWNEETTTLEYDEENKTTRIVITTVNGVTRIE
part2_trp296_1p9m_23_model_1_1	LTEAARTLTPAAFAAAAAAALAAEIAAGRADAAELAAAYRAAIARDVALGLATPEQAAAAAAAIAAAAA

part2_trp296_1p9m_8_model_6_1	ARREAILARLLAAAEARGPLSLAEAAAAAGVSPATIRRLEAAGYLVRQPDGTVVPSAKAL
part2_trp296_1p9m_13_model_5_1	GVLARVEREVPRELALAAALERLAQRPGPLGELARRVLAGRGESELSDEEREAVAAAVLAEAKALAAAA
part2_trp296_1p9m_13_model_6_1	MSEEAARALAREVGVSVAEVERAAAVLRETGWDVEAAVARLVAEGLPEAKARAVTERLAALREAGRP
part2_trp296_1p9m_38_model_6_1	CLCTPTILSGSATVTCTCATCPEADAPTFTYTVSCSSSVPAVVQVTAQVGRAPPVTETVTCS
part2_trp296_1p9m_38_model_7_1	GPPECAAGWAAFQAGDLAAARAHFEAALARDPDHAACRRALAEVYAALGDEAAAAALRAAAT
part2_trp296_1p9m_9_model_4_1	RVTVVDRATGEVVTDITSRSTPTDLPEVTVDYGRPLAPGDEVDTVLER
part2_trp296_1p9m_27_model_7_1	DLEAAVAELAARLSPEELLALADAIRAARDAGDYDAVGALFRAAAARAGVERVTRAEQVPWTAADADAIAALLRAAAA
part2_trp296_1p9m_30_model_7_1	MRVRTVYVDAETGEPVPEADLANYEVLGPALPPEDEPPGVRVVAVRPRG
part2_trp296_1p9m_7_model_8_1	GSRVVLVPASELPRWRALEAEGRAEILGEVETPDGRYLVVRVLE
part2_trp296_1p9m_33_model_0_1	AERAFATREAAEALAALRAAGAVADLVETPDGWVVRWRG
part2_trp296_1p9m_21_model_1_1	PERVTVEGVADPEAALAAARAAGYTDLTPPALVDGLWRFEAVR
part2_trp296_1p9m_32_model_2_1	TTTVLLIPRADLAALEAAAAAAGATVEAAPDGPDWLVRVITS
part2_trp296_1p9m_17_model_0_1	AASALPDAEVRRLAAAAGLDIPEGARFLELPPEQQAALAEAYLAA
part2_trp296_1p9m_30_model_2_1	PSADISPETRARVRAALRANPLTEETLRAIAEELGVPLEASEAVAREVL
part2_trp296_1p9m_15_model_5_1	PERRCAIYEAYLAGLSQAEIAARFGVPRSTVSRVLAEYERALATGHFPPSPPP
part2_trp296_1p9m_1_model_9_1	LLARIDAAIAAVEAAEPGQPISARVAAAAGVSEAAVRRALAAAGVVF
part2_trp296_1p9m_1_model_6_1	GRPLLRTVTGTGRVYEPLGGPGGRVRDVATGEVSTLTPEQAARLTQLYS
part2_trp296_1p9m_1_model_2_1	LAPEFPASAPPNTTVTRTRRVGNNTTFVETRVYDSNGNVSSSSVSGIPA
part2_trp296_1p9m_37_model_0_1	DAVLAAELALAPELPADVRAALAATRDPAARRALLAAAGYPEVA
part2_trp296_1p9m_34_model_4_1	GLTAADRAALDALYAAALAVVSPEAAEALYRAAVAAAAA
part2_trp296_1p9m_16_model_7_1	IRVTVKTSTDPAATTAATAAAAVEAVSPRVKASADPATGTITISASTAEDMAAAVATAEAMGGTVVATSS
part2_trp296_1p9m_13_model_8_1	PVVTPAQLAAIAAALGLDAATTALLRALQAARAAGDAAAAAALWAEIARVGAARAAAIEAALAAAG
part2_trp296_1p9m_8_model_9_1	MDGGVPTDVSALPPLTPEQAARVVAALGLPGVTAEDLLAAWRAGRLSPELLAAVQAVLGW
part2_trp296_1p9m_17_model_9_1	GAPPALVAAILAALATDPALAAATAGLSPEERLALAEAAARALAA
part2_trp296_1p9m_23_model_4_1	MRGRVGDLALEALAERGPLTLAELRAELRRRGVRVSRATLSAVLRELAAEGVLRRSRNARGRTVYSLAA

part2_trp296_1p9m_21_model_7_1	GVEVTVEVRDSGGAVSVTVTVRNLTGESVTVTVVVVTAPGTIP
part2_trp296_1p9m_17_model_7_1	ALLAADLAGRSVAEVAAAAAGVSPATVARELREVLRLPLPPGAPHA
part2_trp296_1p9m_29_model_1_1	VLADPAAADALLATLTPEQRAAAARA AVLALLAGPPVDAAAAAAAAAVAAALEALLPGLTVSPGTAAAIQ RVLAEAA
part2_trp296_1p9m_0_model_2_1	PPLPPAPPGETLVAPPPSEGGVILGAGLPRRV RGETDTVSELVG
part2_trp296_1p9m_39_model_2_1	GSYTTVTVEVG DYVVTVTVAVAYDSNGNVIYFEVSVTVV PKP
part2_trp296_1p9m_27_model_3_1	MGVARVEIEPLEPDAPPGA EVRLRVLTTHPDTPLEAAAREA EAVAAHPEVARVLAVGEPLEREGERVVLEVLAVRR
part2_trp296_1p9m_7_model_2_1	PVAAQAQAAAQAAALAAALGATVTVTVTTTNGSATVTTTATAAP
part2_trp296_1p9m_37_model_6_1	WVVTVTLPDGSVVTLTAPSEEEEEAAALAAAAAAAFTPAAIAIRPA
part2_trp296_1p9m_15_model_1_1	AVRASATVPAEERAAIPPGGSREVTPDWSSDPAARVVVSRDARGRWRLRVEPP
part2_trp296_1p9m_4_model_6_1	GPEALRALAAAAAAAAAAGRARVTLTLTPPEALARMRRLAAEVSAEFPRVTVRVVA
part2_trp296_1p9m_21_model_2_1	TRTLTLTCPGCGRTHTVTLTAAGTYQIPCPCGSDGTLTVTVVF
part2_trp296_1p9m_1_model_3_1	PALTERERRILELRRAGLSQAEIAAALGISPATVSRHLRRAEAAGFPL
part2_trp296_1p9m_8_model_1_1	GVETLRALAAAEADPRVRAVLTAVAELAAAGDHTLA EAAARLGISRATLRRHLARAEAL
part2_trp296_1p9m_18_model_4_1	DGLPTATITVSQARDPEALVTVTTP EGQTYDARLVSDL SAAAAGSTVTARVRVLDSEGRPLEVPVRVTVVAD
part2_trp296_1p9m_34_model_9_1	LTTVTTLRLDGVPTVTSRADGAPAAIARPAGGVTVLSTTVV
part2_trp296_1p9m_39_model_7_1	GGTVAVYDTNGNLAAIVTYDANGNVVRVEGTLVPNVGRVVPL
part2_trp296_1p9m_16_model_5_1	GRTISVTVA AKSTKTLTLTEDRPVTTASAEITTAGDLTVTITASGN TITVTITNNEDTPVTATVRIGT
part2_trp296_1p9m_4_model_5_1	AVAAADALVAAGDAEEAAGDLAAAVAYYEEAARLYRAAGRGAESA ALEARANALAAAA
part2_trp296_1p9m_32_model_8_1	PGTVTVTKDGVTTT TYTVTSLEEEAEALKAEAKAAGGTVTAESA
part2_trp296_1p9m_1_model_4_1	EEEALARLAALLEERAARGITDEEVAAA LGLPPATTRRLRRRVLARAR
part2_trp296_1p9m_28_model_2_1	SLTVDFYLTDAAGRVTLVASRPCTAPGERVTH TTLVDGRPVTQSATCPGRA
part2_trp296_1p9m_30_model_1_1	AAPEVRAYAPGDTITNTDPLAETR TLTFDPETGASTTVTLARGASITL
part2_trp296_1p9m_33_model_2_1	ATAATTTTREETE EVAPGVYRVTVTRTGVNGDTFTEEYLEVEE
part2_trp296_1p9m_28_model_4_1	MEAAIAALEAEIARLEAE LAKASGAEEAARLAAELARARARLARLRAELAAA
part2_trp296_1p9m_16_model_8_1	TTLAVTPAADAVVTLEARPVGSTEWVPVAEEA VAAGETVSLTAPEGTEARVVARTADGAPVAYTATAA

part2_trp296_1p9m_7_model_0_1	TAARPSPEELAAARARYAELARAPRDLPEGEGARLAALLAYAAA
part2_trp296_1p9m_14_model_1_1	SPEAAAAVEALRAAGVELPDVQEEIPGTPGTVVTLPGLTDGVPVTVTFTLSVEDGRARVDVAVERAD
part2_trp296_1p9m_18_model_5_1	GAADAPACRAHRDEILAAVDAAVAAAAPPAVAPAVEAALTAEGRLALEAPDASADEVIARLRVAAAATVAAA
part2_trp296_1p9m_17_model_1_1	ALTEPPVEVRVEGDTIIATTSTHEYVVKLENDLVEEAELEVRPRA
part2_trp296_1p9m_20_model_7_1	GRGREILAAVEAGVPQAEVAARFGVSRATVSRWVAWERARRAA
part2_trp296_1p9m_11_model_8_1	AGLAHGRPAPGETVTLTSETSTQLLTSRVTRAPDGTVTVEDTVTPK
part2_trp296_1p9m_11_model_4_1	GPLLRAVEALARRPVVTDEELAAALAAELGVSAELVRELLAELRARA
part2_trp296_1p9m_22_model_9_1	MSTTTVSGNTVSVTITSSTTTTVTVTLYDENGNALATTTVTVPANATKTVTFTLP
part2_trp296_1p9m_10_model_3_1	MSRIAVAPAEVTTITNTGAEPITVTIEGRAPVTLAPGESITVPPGARYEVRTPDT
part2_trp296_1p9m_22_model_6_1	NPAVTRALEALGLSPEAADRVLAIEIERRGLMLPTRDALEVVREAAAAEGIELPEA
part2_trp296_1p9m_6_model_4_1	DLERLARELAARFGAPLEAVLAVLRRVAARPVLTDEEIAALAAALGTTPEELSRWVLELEEA
part2_trp296_1p9m_26_model_4_1	GAGAAVAAALALGAALGAPALGAAAAAAITAAAAAAAAAAAA
part2_trp296_1p9m_26_model_1_1	PAAARAARAWLAAHPPEEARAAILARYGEVSPEAVEAYLAEA
part2_trp296_1p9m_32_model_1_1	AGSPRARAAAELALAGVSQAEIARRLGIARSTVSRVARAAGV
part2_trp296_1p9m_37_model_1_1	PSVVTLRAPGEEVVLEGVDVAPEVEALEAEYAARGIPVEVEAEA
part2_trp296_1p9m_19_model_9_1	PAATACLLPDGTLVVRLGTPCPPGAACASAVGPTEAEARAAAEARLRALAAALA
part2_trp296_1p9m_13_model_3_1	RRRLTLVVDAANALALLRGACEAAGLPFPEAEVAARGTSFAAMAALAAEVWAKQGVELTVIDAADAP
part2_trp296_1p9m_29_model_8_1	AHAAAEAAAAAEGLDAAALKAAALDRAVAAAYDSPLFSDEEVARITATAIPALVAKQKAEGTITDADVDAAAAALR
part2_trp296_1p9m_2_model_4_1	ATRYVVVDEDGTPLGRFDDLAAAEAAAELEARTGRPYRVVRVED
part2_trp296_1p9m_9_model_0_1	GAARLRAEARAIIAAGDLAALAELIALARELGLDALAAELRAEAAAAA
part2_trp296_1p9m_13_model_9_1	AGTDPADPRGESALAPLVDAVEARGTPEQRARLAEVLAALMAPARAGQAPTAEQRAAALAAVEALLA
part2_trp296_1p9m_7_model_4_1	AAAAAAAAAAAAAAAAAAAAAAGDAAAAARYAELAAAAARAAAA
part2_trp296_1p9m_20_model_4_1	MGAEIRRLRAEGLSDEEIARRLGVSRSSTVREAIRRLLGLEPER
part2_trp296_1p9m_36_model_0_1	PEVERLIRELEAAGAEVLDVRREVGPDSTVTVTITYRRDGEVSTITLTSAESP
part2_trp296_1p9m_34_model_1_1	GRWVLVARSKRDPSRLYYFDNMEEFLASPRREDLEIVGWRL

part2_trp296_1p9m_13_model_2_1	VAVEVSVEDGVPVARLGGGADPTPAQVELARQALADAGVDGRETVVVRGEGAGAEVARALAEAGVARL
part2_trp296_1p9m_22_model_0_1	GRRTAEAQVRAAAGDRAIARVEVVEAPEVVAPGSRVTYRVTLTLEDGEEREVTVTLG
part2_trp296_1p9m_7_model_9_1	TREVRRLDETTAVLVVTDAAAGRTTIYRVDLVTGEETEIAELPEL
part2_trp296_1p9m_18_model_8_1	CDLARAARLVGLTWEEVAARLLAEGHSAADVAAALAAARAADPGPTLSLAGANVVVAVWLEDPARFAALRALA
part2_trp296_1p9m_7_model_1_1	AEETLAVVEAARAYAAAHGIPLAEVAAALAAAGRAAAAAALAAA
part2_trp296_1p9m_33_model_4_1	GRLYTVTDAAGNLLATATSYAAAVAAPAGTAYTVTVTPL
part2_trp296_1p9m_39_model_5_1	MWEEALRRIEELRAEGKPREEIIREAARLLGISEEEVRRALR
part2_trp296_1p9m_27_model_4_1	SLREAVRAEALAAGLSEEDAEHAAEAIEEAVTLARNGRLDDEARAAVAAAAAALPDPEAAAWLRERGKALIAEAVAAAA
part2_trp296_1p9m_17_model_4_1	GRARARRAVLEAARTGVARDARGRPVTVDSPLERAVARELG LPIP
part2_trp296_1p9m_22_model_1_1	DPARVVLTYSAPVDAATRAALAAEIRAVAVRHADASAEIDRLAAPHGVTDWTAEQ
part2_trp296_1p9m_36_model_7_1	GLRDTVLLRRPDGSSVPVEAVAESTGTRERVVVRYVALGPERRVVAEEVVT
part2_trp296_1p9m_31_model_0_1	GDRARLAALRAEAAALRAAVLAAPRLSPEELAAALRARLEALAAERRAIAPGLSDEELRALGEALQALAAALDAAAARL
part2_trp296_1p9m_31_model_4_1	SGTVTVTLDGKTVGTGAYSDAGASGVTMTGDIPAGATVTITLTVTGGDSVTRTVTHTGADIAAGSTITVTVNSGGTV
part2_trp296_1p9m_26_model_7_1	DPAALAAAEAAAAALRAAGRAEEAERVRAALLAAVPSYYPL
part2_trp296_1p9m_17_model_5_1	GGGAAVAATLAALPPSLPAAERAAIAAAAEAAARAANPSTVLDSL
part2_trp296_1p9m_34_model_2_1	PSLAEWRATMAAAAALAAALGGRVLSVRYDAATRRLVVTIDF
part2_trp296_1p9m_30_model_8_1	PAEEAVRLLEAGVPPEEVVARVRARPGVLAAEVS LDEETGRWVARVERE
part2_trp296_1p9m_35_model_7_1	TAAARATLEAEIARLERLLAEKSAELSPEELARLRERLARARRRLASL
part2_trp296_1p9m_21_model_9_1	EGGTFTVEVSAAPGETVVLTSSYGGFLTRYNLATGEVTVEDLR
part2_trp296_1p9m_37_model_5_1	GMSAEQRARTAALIRAHPDWSVAEVAAATGVSRAAASAVRRELA
part2_trp296_1p9m_32_model_9_1	CQAAADAAARAVVDALRAAGLGDDPAVDIAIDAVRAATLAAC
part2_trp296_1p9m_6_model_9_1	GTALDEAILALIREEGELTVDEIAARFGISRATASARVRTLEARGLVVRRVVDGRVVVSPAP
part2_trp296_1p9m_28_model_3_1	PVRLRV TIRAPDGT LHTVTVTAESEAAAAAAAEIAAAIHDPARIVAVEPV
part2_trp296_1p9m_20_model_3_1	AVVTETVTSALDATTRATTTVATAVDGDTVLRVTVSAPVVT
part2_trp296_1p9m_10_model_7_1	ASCTATVTVELSPGGRVLSVRESWVL RDGRLYLVREV VVEDRGRRRVYAVVECP

part2_trp296_1p9m_2_model_2_1	MVTVLTTNLETGETTEEVRPLPPGATLEEARAAAVSPPGFVEVRVE
part2_trp296_1p9m_35_model_6_1	AVRVEEVLRRRERLPDGSTLVVGRASDGS�VEALYDAAGRLLSRRRAP
part2_trp296_1p9m_30_model_5_1	ADVAAARAATAAERAALAAASAAAGPDPAAAAEIAALEARLAAREAALA
part2_trp296_1p9m_31_model_6_1	GALRALVAALERVLGPGARPGASARLADVLAEVQAAHGVSRREEAEAAALAAL EEAGFLRREGDTITLLRPLAEIRAWAA
part2_trp296_1p9m_16_model_1_1	GALERAIRARDVDAAAAAAAAAAEVRRRAAE EGRDADAFAEVAAYLRERLPPALAEAVIARARARLAAA
part2_trp296_1p9m_31_model_2_1	PVTTTAAPVTVAPGETATVQAGDTTLTITAPAAEDTVLGTRVERLPLPQPGGWARVQITTSVVDGVVRHTVTVVTETP
part2_trp296_1p9m_31_model_7_1	AGPAARAVAAELRAAGATDVEVLSAEPLPGGLVRVTVTYVDPAGVRREV VYVCRVSEDEDGRPVVEVVAREERPIAAP
part2_trp296_1p9m_28_model_1_1	VAACAAAGAAATAATAARLAAANLDPSVTAAERAAAAAAVGAATTAACLAA
part2_trp296_1p9m_7_model_6_1	AGLALDRYLD AHP LATVDQRELAARFGVSEATVSRAIRRRRAEL
part2_trp296_1p9m_6_model_7_1	GTVTVVAVDENGNRSTPVEVDLDALSPAERARILARLAATGSDTLVATGEEARRIARAYAAA
part2_trp296_1p9m_24_model_6_1	SRLPPRVTL EVPADASDEEIEARIAALARSYGATDVTVEWHDT P SPVPGVRLATATGTIDGIPVTSEVEVRLL
part2_trp296_1p9m_2_model_1_1	PVVLSRTVTTTGNTTTTVTEVVAYVSTDSSGRTTVERVTRITVTTF
part2_trp296_1p9m_14_model_3_1	AAAQAAALAAQG AALAAAGDLEAAAAAYEEAAALAPDNAAIWAALA AVYAALGREDLAAGARAHAAA
part2_trp296_1p9m_0_model_0_1	REAAALAAALARVGGELAAVEETPEGLLV TIVDPETGRRTRVLLT
part2_trp296_1p9m_34_model_6_1	GLEVLPDRVVVSAPTLEAARAAAPAVAAELGVPLERVVVRAA
part2_trp296_1p9m_35_model_5_1	EGQRLVEEVRARKAALGLERLTEVRYVPSLQANVRITVEGDRVTVEVV
part2_trp296_1p9m_26_model_0_1	MALREALAAAPAVRVGDLLVRVAEAVENGRRHV E VVVEPLA
part2_trp296_1p9m_20_model_8_1	AERVARLRELAAQGLSVREIAAALGLSES AVSALARREGIELP
part2_trp296_1p9m_29_model_6_1	MTAAQAAALAARAAA E HSAADAVAAFKAAAAAE CNVIPIDSTEGLTASEEPGRTVLRALVAENGRLVEVVATTAV
part2_trp296_1p9m_39_model_3_1	GSTTTCTATAPVATPGVTVTETCTTT SNGVTTTTTRQAIPSP
part2_trp296_1p9m_38_model_2_1	ALEAALAAVGQEAARLITPEQALELAALVAAQPTLEAALRVLAERL RALGLDDAAARAEAAA
part2_trp296_1p9m_4_model_8_1	MGGRVLALLAQGPRSLAEIAAATGRNASALLARLEALGLVERDPATRLYSVPAGRAA
part2_trp296_1p9m_0_model_9_1	SRGVATVTCPVCGYTFERPIPEEDGVTLDDIVCPVCGSTELARA
part2_trp296_1p9m_21_model_0_1	SGRLLEALVRLDGVERLLRG TREELDAELERLGGEVLAVREAP
part2_trp296_1p9m_5_model_4_1	GTVRVTFADGRVVEVPGGEEGVREVARLEAEHGPPVVAADV

part2_trp296_1p9m_18_model_6_1	TLVRVRVARGDEAAVAAVVAAFAALGFAAEVEETPDGALVVVRVPDERLPEARAVADELVARGLALGYTALP
part2_trp296_1p9m_17_model_3_1	DTAALLARVAAEESGEARAALVAALEAAGRTEAAAAAAEAAALAKA
part2_trp296_1p9m_7_model_5_1	PDVRTREEALAAELDPELAAELRARNASLDELAALTALRAA
part2_trp296_1p9m_12_model_0_1	DLPARVDALAAALAAEAPPDVEVTWRGLPTTPGTPTQEGVGRYELLAVDKATGEATSLRVLTGEEVRA
part2_trp296_1p9m_19_model_4_1	GFRVEVEGDRARGVATRDAAEALAEELRARGFRLVREVPGVAGRPPRTVLEFERE
part2_trp296_1p9m_29_model_0_1	GFRAAVPLRVGDPVEIRLDDGTVVPGRVDRVEGDVVSVALEPLTPGVHRVTVRFLTEPPQTSEPRLVEFPFRVP
part2_trp296_1p9m_24_model_3_1	MAEVAAILAAAEARMRTDPAADVDAAAAAEAAADAALDAAVAAGLLTPEEAAAQRAAFRAELARVAAAAARAA
part2_trp296_1p9m_20_model_9_1	GPIAAAAAAVRDALIAAGVPEAEAAAAAAAVEEALLAAQNAAN
part2_trp296_1p9m_28_model_6_1	RLVVVGDRAAALERAVALDPSTVEVAGTVAAKDAYIRARCAAHGLPPVALAE
part2_trp296_1p9m_12_model_4_1	GVTVDVAGLSPAAAAALAAAVRATLEALGLTNVVTVADGVLTAGAEALDVSPAAVRAAVAAAAVAVR
part2_trp296_1p9m_34_model_0_1	ATTTTTTSTPDGSAAGSSTVTTTTTAAGTTTTTSAPTSTVT
part2_trp296_1p9m_34_model_7_1	GDLSALADVYLERGDVVTPDGTTRYRALDAGPLAEVRLEPLD
part2_trp296_1p9m_31_model_9_1	ALAAAAAAAAAAAAAATTPDALVAAAEALRAALPGASVTDLRVTTSADGRTITLTATVRWYDGDTLHEAPLTITRTV
part2_trp296_1p9m_23_model_9_1	TATAQTLALLAETLAATPWREVDSVVVESGDAARLVATLAALGIEAEETATPGTVRLSRESAARAAELI
part2_trp296_1p9m_31_model_3_1	ALELAAAGRTAEALARARAAVAAGAPAEAVAALARACLASGHPDAFAAAAELTALAAALDPAAGAAAAAALAAKAAA
part2_trp296_1p9m_19_model_5_1	VTESETRELPLAEVLERYGSLEALAAALRAAHPDYRVELHVDEERGLVRVRLSR
part2_trp296_1p9m_6_model_0_1	SNVLTLEDMTPEEIAARLRAELEAVAGVAVAAVETAEDGAVRVTTITLADGERATVTVRNV
part2_trp296_1p9m_21_model_3_1	AAARELEALYRRLVEEEGLSPEEAAARVAERRAALEAAAAAA
part2_trp296_1p9m_32_model_4_1	SVTRTLTLQAGQRVTTATVTRPDGSTVTRTFTAQVTGHTITL
part2_trp296_1p9m_10_model_5_1	ALAAAVAAAAAAHPDLTVVAATASVDAAGRPVVRVTLRDAAGRRLVVEVRLPGRA
part2_trp296_1p9m_30_model_4_1	DPDRCVIVYRAPDGREAAVVVPASTPAECAAAAAAALEAAGYTVLDVRLT
part2_trp296_1p9m_11_model_5_1	GVTSKTLTGAGTSTTATVLSRDPAANTCLVSVTTNGATSTETVPCP
part2_trp296_1p9m_18_model_3_1	LPPQEDPLDAAVAELAAYEAGNVTLSPHEELAALRAELDRLVAEEAARRGLPLAEVRARVEALRARYLALLAL
part2_trp296_1p9m_2_model_7_1	RIVEAVRRGEPFTVEAPPGSRLRTITFADGTVVRITSEELEALLA
part2_trp296_1p9m_36_model_6_1	GEARYRAALAAARAVLAADPEDVEGAVAVLVAHGVDEAHAREEAERLAAEVA

part2_trp296_1p9m_5_model_0_1	SGETRAAVAALLDAGLSQREVAARLGLSPATVSRIVREL
part2_trp296_1p9m_13_model_1_1	MGAALLDAALAATTDVAEVTVLVDGELRTIRLPVSADDDPAAIAARLDALFAAHPNVVAARLVPLP
part2_trp296_1p9m_33_model_3_1	LAVRVAREVLGVEVTDPREARELLAAHAAEVEAAAGVDTPP

6. AI Use Statement:

AI-assisted sequence design was performed. Computational predictions were distinguished from experimental validation.